

GHANA GOLD BOARD (GOLDBOD)



REQUEST FOR PROPOSALS (RFP)

**PROCUREMENT OF TECHNICAL SERVICES FOR DESIGN,
DEVELOPMENT, IMPLEMENTATION AND SUPPORT OF
TRACEABILITY AND ASSURANCE SYTEMS (TAS)**

Ref. No.: GR/GGB/TS/NCT/042/2026

JUNE 2026

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PART 1 – Tendering Procedures

Section I. Instructions to Tenderers

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REQUEST FOR PROPOSALS (RFP)**PROCUREMENT OF TECHNICAL SERVICES FOR THE DESIGN, DEVELOPMENT,
IMPLEMENTATION AND SUPPORT OF A TRACEABILITY AND ASSURANCE SYSTEM
REF. NO.: GR/GGB/TS/NCT/042/2026**

1. The Ghana Gold Board (GoldBod) intends to apply part of its Internally Generated Funds for the Procurement of Technical Services for the Design, Development, Implementation and Support of a Traceability and Assurance System (TAS).
2. The Ghana Gold Board (GoldBod) now requests sealed Technical and Financial Proposals from eligible shortlisted Firms. This Request for Proposals (RFP) follows your Firm's response to the Expressions of Interest (EOI) published in the Ghana Electronic Procurement System (GHANEPS) on 25th April 2026, the Ghanaian Times Newspaper on 27th April, 2026 and the Daily Graphic Newspaper on 1st May, 2026 respectively.
3. Shortlisted Firms are not permitted to transfer this invitation to any other Firm. The Selection of Firms will be based on the **Quality and Cost Based Selection (QCBS)** procedures specified in the Public Procurement Act, 2003 (Act 663) and its Amendment Act, 2016 (Act 914) and in the guidelines of the Public Procurement Authority (PPA) of the Republic of Ghana.
4. Firms may obtain further information from the Procurement Department, Room 406 from 9:00am to 12:00pm and 2:00pm to 5:00pm, Monday to Friday and download the Request for Proposal (RFP) document in the GHANEPS.
5. Proposals must be accompanied by a **Tender Security of One Million Ghana Cedis (GHS1,000,000.00)** from an Insurance Company in Ghana regulated by the National Insurance Commission (NIC) and must be uploaded on the GHANEPS.
6. Proposals (RFP) shall be in English and must be uploaded on the Ghana Electronic Procurement System (GHANEPS). The Technical and Financial Proposals **MUST** be uploaded **separately** and either marked Technical or Financial Proposal. Proposals shall be valid for 182 days. Firms are to note that, there will be a Proof of Concept (PoC) demonstrating proposed solution as part of evaluation of proposals.
7. Proposals must be submitted through the Ghana Electronic Procurement System (GHANEPS) at or before **16:00 hours GMT** on or before **Monday 13th July, 2026**.

Address:

**The Chief Executive Officer
Ghana Gold Board
P. O. Box M 108
1 Thorpe Road, High Street - Accra
Tel: +233 244 895 900
Email: frank.ademan@goldbod.gov.gh**

List Of Shortlisted Firms:

1. *Messrs Virtual Infosec Africa Limited*
2. *Messrs G20 Tech. Ltd.*
3. *Messrs Paramount Project Management Services*
4. *Messrs Persol Systems Limited*
5. *Messrs Samir Engineering and Trading Company Ltd.*
6. *Messrs Margins Systems Applications Limited*
7. *Messrs Tempdata Ghana Limited*
8. *Messrs The Digicoast Limited*
9. *Messrs Kollateral Investments and Advisory*
10. *Messrs EED Consult Limited*
11. *Messrs Red Mango Limited*
12. *Messrs Fauree Link (GH) Ltd.*
13. *Messrs Dapaul Company Limited*
14. *Messrs Webber Mill Limited*
15. *Messrs TechNet ProSol Limited*
16. *Messrs Smart Innovations Ghana Limited*
17. *Messrs AppsnMobile Solutions Limited*
18. *Messrs HFFields Limited*
19. *Neurix Data Centres Limited*

Section I. Instructions to Tenderers

A. General

- 1. Scope of Tender**
 - 1.1 In support of the Invitation for Tenders (IFT) indicated in the Tender Data Sheet (TDS), the Procurement Entity named in the TDS, hereinafter referred to as “the Purchaser,” issues this Tender Document for the Procurement of Information Technology Products and Related Services as specified in Part 2 (Supply Requirements). The name, identification, and number of lots of the International Competitive Tender (ICT) are provided in the TDS.
 - 1.2 Throughout this Tender Document the terms defined Clause 1.1 of Section VIII, General Conditions of Contract (GCC) shall have the meanings defined therein and, if the context so requires, singular means plural and vice versa.
 - 1.3 Tendering will be conducted through the competitive tendering procedures specified in the Public Procurement Act, 2003, Act 663 of the Republic of Ghana.
- 2. Source of Funds**
 - 2.1 The Purchaser named in the **Tender Data Sheet** shall fund this procurement from part of its budgetary allocation¹ toward the realization of the project named in the TDS.
 - 2.2 Payments will be made only at the request of the Purchaser and upon approval by a designated official of the Republic of Ghana in accordance with the terms and conditions of the contract agreement between the Purchaser and the Firm (hereinafter called the Contract Agreement or Contract), and will be subject in all respects to the Financial Administration Act, 2003, Act 654 and the Financial Administration Regulations, 2004 of the Republic of Ghana. No party other than the Firm shall derive any rights from the Contract Agreement or have any claim to the funds.
- 3. Corrupt Practices**
 - 3.1 The Government of Ghana (GOG) requires that all Procurement Entities as well as Tenderers, Firms, Contractors and Consultants participating in contracts financed from the public funds of the Republic of Ghana, adhere to the highest ethical standards, both during the tendering process and throughout the execution of such contracts. The list of definitions set forth below in conjunction with Section 92 and 93 of the Public Procurement Act 2003, Act 663 involves the most common

¹ Where source of fund is different, this is to be specified in addition to any associated terms and conditions of the financing arrangement in the Tender Data Sheet (TDS)

types of corrupt practices, but is not exhaustive. For this reason, the Public Procurement Authority will also consider claims of similar nature involving alleged acts of corruption, in accordance with the established procedure.

- (a). “Bribery” means the act of unduly offering, giving, receiving or soliciting anything of value to influence the process of procuring goods or services, selecting consultants, or executing contracts.
- (b). “Extortion” or “Coercion” means the act attempting to influence the process of procuring goods or services, selecting consultants, or executing contracts by means of threats of injury to person, property or reputation.
- (c). “Fraud” means the misrepresentation of information or facts for the purpose of influencing the process of procuring goods or services, selecting consultants, or executing contracts, to the detriment of the Procurement Entity/Purchaser or other participants.
- (d). “Collusion” is an agreement between tenderers designed to result in tenders at artificial prices that are not competitive.

3.2 If, in accordance with the administrative procedures of the Public Procurement Authority, it is demonstrated that a government/public official, or anyone acting on his or her behalf, and/or a Tenderer in a procurement process or Firm/contractor during the execution of the contract carried out in connection with a project financed from the public funds of the Republic of Ghana has committed corrupt practices, the Public Procurement Authority or the appropriate Tender review Authority will:

- (a). reject a proposal to award a contract in connection with the respective procurement process; and/or
- (b). declare a firm and/or its personnel directly involved in corrupt practices, temporarily or permanently ineligible to be awarded future contracts financed from the public funds of the Republic of Ghana.

3.3 The Tenderer shall disclose any commissions or fees that may have been paid or are to be paid to agents, representatives, or commission agents with respect to the tendering process or execution of the contract. The information disclosed must include the name and address of the agent, representative, or commission agent, the

amount and currency, and the purpose of the commission or fee. The information must be included in the Tender Submission Sheet. Furthermore, Tenderers shall be aware of the provision stated in Clauses 13.6 and 35.1 (c) of the General Conditions of Contract.

3.4 Any communication between the Tenderer and the Purchaser related to matters of alleged fraud or corruption must be made in writing

3.5 Any willful misrepresentation in the Tender Form regarding legal ownership, license or authorization to the Intellectual Property Rights in any Products included in the tender shall be considered a fraudulent practice

4. Eligible Tenderers

4.1 A Tenderer may be a natural person, private, public or government-owned legal entity, subject to ITT Clause 4.7, or any combination of them with a formal intent to enter into an agreement or under an existing agreement in the form of a joint venture, consortium, or association, all parties shall comply with the requirements of GCC Clause 4.6.

4.2 A Tenderer, and all parties constituting the Tenderer, shall have the nationality of an eligible country, in accordance with Section V, Eligible Countries. A Tenderer shall be deemed to have the nationality of a country if the Tenderer is a citizen or is constituted, incorporated, or registered and operates in conformity with the provisions of the laws of that country. This criterion shall also apply to the determination of the nationality of proposed subcontractors or Firms for any part of the Contract including related services.

4.3 A Tenderer should not be associated, or have been associated in the past, directly or indirectly, with a firm or any of its affiliates which have been engaged by the Purchaser to provide consulting services for the preparation of the design, specifications, and other documentations to be used for the procurement of the Products to be purchased under this Invitation for Tender.

4.4 A Tenderer that is under a declaration of ineligibility by the Public Procurement Authority in accordance with ITT Clause 3, at the date of the deadline for Tender submission or thereafter, shall be disqualified.

4.5 Tenderers shall provide such evidence of continued eligibility satisfactory to the Purchaser as the Purchaser may reasonably request.

4.6 Tenderers of an eligible country may be excluded if:

- (a) as a matter of law or official regulation, Ghana prohibits commercial relations with that country.
- (b) by an act of compliance with a decision of the United Nations Security Council taken under Chapter VII of the Charter of the United Nations, Ghana prohibits any import of goods from that country or any payments to persons or entities in that country.

- 4.7 Government-owned enterprises shall be eligible only if they can establish that they are legally and financially autonomous and operate under commercial law.
- 4.8 Unless otherwise specified in the **TDS**, Tenderers may submit more than one (separate and complete) tender in this tendering process, whether for the same or different lots, and may also participate as subcontractors in other tenders. Subcontractors may participate in more than one tender.
- 4.9 If a pre-qualification process has been undertaken for this procurement, as indicated in the Purchaser's Invitation for Tenders, only pre-qualified Tenderers meeting all other eligibility criteria specified in ITT Clause 4 are eligible to tender.

5. Eligible Products and Services

- 5.1 All goods and services to be supplied under the Contract shall have their origin in eligible source countries as specified in Section V, Eligible Countries, and all expenditures made under the contract will be limited to such products and services. Products and services originating from an otherwise eligible country may be excluded if that country is subject to the conditions specified in ITT Clause 4.6.
- 5.2 For purposes of this clause, "origin" means the place where the products are produced or from where the Related Services are supplied. Products are produced when, through manufacturing, processing, software development or substantial and major assembly or integration of components, a commercially recognized product results that differ substantially in its basic characteristics or in purpose or utility from its components.
- 5.3 The origin of the products and services is distinct from the nationality of the Tenderer and may be different.
- 5.4 By signing the Tender Form, the Tenderer confirms that it has the right or proper authorization to offer the proposed Products to the Purchaser under the terms of the Contract. The Tenderer shall demonstrate in accordance with ITT Clause 18.2 that it has been duly authorized by the

Producer of the Products to supply, in Ghana, the key Products included in its tender that the Tenderer does not itself produce.

B. The Tender Document

6. Contents of Tender Document

- 6.1 The required products and related services, tender procedures, and contract terms are prescribed in the tender document. The Tender Document consists of Parts 1, 2, and 3, which include all the Sections indicated below, and should be read in conjunction with any Addenda issued in accordance with ITT Clause 9.

PART 1 Tendering Procedures

- Section I. Instructions to Tenderers (ITT)
- Section II. Tender Data Sheet (TDS)
- Section III. Evaluation and Qualification Criteria
- Section IV. Tender Forms
- Section V. Eligible Countries

PART 2 Supply Requirements

- Section VI. Schedule of Supply
- Section VII. Technical Specifications

PART 3 Contract

- Section VIII. General Conditions of Contract (GCC)
- Section IX. Special Conditions of Contract (SCC)
- Section X. Contract Forms

- 6.2 The Invitation for Tenders issued by the Purchaser is not part of the Tender Documents.
- 6.3 The Purchaser is not responsible for the completeness of the Tender Documents and their addenda, if they were not obtained directly from the Purchaser.
- 6.4 The Tenderer is expected to examine all instructions, forms, terms, and specifications in the Tender Documents. Failure to submit a tender substantially responsive to the tender documents in every respect or failure to furnish all information or documentation required by the Tender Documents will be at the Tenderer's risk and may result in the rejection of its tender.

7. Clarification of Tender Documents

- 7.1 A prospective Tenderer requiring any clarification of the Tender Document shall contact the Purchaser in writing at the Purchaser's address indicated in the TDS. The Purchaser will respond in writing to any request for clarification no later than the number of days specified

in TDS prior to the deadline for submission of tenders, provided that such request is received no later than the number of days specified TDS prior to the deadline for submission of tenders. The Purchaser shall forward copies of its response to all those who have acquired the Tender Document directly from it, including a description of the inquiry but without identifying its source. Should the Purchaser deem it necessary to amend the Tender Document because of a clarification, it shall do so following the procedure under ITT Clause 9.

- 7.2 When specified in the **TDS**, the Purchaser will organize and Tenderers may attend a pre-tender meeting at the time and place indicated in the **TDS**. The purpose of the meeting will be to clarify issues and answer questions on any aspect of the Tendering Document. Tenderers are requested, as far as possible, to submit any questions in writing, to reach the Purchaser not later than one week before the meeting. Minutes of the meeting, including the questions raised and responses given, together with any responses prepared after the meeting, will be provided in writing without delay to all those who acquired the Tender Document directly from the Purchaser. Any amendment of the Tender Document resulting from the pre-tender meeting shall be made exclusively by issuing an amendment in accordance with ITT Clause 9 and not through the minutes of the pre-tender meeting.

8. Site Visit

- 8.1 The Tenderer may wish to visit the Project Sites specified in the Implementation Schedule in Section VI, Schedule of Supply, for installation of the Products and obtain for itself, at its own responsibility, expense and risk, all information necessary for preparing its tender and entering into the Contract.
- 8.2 The Purchaser will organize the site visits concurrently with the pre-tender meeting referred to in ITT Clause 7.2, as specified in the **TDS**. Alternatively, the Purchaser may arrange for the Tenderer and any of its personnel or agents to visit the sites separate from the pre-tender meeting, provided that the Tenderer gives the Purchaser advance notice of a proposed visit of at least the number of days specified in the **TDS**.

9. Amendment of Tender Documents

- 9.1 At any time prior to the deadline for submission of Tenders, the Purchaser may amend the Tender Documents by issuing addenda.
- 9.2 Any addendum issued shall be part of the Tender Documents and shall be communicated in writing to all who have obtained the Tender Documents directly from the Purchaser.
- 9.3 To give Tenderers reasonable time to take an addendum into account in preparing their Tenders, the Purchaser may, at its discretion, extend the deadline for the submission of Tenders, pursuant to ITT Sub-Clause 23.2

C. Preparation of Tenders

- | | |
|--|--|
| 10. Cost of Tendering | 9.1 The Tenderer shall bear all costs associated with the preparation and submission of its Tender, and the Purchaser shall not be responsible or liable for those costs, regardless of the conduct or outcome of the Tendering process. |
| 11. Language of Tender | 10.1 The Tender, as well as all correspondence and documents relating to the Tender exchanged by the Tenderer and the Purchaser, shall be written in the language specified in the TDS. Supporting documents and printed literature that are part of the Tender may be in another language provided they are accompanied by an accurate translation of the relevant passages into the language specified in the TDS, in which case, for purposes of interpretation of the Tender, such translation shall govern. Information supplied in any language other than the language of the tender without the proper translation shall be rejected. |
| 12. Documents Comprising the Tender | <p>11.1 The Tender shall comprise the following:</p> <ul style="list-style-type: none"> (a) Tender Form furnished in Section IV, Tender Forms, duly completed and signed in accordance with ITT Clauses 21.2. This form must be completed without any alterations to its format, and no substitutes shall be accepted. All blank spaces shall be filled with the information requested; (b) Written confirmation authorizing the signatory of the Tender to commit the Tenderer, in accordance with ITT Clause 21.2; (c) Tender Security furnished in accordance with ITT Clause 20; (d) Price Schedules furnished in Section IV, Tender Forms, duly completed in accordance with ITT Clauses 13 and 14 and the General Instructions for the Tender Forms, signed in accordance with ITT Clause 21.2 and including the declarations of the country of origin required by ITT Clause 16 to confirm that the Products and Related Services offered by the Tenderer are of eligible origin; (e) Unless otherwise specified in the TDS under ITT Clause 13.4(d), the Supplemental Agreements required pursuant to GCC Clause 8 for provision of Maintenance Period Products and Services in accordance with GCC Clause 9.2; (f) A list of all key items of the Products or Services that the Tenderer proposes to purchase or subcontract from others, and the full name, address and nationality of the proposed |

Subcontractor for each of those items in the Schedule of Key Subcontractors;

- (g) Documentary evidence in accordance with ITT Clause 15 establishing the Tenderer's eligibility to tender;
- (h) Documentary evidence in accordance with ITT Clause 18 establishing the Tenderer's qualifications to perform the contract if its Tender is accepted;
- (i) The Tenderer's committed Implementation Schedule for arrival and Installation of the Products and delivery of the Services specified in the Schedule of Supply;
- (j) The Technical Proposal in accordance with ITT Clauses 17 and 29, which shall include the Project Plan and documentary evidence that the proposed Products and related services conform to the requirements of the Tender Document;
- (k) A list of deviations in the tender to the contractual terms or to the Supply Requirements in the Tender Document. For any of the deviations to be considered, the Tenderer must specify the increase in Tender Price that the Tenderer would accept for the withdrawal of the deviation and indicate the exact specification or contractual provision in the Tender Document to which the deviation applies. Tenderers are reminded that their tender may be rejected as non-responsive, pursuant to ITT Clause 29, if material deviations are taken or critical requirements are not met.
- (l) any other documentation specified in the TDS.

13. Tender Prices and Discounts

- 13.1 The Tenderer shall identify and express the prices of the proposed Products and Related Services in the Price Schedules and in the Tender Form furnished in Section IV, Tender Forms, in accordance with the instructions therein and the requirements specified below.
- 13.2 The terms EXW, CIP, etc., shall be governed by the rules prescribed in the INCOTERMS 2000 edition published by the International Chamber of Commerce, Paris.
- 13.3 All items in the Required Products and Services Schedules in the Schedule of Supply must be listed and priced separately in the appropriate Price Schedules in accordance with the General

Instructions for the Tender Forms in Section IV, using separate Price Schedules for each lot. If a Price Schedule shows items listed but not priced, their prices shall be assumed to be included in the prices of other items. Items not listed in the Price Schedule shall be assumed to be not included in the tender and, provided that the tender is substantially responsive, the corresponding adjustment shall be applied in accordance with ITT Sub-Clause 30.4.

13.4 Prices for the proposed Products and Related Services shall be entered separately in the Price Schedules, as follows:

- (a) Unless otherwise specified in the **TDS**, unit and total prices of Products offered from outside Ghana shall be quoted CIP (named place of destination) to the Purchaser's Project Sites specified in the Implementation Schedule in Section VI, Schedule of Supply, on the Imported Products Price Schedule, as follows:
 - (i) Unit and total cost of the Products, exclusive of all taxes and duties in Ghana. The Tenderer shall be free to use transportation through carriers registered in any eligible country and to obtain insurance from any eligible source country. Any specific requirements for the contract of carriage shall be specified in **SCC** Clause 18.3(a).
- (b) Prices of Products offered from within Ghana, on the Locally Supplied Products Price Schedule, as follows:
 - (i) Unit and total prices of Products quoted EXW (ex works, ex factory, ex warehouse, ex showroom, or off-the shelf, as applicable), including all customs duties and sales and other taxes already paid or payable on the components and materials used in the production of the Products, or on the previously imported Products, but excluding sales or other taxes assessed on the Products in Ghana if the contract is awarded.
 - (ii) the price for transportation, insurance, and other local costs incidental to delivery of the EXW components of each Installation Batch of Products specified in the Implementation Schedule in the Schedule of Supply, to its designated Project Site, unless otherwise specified in the **TDS**, listed separately for each Installation Batch on the Related Services Price Schedule in accordance with ITT Clause 13.4 (c).

- (c) Prices for Services required by the Tender Document related to the supply and Installation of the Products during the Implementation Period (e.g., delivery charges for EXW Products and Project Implementation Services as described in GCC Clause 15) and for the Warranty Period (e.g., maintenance and support services as described in GCC Clauses 15, 16 and 17), including all incidental costs associated with the provision thereof, and all taxes, duties, levies and fees imposed on the Tenderer, its subcontractors, or employees on account of the Services in any country, shall be separated into the local and foreign currency components on the Related Services Price Schedule.
 - (d) Unless otherwise specified in the **TDS**, maximum costs (including provision for any appropriate price escalation or adjustments over time) for the Maintenance Period (as defined in GCC Clause 1.1(q)), on the Maintenance Costs Schedule with separate Products and Services subtotals, as follows:
 - (i) The cost of all supplies, software license renewals and new versions, spare parts and other Products required during the Maintenance Period or necessary for continued successful and proper operation of the Products for the Maintenance Period, quoted CIP or EXW in accordance with ITT Clause 13.4(a) or 13.4(b) for foreign or locally supplied Products, respectively; and
 - (ii) The cost of all (recurrent or subscription) maintenance and support services required during the Maintenance Period as indicated in GCC Clause 9.2, separated into the local and foreign currency components in accordance with ITT Clause 13.4(c).
 - (e) Totals of the Prices for Products and Related Services, and for Maintenance Costs, subtotaled by currency and delivery basis (EXW or CIP), on the Tender Price Summary Schedule.
- 13.5 The Tenderer's separation of price components in accordance with ITT Clause 13.4 above will be solely for the purpose of facilitating the comparison of tenders and will not in any way limit the Purchaser's right to contract on any of the terms offered.
- 13.6 The price to be quoted in the Tender Form, in accordance with ITT Clause 12.1(a), shall be the total price of the tender, excluding any discounts offered. The Tenderer shall quote any unconditional

discounts and the methodology for their application separately in the Tender Form.

- 13.7 Prices quoted by the Tenderer shall be fixed during the Tenderer's performance of the contract and not subject to variation on any account, unless otherwise specified in the **TDS**. A tender submitted with an adjustable price quotation will be treated as non-responsive and will be rejected, pursuant to ITT Clause 29. If, however, in accordance with the **TDS**, prices quoted by the Tenderer shall be subject to adjustment during the performance of the contract, a tender submitted with a fixed price quotation will not be rejected, but the price adjustment would be treated as zero.
- 13.8 If so specified in the **TDS** under ITT Clause 1.1, tenders are being invited for individual lots (contracts) or for any combination of lots. Tenderers wishing to offer any discount or price reduction for the award of more than one Contract (lot) shall specify in their tender the price reductions applicable to individual lots or to a combination of lots. Price reductions or discounts shall be submitted in accordance with ITT Sub-Clause 13.6, provided that the tenders for all lots are submitted and opened at the same time.

14. Currencies of Tender

14.1 Tender Prices shall be quoted in the following currencies:

- (a) for Products and Related Services originating in Ghana, including delivery and insurance for EXW products, the Tender prices shall be quoted in the currency of Ghana, the Cedi (GHC) unless otherwise specified in the **TDS**; and
- (b) For Products and Services to be supplied from outside Ghana, in the currency of the Tenderer's home country or in another currency widely used in international trade. If the Tenderer wishes to be paid in a combination of different currencies, it must quote its price accordingly, but no more than three foreign currencies may be used.

15. Documents Establishing the Eligibility of the Tenderer

- 15.1 To confirm their eligibility in accordance with ITT Clause 4, Tenderers shall complete the eligibility declarations in the Tender Form, included in Section IV, Tender Forms.
- 15.2 In the absence of pre-qualification, documentary evidence of the Tenderer's eligibility to tender shall establish to the Purchaser's satisfaction that the Tenderer and all parties constituting the Tenderer are, at the time of tender submission, eligible to participate in the Tender process as defined in ITT Clause 4, including but not limited to documentary evidence that the Tenderer is legally

incorporated or registered in a territory of an eligible country as defined in ITT Clause 4.2 and operates in conformity with the provisions of the laws of that country.

- 15.3 If a Tenderer intends to subcontract the supply of any Products or Services designated as a “Key Item” in the Schedule of Supply, it shall include in the tender details of the name and nationality of the proposed Subcontractor, ensure that the Subcontractor complies with the requirements of ITT Clause 4 and that the Products or Services to be provided by the Subcontractor comply with the requirements of ITT Clause 5, and shall be responsible for submitting the related evidence required by ITT Clause 15.2 with the tender.
- 15.4 In the case of a joint venture, consortium, or association, unless otherwise specified in the **TDS**, all parties shall comply with and formally agree to the requirements of ITT Clause 4 and GCC Clause 4.6, which agreement shall be evidenced by including a statement to this effect in the authorization mentioned under ITT Clauses 12.1(b) and 21.2, as well as in the tender.

**16. Documents
Establishing
Eligibility of
Products and
Services**

- 16.1 To establish the eligibility of the Products and Related Services in accordance with ITT Clause 5, Tenderers shall complete the country of origin declarations in the Price Schedules furnished in Section IV, Tender Forms. Countries should be identified using official ISO standard 3166 country codes.

17. Technical Proposal

- 17.1 Standards for workmanship, process, material, and equipment, as well as references to brand names or model numbers or national or proprietary standards specified by the Purchaser in Part 2 — Supply Requirements, are intended to be descriptive only and not restrictive. Except where explicitly prohibited in the **TDS** for specific items or standards, the Tenderer may substitute alternative standards, brand names, and/or model numbers, provided that it demonstrates to the Purchaser's satisfaction, that the substitutions ensure substantial equivalence or are superior to those specified in the Supply Requirements.
- 17.2 To establish the conformity of the Products and Related Services to the Tender Document, the Tenderer shall furnish as part of its tender the documentary evidence constituting the Technical Proposal, which should follow the outline and structure (including section and specification numbering) of Section VII, Technical Specifications, including the following:
- (a) A point-by-point response to the Purchaser's Technical Specifications, including a detailed description of the essential technical and performance characteristics of the proposed Products and Related Services, demonstrating their substantial responsiveness to the specifications and the underlying business requirements;
 - (b) A detailed and comprehensive Project Plan for successful and timely project implementation in accordance with the Implementation Schedule and the project planning and management requirements in the Technical Specifications;
 - (c) Annex I, consisting of a description of any deviations and exceptions to the Technical Specifications; and
 - (d) Annex II, consisting of any additional literature, documentation, drawings, data or other information or documentary evidence specified in **TDS** or offered by the Tenderer in support of the Tender.

18. Documents Establishing the Qualifications of the Tenderer

- 18.1 To establish its qualifications to perform the Contract, the Tenderer shall submit the evidence indicated for each qualification criteria specified in Section III.C, Mandatory Qualification Criteria.
- 18.2 By submission of documentary evidence in its tender, the Tenderer must also establish the following to the Purchaser's satisfaction:

- (a) that it has the financial, technical, and production capability necessary to perform the Contract and has a successful performance history with similar projects.
- (b) that, in the case of a Tenderer offering to supply those items of the Products or recurrent supply Services designated as “Key Items” in Section VI, Schedule of Supply which the Tenderer did not itself produce, the Tenderer has been duly authorized by the Producer or the Service provider to offer and supply those Products and/or Services in Ghana. This will be accomplished by submission of a Producer’s Authorization Form furnished in Section IV, Tender Forms for each key item.
- (c) that, in the case of a Tenderer not doing business within Ghana, the Tenderer is or will be (if awarded the Contract) represented by an Agent in Ghana who is equipped, and able to carry out the Tenderer’s Implementation, Warranty and Maintenance Period service obligations specified in GCC and the Supply Requirements.

18.3 Tenders submitted by a joint venture, consortium or association shall also comply with the requirement that the party or combination of parties responsible for provision of a specific component or aspect of the project and Contract must meet the relevant minimum qualification criteria for that component or aspect.

18.4 If a pre-qualification process has been undertaken for this procurement, as indicated in the Purchaser’s Invitation for Tenders, the Tenderer shall, as part of its tender, update any information submitted with its application for pre-qualification.

19. Period of Validity of Tenders

19.1 Tenders shall remain valid for the period specified in the TDS after the Tender submission deadline date prescribed by the Purchaser. A Tender valid for a shorter period shall be rejected by the Purchaser as non-responsive.

19.2 In exceptional circumstances, prior to the expiration of the tender validity period, the Purchaser may request Tenderers to extend the period of validity of their tenders. The request and the responses shall be made in writing. A Tenderer may refuse the request without forfeiting its tender security. A Tenderer that has submitted more than one tender must either provide the requested extension for all its tenders or refuse the extension for all its tenders. A Tenderer agreeing to the request will not be required or permitted to modify its tender, but will be required to extend the validity of its tender

security for all of its tenders for a corresponding period. A Tenderer may refuse the request without forfeiting its tender security.

20. Tender Security

20.1 Pursuant to ITT Clause 12.1, the Tenderer shall furnish, as part of its tender, a tender security in the amount and currency specified in the **TDS**. The Tender security shall be valid for thirty days (182) beyond the end of the period of validity of the Tender specified in ITT Clause 19, including any extensions thereto.

20.2 The Tender security shall be, at the Tenderer's option, in the form of:

- (a) a bank guarantee;
- (b) an irrevocable letter of credit; or
- (c) a cashier's or certified check;
- (d) an insurance bond

all from a reputable bank or insurance/bonding company acceptable to the Purchaser, from an eligible country. The tender security shall be submitted either using the Tender Security Form included in Section IV, Tender Forms, or in another substantially similar format. In any case, the form must include the complete name of the Tenderer.

20.3 The tender security of a joint venture, consortium or association shall be issued in the name of the entity submitting the Tender. In the event that the joint venture, consortium or association agreement is not to be executed until notification of award, the Tender security shall be issued so as to commit fully all parties thereto.

20.4 Any tender not accompanied by an acceptable tender security shall be rejected by the Purchaser as non-responsive.

20.5 The tender security is required to protect the Purchaser against the risk of the following conduct by a Tenderer which would warrant and may result in the security's forfeiture for all of the Tenderer's Tenders:

- (a) if a Tenderer withdraws any of its Tenders after the deadline for Tender submission but during the period of Tender validity specified under ITT Clause 19, except as provided in ITT Clause 19.2; or

(b) fails to accept the Purchaser's adjustments to Tender Price, pursuant to ITT Clause 34.

(c) in the case of the successful Tenderer, if the Tenderer fails to:

(i) sign the Contract or any Supplemental Agreements in accordance with ITT Clause 45; or

(ii) furnish performance security in accordance with ITT Clause 46; or

20.6 The tender security of unsuccessful Tenderers shall be returned as promptly as possible upon the successful Tenderer's furnishing of the performance security pursuant to ITT Clause 46, but not later than thirty (30) days after the expiration of the period of Tender validity.

20.7 The tender security of the successful Tenderer shall be returned as promptly as possible once the successful Tenderer has signed the Contract and furnished the required performance security.

21. Format and Signing of Tender

21.1 The Tenderer shall prepare one original and the number copies of the tender specified in the TDS clearly marking each one as "ORIGINAL TENDER," "COPY 1," "COPY 2," etc., as appropriate. In the event of any discrepancy between the original and the copies, the original shall govern.

21.2 The original and all copies of the tender shall be typed or written in indelible ink and shall be signed by a person or persons duly authorized to sign on behalf of the Tenderer. This authorization shall consist of a written confirmation as specified in the **BDS** and shall be attached to the Tender. In the case of a joint venture, consortium, or association, authorizations are also required from all parties therein in accordance with ITT Clauses 4.1 and 15.4. The name and position held by each person signing an authorization must be typed or printed below the signature.

21.3 All pages of the tender, except for unamended printed literature, shall be signed or initialed by the person or persons signing the tender. Any interlineation, erasures, or overwriting shall be valid only if they are signed or initialed by the person or persons signing the Tender.

21.4 The Tenderer shall furnish information as described in the Tender Form on commission or gratuities, if any, paid or to be paid to agents relating

to this Tender, and to contract execution if the Tenderer is awarded the contract.

D. Submission and Opening of Tenders

22. Sealing and Marking of Tenders

- 22.1 The Tenderer shall seal the original and each copy of the bid in separate envelopes, duly marking the envelopes as “ORIGINAL” and “COPY NO.” These envelopes containing the original and the copies shall then be enclosed in one single outer envelope.
- 22.2 The inner and outer envelopes shall:
- (a) be addressed to the Purchaser at the address specified in **TDS** under ITB Clause 23.1;
 - (b) bear the Project name and the ICB name and number specified in **TDS** under ITB Clauses 2.1 and 1.1, respectively; and
 - (c) bear a warning “DO NOT OPEN BEFORE _____,” to be completed with the time and the date of bid opening specified in **TDS** under ITB Clause 26.
- 22.3 The inner envelopes shall also bear the name and address of the Tenderer so that the tender can be returned unopened in case it is declared late under ITT Clause 24
- 22.4 If the outer envelope is not sealed and marked as required, the Purchaser will assume no responsibility for the tender’s misplacement or premature opening. If the outer envelope discloses the Tenderer’s identity, the Purchaser will not guarantee the anonymity of the tender submission, but this disclosure will not constitute grounds for tender rejection.

23. Deadline for Submission of Tenders

- 23.1 Tenders must be received by the Purchaser at the address and no later than the date and time indicated in the TDS.
- 23.2 The Purchaser may, at its discretion, extend the deadline for the submission of Tenders by amending the Tender Document in accordance with ITT Clause 9, in which case all rights and obligations of the Purchaser and Tenderers previously subject to the deadline shall thereafter be subject to the deadline as extended.

24. Late Tenders

- 24.1 The Purchaser shall not consider any Tender that arrives after the deadline for submission of Tenders, in accordance with ITT Clause 24. Any Tender received by the Purchaser after the deadline for submission of Tenders shall be declared late, rejected, and returned unopened to the Tenderer.

25. Withdrawal and Modification of Tenders

- 25.1 The Tenderer may modify or withdraw its tender after submission, provided that written notice of the modification or withdrawal is received by the Purchaser prior to the deadline prescribed for submission of tenders. No tender may be modified after the deadline for submission of tenders.
- 25.2 The Tenderer's modification or withdrawal notice shall be prepared, sealed, marked, and dispatched in accordance with the provisions of ITT Clauses 21 and 22. A withdrawal notice may be sent electronically, but must be followed by a signed confirmation copy, postmarked not later than the deadline for submission of tenders. Notices received after the tender submission deadline will be ignored, and the submitted tender will be deemed to be a valid tender as originally submitted. Modifications shall include the same number of copies as required for the tender.
- 25.3 No tender may be withdrawn in the interval between the deadline for submission of tenders and the expiration of the period of tender validity specified by the Tenderer on the Tender Form. Withdrawal of a tender during this interval may result in the forfeiture of the tender security for all of the Tenderer's tenders, pursuant to ITT Clause 20.5.

26. Tender Opening

- 26.1 The Purchaser open all tenders in the presence of Tenderers' representatives who choose to attend, at the place, date and time specified in the **TDS**. Tenderers' representatives who attend shall sign a register as proof of their attendance
- 26.2 First, envelopes marked "WITHDRAWAL" shall be opened and read out and the corresponding tender envelope shall be returned unopened to the Tenderer. Next, envelopes marked "MODIFICATION" shall be opened and read out with the corresponding tender. No tender withdrawal or modification shall be permitted unless the corresponding notice contains a valid authorization to request the withdrawal or modification and the notice is read out at tender opening.
- 26.3 All tenders shall be opened one at a time, reading out: the name of the Tenderer and whether there is a modification; the Tender Prices, including any discounts offered; the presence of any required tender security; the presence or absence of required authorizations; and any other details as the Purchaser may consider appropriate.
- 26.4 Only envelopes that are opened and read out at tender opening shall be considered further, irrespective of the circumstances. No tender shall be rejected at tender opening, except for late tenders as defined in ITT Clause 24.

- 26.5 During the tender opening, the Purchaser shall prepare minutes of the proceedings that shall include, as a minimum: the name of the Tenderer and whether there is a withdrawal or modification; the Tender Price, per lot if applicable, including any discounts offered; and the presence or absence of the required tender security and authorization.
- 26.6 The Tenderers' representatives who are present shall be requested to sign the minutes. The omission of a Tenderer's signature on the record shall not invalidate the contents and effect of the record. A copy of the record shall be distributed to all Tenderers who request them.

E. Evaluation and Comparison of Tenders

- 27. Confidentiality**
- 27.1 Information relating to the examination, evaluation, comparison, and post-qualification of tenders, and recommendation of contract award, shall not be disclosed to tenderers or any other persons not officially concerned with such process until information on Contract award is communicated to all tenderers.
- 27.2 Any effort by a Tenderer to influence the Purchaser in the examination, evaluation, comparison, and post-qualification of the tenders or Contract award decisions may result in the rejection of its tender.
- 27.3 From the time of tender opening to the time of Contract award, if any Tenderer wishes to contact the Purchaser on any matter related to the Tender process, it should do so in writing.
- 28. Clarification of Tenders**
- 28.1 To assist in the examination, evaluation, comparison and post-qualification of the Tenders, the Purchaser may, at its discretion, ask any Tenderer for a clarification of its Tender. Any clarification submitted by a Tenderer in respect to its Tender and that is not in response to a request by the Purchaser shall not be considered. The Purchaser's request for clarification and the response shall be in writing. No change in the prices or substance of the Tender shall be sought, offered, or permitted, except to confirm the correction of arithmetic errors discovered by the Purchaser in the evaluation of the Tenders, in accordance with ITT Clause 30.

29. Responsiveness of Tenders

- 29.1 The Purchaser's determination of a tender's responsiveness will be based on the contents of the tender itself and any clarifications thereto obtained in accordance with ITT Clause 28.
- 29.2 A substantially responsive tender is one that conforms to all the terms, conditions, and mandatory requirements of the Tender Document without material deviation, reservation, objection, or omission. A material deviation, reservation, objection, or omission is one that:
- (a) affects in any substantial way the scope, quality, or performance of the Products and Related Services specified in the Tender Document; or
 - (b) limits in any substantial way, inconsistent with the Tender Document, the Purchaser's rights or the Tenderer's obligations under the Contract; or
 - (c) the acceptance of which would unfairly affect the competitive position of other Tenderers presenting substantially responsive tenders.
- 29.3 If a tender is not substantially responsive, it shall be rejected by the Purchaser and may not subsequently be made responsive by the Tenderer by correction of the nonconformity.

30. Non-conformities, Errors, and Omissions

- 30.1 Provided that a tender is substantially responsive, the Purchaser may waive any non-conformity, error or omission in the tender that does not constitute a material deviation.
- 30.2 Provided that a tender is substantially responsive, the Purchaser may request that the Tenderer submit necessary information or documentation within a reasonable period of time, to rectify nonmaterial non-conformities or omissions in the tender related to documentation requirements. Such omission shall not be related to any aspect of the price of the tender. Failure of the Tenderer to comply with the request may result in the rejection of its tender
- 30.3 Provided that a tender is substantially responsive, the Purchaser shall correct arithmetical errors on the following basis:
- (a) if there is a discrepancy between the unit price and the total price that is obtained by multiplying the unit price and quantity, the unit price shall prevail and the total price shall be corrected, unless in the opinion of the Purchaser there is an obvious

misplacement of the decimal point in the unit price, in which case the total price as quoted shall govern and the unit price shall be corrected;

- (b) if there is an error in a total corresponding to the addition or subtraction of subtotals, the subtotals shall prevail and the total shall be corrected; and
- (c) if there is a discrepancy between words and figures, the amount in words shall prevail, unless the amount expressed in words expresses an arithmetic error.

30.4 Provided that a tender is substantially responsive, the cost of any Product or Related Service that has been omitted or is necessary to correct a minor deviation will be added to the Tender Price, for comparison purposes only, using the highest unit cost from among the other responsive tenders, or, in the absence of such costs, from the source indicated in the **TDS**. If the missing Product or Service corresponds to a technical evaluation criteria, it will be scored at zero points.

30.5 If the Tenderer that submitted the lowest evaluated tender does not accept the correction of errors, all of its tenders shall be disqualified and its tender security may be forfeited.

31. Preliminary Examination of Tenders

31.1 The Purchaser shall examine the tenders to confirm that all documents, securities and authorizations requested in ITT Clause 12 have been provided, and to determine the completeness of each document submitted. If any of the following documents or information are missing, the tender shall be rejected:

- (a) Complete and properly signed Tender Form and Price Schedules, valid for the requisite period;
- (b) Written confirmation of authorization to commit the Tenderer; and
- (c) The requisite Tender Security

31.2 The Purchaser shall examine the tenders, including the list of deviations defined in ITT Clause 12.1(k), to confirm that all terms and conditions specified in the GCC and the SCC have been accepted by the Tenderer without any material deviation or reservation.

31.3 The Purchaser shall examine the tenders to verify their eligibility and qualifications in accordance with ITT Clauses 15, 16 and 18, and

Section III.C, Mandatory Qualification Criteria. In the case of pre-qualification, the Purchaser will ensure that each tender is from a pre-qualified Tenderer.

- 31.4 The Purchaser shall examine the Annex to the Technical Proposal described in ITT Clause 17.2© to verify that each tender fulfills the mandatory technical specifications in Section VI, Schedule of Supply.
- 31.5 If, after the examination of a tender's acceptance of the terms and conditions, eligibility and qualifications, and compliance with the mandatory technical specifications, the Purchaser determines that the tender is not substantially responsive in accordance with ITT Clause 29, it shall reject the tender.

32. Evaluation of Technical Merit

- 32.1 The Purchaser shall conduct a detailed evaluation of the Technical Proposals of all tenders which remain responsive following the Preliminary Examination to confirm that they fulfill all the mandatory requirements of the Tendering Document.
- 32.2 Unless otherwise specified in Section III.B, Technical Merit Evaluation Criteria, the Purchaser's detailed evaluation of responsive tenders will take into account technical factors, in addition to cost factors, by calculating a technical merit discount factor (Dt), employed as described in ITT Clause 38 and calculated as follows:

Dt = 1 – ((St / 100) x (T / 100)), where:

St = The technical score, up to 100 points, awarded to the tender as prescribed in Section III.B, Technical Merit Evaluation Criteria.

T = The percentage weight for technical merit, as specified in Section III.B, Technical Merit Evaluation Criteria.

33. Conversion to Single Currency

- 33.1 The currency and the date and source of the exchange rate to be used for converting tender prices from the various currencies in which they are quoted by Tenderers to a common base for the purpose of evaluation and comparison of tenders is specified in the **TDS**, which shall be either
- (a) Ghanaian Cedi (GHC) and the selling exchange rate established for similar transactions by the Central Bank or a commercial bank in Ghana; or

- (b) a currency widely used in international trade, such as US dollars, and the selling rate of exchange published in the international press for the amount payable in foreign currency; and the selling exchange rate established for similar transactions by the Bank of Ghana for the amount payable in the Ghanaian Cedi (GHC).

**34. Adjustments
to Tender
Prices**

- 34.1 The Purchaser will calculate the Adjusted Tender Price (P) of each tender on the basis of the total prices for Products and for Related Services (but initially excluding the Total Maintenance Costs) quoted in the Tender Price Summary Schedule provided in Section IV, Tendering Forms, and completed in accordance with ITT Clause 13, plus or minus the following adjustments.
- 34.2 Correction of arithmetic errors and omissions determined pursuant to ITT Clause 30;
- 34.3 Adjustments due to discounts offered pursuant to ITT Clause 13.6. If specified in the TDS under ITT Clause 1.1 that tenders are being invited for individual lots (contracts) or for any combination of lots, the methodology of evaluation to determine the lowest evaluated lot combinations, including any multi-lot discounts offered in the Tender Form, is specified in Section III, Evaluation and Qualification Criteria.;
- 34.4 Adjustments for Implementation Schedule deviations, if permitted in the **TDS**, as follows: If a Tenderer offers an Implementation Schedule for some or all of the items that deviates from the Schedule of Supply, a pro rata increase or decrease of the total price of the affected items will be added or subtracted using the percentage(s) specified in the **TDS** for each week of delay or advance. Tenders offering Installations beyond the maximum permissible delay specified may be rejected.;
- 34.5 Adjustments for deviations taken to the Contract payment schedule specified in the **SCC**. If deviations are permitted in **TDS**, the Adjusted Tender Price will be increased pro rata by the amount of interest that could otherwise be earned on the amount of any payments that would fall due under the proposed schedule earlier than the schedule stipulated in the Tendering Document, at the annual interest rate specified in the **SCC**;
- 34.6 The price increases associated with the withdrawal of the deviations or exceptions included the tender, in accordance with ITT Clause 12.1(k);
- 34.7 Adjustments due to the application of a margin of Domestic Preference in accordance with ITT Clause 36

35. Exclusions

35.1 For the purpose of evaluation, the Purchaser will exclude the following from the Adjusted Tender Price (P) and also from the evaluation of Technical Merit:

- (a) any Ghanaian taxes or duties due upon delivery of the Products or Services if the Contract is awarded to the Tenderer that are not included in tender prices pursuant to ITT Clause 13.4;
- (b) any allowance for price adjustment during the period of execution of the Contract, if provided in the tender; and
- (c) any unsolicited options or alternative offers included in the tender.

36. Domestic Preference

36.1 Unless otherwise specified in the **TDS**, the Purchaser will not grant a margin of preference for supply of domestic Products. Under this preference, for Products supplied from outside Ghana, the total CIP price of such Products shall be increased by the applicable import tariff (custom duties and other import taxes) or fifteen (15) percent, whichever is less. If duties vary from item to item, the appropriate tariff for each item shall apply. No preference shall be applied for any Related Services or works components of the tender.

36.2 No margin of preference will be granted for any other price component, and Tenderers will not be permitted or required to modify the source of any hardware, software, related equipment, materials, or other Products, after tender opening

37. Maintenance Costs

37.1 The Maintenance Costs (R) are reduced to present value using the following formula:

$$R = \sum_{X=1}^L \frac{R_x}{(1+I)^X}$$

where:

L = total number of years spanning the Useful Life Period (per GCC Clause 1.1 (gg)) of all Products and including any additional years due to phased installations.

X = an index number 1, 2, 3, L, representing each year for which Maintenance Costs are quoted.

R_x = total Maintenance Costs for year 'x', from the Maintenance Costs Schedule.

I = discount rate for the present value calculation specified in Section III, Evaluation and Qualification Criteria.

38. 38.1 For comparison purposes, the Purchaser will calculate the Evaluated Tender Price (P_1) for each tender, as follows:

P_1 = Adjusted Tender Price (P), per ITT Clause 34, minus any Price Exclusions, per ITT Clause 35, plus the Maintenance Costs present value (R), per ITT Clause 37

39. Comparison of Tenders

- 39.1 To evaluate a tender, the Purchaser shall only use all the criteria and methodologies defined in the Instructions to Tenderers and in Section III, Evaluation and Qualification Criteria. No other criteria or methodology shall be permitted.

- 39.2 In the final step of the evaluation and comparison of responsive tenders, the Purchaser selects as the *Lowest Evaluated Tender* and eligible for contract award, the tender with the lowest Evaluated Tender Cost (C). The formula for the Evaluated Tender Cost (C), calculated using the technical merit discount factor (D_t) per ITT Clause 32.2, is as follows:

$$C = P_1 \times D_t$$

F. Post-Qualification and Award of Contract

40. Postqualification of the Tenderer

- 40.1 The Purchaser will determine at its own cost and to its satisfaction whether the Tenderer selected as having submitted the Lowest Evaluated Tender is qualified to perform the Contract satisfactorily. This determination may include visits interviews with the Tenderer's clients referenced in its tenders, site inspections, and any other measures specified in **Section III.D, Post-Qualification Measures**. In cases of pre-qualification, the Purchaser shall also verify that no changes in the Tenderer's qualifications have occurred since pre-qualification.

- 40.2 The Purchaser will evaluate the documentary evidence of the conformity of the Products and Services offered to the Tendering Document, through detailed examination of the Tender, as well as other information the Purchaser deems necessary and appropriate. Factors not included in Section III shall not be used in the evaluation of the Tenderer's qualifications. If specified in **Section III.D, Post-Qualification Measures**, at the time of post-qualification the Purchaser may also carry out tests to determine that the performance or

functionality of the Products offered meets those specified in the Supply Requirements.

40.3 If supply of Products or Services for the Maintenance Period are required by the TDS for ITT Clause 13.4(d), the determination of the capabilities of the lowest evaluated Tenderer will take into account its ability to execute and successfully perform the Supplemental Agreements for Maintenance Cost items with terms substantially equivalent to those offered in the tender, the signing of which is a prerequisite for the release of the Tender Security and the project implementation Start Date pursuant to GCC Clause 3.

40.4 An affirmative determination shall be a prerequisite for award of the Contract to the Tenderer. A negative determination shall result in disqualification of the tender, in which event the Purchaser shall proceed to the next lowest evaluated tender to make a similar determination of that Tenderer's capabilities to perform satisfactorily.

**41. Purchaser's
Right to
Accept Any
Tender, and
to Reject Any
or All
Tenders**

41.1 The Purchaser reserves the right to accept or reject any tender, and to annul the tendering process and reject all tenders at any time prior to contract award, without thereby incurring any liability to Tenderers.

**42. Award
Criteria**

42.1 The Purchaser shall award the Contract to the Tenderer whose offer has been determined to be the lowest evaluated tender and is substantially responsive to the Tender Documents, provided further that the Tenderer is determined to be qualified to perform the Contract satisfactorily.

**43. Purchaser's
Right to
Vary
Quantities at
Time of
Award**

41.1 At the time the Contract is awarded, the Purchaser reserves the right to increase or decrease the quantity of Products and Services originally specified in the Tender Document by an amount up to the percentage indicated in the TDS, and without any change in the unit price or other terms and conditions.

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| 44. Notification of Award | <p>44.1 Prior to the expiration of the period of tender validity, the Purchaser shall notify the successful Tenderer in writing and to be confirmed in writing, that its tender has been accepted. The notification of award will constitute the formation of the Contract.</p> <p>44.2 Upon the successful Tenderer's furnishing of the signed Contract Form, any Supplemental Agreements and the performance security pursuant to ITT Clauses 45 and 46, the Purchaser will promptly notify each unsuccessful Tenderer in writing and will discharge its tender security, pursuant to ITT Clause 20.</p> <p>44.3 If, after notification of award, a Tenderer wishes to ascertain the grounds on which its tender was not selected, it should address its request to the Purchaser. The Purchaser will promptly respond in writing to the unsuccessful Tenderer.</p> |
| 45. Signing of Contract | <p>45.1 At the same time as the Purchaser notifies the successful Tenderer that its tender has been accepted, the Purchaser will send the Tenderer the Contract Form provided in the Tendering Documents, incorporating all agreements between the parties.</p> <p>45.2 As soon as practically possible, but no more than twenty eight (28) days following receipt of the Contract Form, the successful Tenderer shall sign and date the Contract Form and, unless additional days are allotted therefor in the TDS, any Supplemental Agreements, and return them to the Purchaser.</p> |
| 46. Performance Security | <p>46.1 Within twenty eight (28) days following receipt of notification of award from the Purchaser, the successful Tenderer shall furnish the performance security in accordance with the GCC, using the Performance Security Form provided in Section X, Contract Forms, or another form acceptable to the Purchaser.</p> <p>46.2 Failure of the successful Tenderer to comply with the requirements of ITT Clauses 45 or 46.1 shall constitute sufficient grounds for the annulment of the award and forfeiture of the tender security for all the Tenderer's tenders. In that event the Purchaser may award the Contract to the next lowest evaluated responsive tender submitted by a qualified Tenderer, or call for new tenders</p> |
| 47. Protests or Claims | <p>47.1 Tenderers may submit a protest or claim to the Purchaser, in writing, at anytime but not later than ten (10) days after the Notification of award date, pursuant to ITT Clause 44.</p> |

Section II. Tender Data Sheet

The following information and provisions shall complement, supplement, or amend the provisions in the Instructions to Tenderers (ITT). Whenever there is a conflict, the provisions herein shall prevail over those in the ITT.

	A. General
ITT 1.1	The number and title of the Invitation for Tenders is : GR/GBB/TS/NCT/042/2026 Request for Proposals (RFP)
ITT 1.1	The Purchaser is: Ghana Gold Board (GOLDBOD)
ITT 1.1 22.2(b)	The identification number of the ICT is: GR/GGB/TS/NCT/042/2026
ITT 2.1	The Source of Funds is: Internal Generated Funds
ITT 2.1 22.2(b)	The name of the Project is: Procurement of Technical Services for the Design, Development, Implementation and Support of a Traceability and Assurance System
ITT 2.1	The name of the Contract(s) is: Procurement of Technical Services for the Design, Development, Implementation and Support of a Traceability and Assurance System
ITT 4.8	Proposals (RFP) shall be in English and must be uploaded on the Ghana Electronic Procurement System (GHANEPS). The Technical and Financial Proposals <u>MUST</u> be uploaded <u>SEPARATELY</u> and either marked Technical or Financial Proposal. Technical Proposal – Procurement of Technical Services for the Design, Development, Implementation and Support of Traceability and Assurance System (TAS) <u>AND</u> Financial Proposals – Procurement of Technical Services for the Design, Development, Implementation and Support of Traceability and Assurance System (TAS) Firms are to note that, there will be a Proof of Concept (PoC) demonstrating proposed to solution to the evaluation Panel.
	B. Contents of Tender Document

ITT 7.1	For <u>tender clarification purposes</u> only, the Purchaser's address is: The Chief Executive Officer Ghana Gold Board P. O. Box M 108 1 Thorpe Rd. Accra, Ghana Accra - Ghana Attention: Head, Procurement Department 4th Floor, Room Number: 406 City: High Street, Accra Region: Greater Accra, Ghana. Telephone: +233 20998 9449 Email address: frank.ademan@goldbod.gov.gh
ITT 7.1	The GoldBod may receive requests for clarification through the GHANEPS at least 12 days prior to the deadline for submission of proposals and shall provide responses to clarification to all Firms no later than 5 days prior to the deadline for submission of tenders.
ITT 7.2	The pre-proposal meeting is scheduled for the following date, time and place: Date: 30th June, 2026 Time: 11:00 am Location: Virtual
ITT 8.2	Tenderers are required to engage stakeholders across gold trade eco system to enhance the quality of their Technical Proposal
ITT 9.1, 9.2, 9.3	Issuance of addendum is permissible.
	C. Preparation of Tenders
ITT 11.1	The language of the Tender is: <u>English.</u>
ITT 12.1 (I)	The Tenderer shall submit with its Tender the following additional documents: <i>Tender Security of GHC 1,000,000.00 from Bank regulated by Bank of Ghana (BoG) or Insurance Company regulated by National Insurance Commission (NIC). (In case of Joint Venture or Partnership, it should be provided in the name of the lead Firm)</i> <i>Technical Proposal which includes Firms Experience with references, Work Plan, Methodology, Key Personnel etc.</i>

	3. <i>Price Schedule in accordance with Section IV, Tender Format, duly completed in accordance with ITT Clause 13 and 14 and the General Instructions for the Tender</i> 4. <i>Power of Attorney (For all companies in the case of Joint Venture/Partnership)</i> 5. <i>Joint Venture/Partnership Agreement (Where Applicable)</i> 6. <i>Supplementary agreements required pursuant to GCC Clause 8 for provision of maintenance services in accordance with GCC Clause 9.2 (This supplementary system maintenance cost should not be part of the Total Cost Build-up, should be attached but separated from the total tender price.)</i>
ITT 13.4(a)	<u>No changes</u> to ITT Incoterm provisions for Imported Products.
ITT 13.4(b)(ii)	<u>No changes</u> to ITT requirement for quotation of transportation, insurance, and other local costs incidental to delivery of the EXW components of each Installation batch of Products to its Project Site.
ITT 13.4(d)	Maintenance Costs for five (5) years <u>shall</u> be provided separately from the Financial Proposal. However, they should be attached and submit together. (Refer to page 50)
ITT 13.7	Tender prices shall be <u>fixed</u> during contract execution.
ITT 14.1(a)	<u>No changes</u> to ITT local currency requirement.
ITT 15.4	<u>No changes</u> to the ITT provisions concerning joint ventures, partnership consortiums, and associations.
ITT 17.1	<u>No changes</u> to ITT provisions concerning equivalent standards, brand names and model numbers.
ITT 17.2(d)	<u>Additional relevant information</u> may be submitted with the Technical Proposal.
ITT 19.1	The tender validity period shall be <u>182 days</u> after submission of Proposals and shall be <u>valid</u> until award of contract.
ITT 20.1	A tender security of GHC 1,000,000.00 in Ghana Cedis (GHC) from a Bank regulated by Bank of Ghana or an Insurance Company regulated by National Insurance Commission.
ITT 21.1	The original proposals should be submitted through GHANEPS
ITT 21.2	The written confirmation of authorization to sign on behalf of the Tenderer shall consist of a <u>notarized power of attorney signed by legally authorized signatories of all parties comprising the Tenderer.</u>

	D. Submission and Opening of Tenders
ITT 23.2 (c)	GoldBod have the discretion to extend the deadline of submission.
ITT 23.1	Proposals must be submitted not later than Tuesday 30 th June, 2026 Firms are required to separately upload their Technical and Financial Proposals in the GHANEPS Note: No electronic submission via email shall be permitted.
ITT 26.1	The Tender Opening shall be done through the GHANEPS
	E. Evaluation and Comparison of Tenders
ITT 30.4	The alternate source for pricing missing or nonconforming items for evaluation purposes is: Not applicable.
ITT 33.1	Firms are strictly advised to price in Ghana Cedis
ITT 34.4	Implementation Schedule deviations and adjustments are <u>not permitted</u>
ITT 34.5	Payment schedule deviations and adjustments are <u>not permitted</u>.
	F. Award of Contract
ITT 43.1	The maximum percentage by which quantities may be increased is: 10% The maximum percentage by which quantities may be decreased is: 10%

Section III. Evaluation and Qualification Criteria

The Evaluation and Qualification Criteria specify the criteria that the Purchaser will use to evaluate the tenders and post-qualify the lowest-evaluated Tenderer. This section specifies the parameters for and any other adjustments to the ITT Clauses related to the evaluation of tenders and the qualification of Tenderers and the proposed IT products, similar to the role of the TDS.

III.A. Financial Evaluation

FINANCIAL	POINTS
1. Financial Proposals a. Cost competitiveness b. Payments terms c. Breakdown of cost structure	20pts

Note: Financial Proposals must include all relevant taxes of the Republic of Ghana and all prices shall be quoted in Ghana Cedis (GHC)

III.B. Technical Merit Evaluation Criteria

As part of the determination and comparison of Evaluated Tender Prices, responsive tenders will be evaluated by the Purchaser against technical and qualifications criteria using the procedures specified herein and in ITT Clauses 32 and 38. The Technical Score (st) for each tender will be calculated as the sum of scores given by the Purchaser's Technical Evaluation Team to the criteria shown in the following table. The maximum value of the discount given to the Evaluated Tender Price for technical merit (T) is **80 %**, as defined in ITT Clause 32.

TECHNICAL EVALUATION CRITERIA		PTS
A. Experience of the Firm		15pts
a) General Experience in Software Deployment		8pts
b) Similar Experience Relevant to the Assignment (4 points each for any Traceability Experience)		5pts
c) Specific Experience Relevant to the Assignment (4 points each for any Gold or Precious Metal Traceability Experience)		2pts
B. Adequacy of the Proposed Work Plan and Methodology in responding to the Terms of Reference (ToR)		20pts
a) Demonstrate clear understanding of assignment deliverables, outputs, and outcomes		5pts
b) Firms' mobilization of team, planning and execution		2pts
c) Firms reporting obligation and clearly defined deliverables		2pts
d) Comments and value added to the Terms of Reference (TOR)		5pts
e) Clearly defined relevant activities and sub activities		3pts
f) Specific durations of implementation (proposed work plan)		3pts
C. Proof of Concept (Demonstration of Proposed Solution)		30pts
a. Live presentation of proposed solutions, system, equipment and technology		10
b. Functional compliance of integration, securities features, reporting and audit		5
c. Performance validation of speed and response times, scalability, stability and compatibility with existing systems		5
d. Data protection requirements, training, regulatory obligations		5
e. Look and feel of the system		5
D. Qualifications and competence of the key staff for the Assignment (including membership of professional bodies)		35pts
a) Project Director/Team Lead's Minimum Qualification and Experience:		5pts
i Master's degree in computer science, Engineering, Information Technology, Project Management, Business Administration, or related discipline.		
ii Minimum fifteen (15) years of senior programme delivery experience in large-scale public-sector or enterprise technology programmes.		
iii Demonstrated leadership of at least three (3) comparable national-scale digital transformation, traceability, enterprise systems, or projects of similar scale and complexity.		
iv PMP, PRINCE2 Practitioner, or equivalent senior project/programme management certification (mandatory).		

- v Demonstrated experience leading multi-disciplinary teams of at least twenty (20) personnel across multiple operational locations. - vi Africa or Ghana implementation experience strongly preferred.	
b) Project Manager's Minimum Qualification and Experience: i. Bachelor's degree (Master's preferred) in Information Technology, Engineering, Computer Science, or Project Management. ii. Minimum ten (10) years of project management experience, including at least five (5) years managing complex enterprise ICT implementations. iii. PMP, PRINCE2 Practitioner, PMI-ACP, or equivalent certification (mandatory). iv. Demonstrated experience managing phased or gated implementation methodologies and milestone-linked delivery frameworks. v. Experience coordinating multi-vendor and multi-stakeholder programmes preferred. vi. Africa or Ghana delivery experience strongly preferred.	4pts
c) Enterprise Blockchain Architect Minimum Qualification and Experience: i. Master's degree in Computer Science, Software Engineering, Distributed Systems, or related field. ii. Minimum eight (8) years of enterprise systems architecture experience with at least five (5) years in permissioned blockchain implementation. iii. Demonstrated experience designing and deploying at least two (2) enterprise-grade blockchain or distributed ledger platforms. iv. Proven expertise with one or more enterprise blockchain platforms including Hyperledger Fabric, Hyperledger Besu, R3 Corda, Quorum, or equivalent enterprise-grade permissioned DLT platforms. v. Demonstrated experience with: - o smart contracts / chaincode; - o consensus architecture; - o node governance; - o blockchain security; - o cryptographic integrity controls; - o on-chain/off-chain architecture. vi. Experience integrating blockchain platforms with enterprise APIs, mobile systems, and regulatory systems. vii. Relevant blockchain certifications strongly preferred.	3pts
d) Senior Software Developer & Enterprise Architect Minimum Qualification and Experience:	3pts

i. Master's degree in computer science, Software Engineering, Information Systems, or related field. ii. Minimum ten (10) years of experience in software development, enterprise architecture, systems design, and solution delivery. iii. Demonstrated experience delivering at least three (3) large-scale enterprise platforms involving multiple stakeholders, system integrations, or high transaction volumes. iv. Proven experience in secure solution design aligned with ISO/IEC 27001 and OWASP Application Security Verification Standard (ASVS). v. Strong experience in: - o API integrations; - o microservices; - o event-driven architecture; - o cloud-native and distributed systems; - o enterprise integration patterns.	
e) Cybersecurity/Information Security Expert Minimum Qualification and Experience: i. Master's degree in Cybersecurity, Information Security, Computer Science, or related field. ii. Minimum ten (10) years of information security experience with at least five (5) years leading cybersecurity for regulated or critical systems. iii. CISSP, CISM, or CISA certification (mandatory). iv. ISO/IEC 27001 Lead Implementer or Lead Auditor certification (mandatory). v. Demonstrated experience with: - o enterprise security architecture; - o identity and access management; - o blockchain security; - o mobile application security; - o incident response; - o penetration testing oversight. vi. Experience implementing cybersecurity controls aligned with NIST and OWASP frameworks preferred.	3pts
f) Technology, Network & Infrastructure Expert Minimum Qualification and Experience: i. Bachelor's degree (Master's preferred) in Computer Engineering, Telecommunications, Network Engineering, Information Systems, or related field.	3pts

ii. Minimum ten (10) years of enterprise infrastructure, network, and data-centre operations experience, including at least five (5) years in leadership roles. iii. CCNP, CCNA, or equivalent enterprise network certification (mandatory). iv. ITIL v4 Foundation or higher certification preferred. v. Familiarity with Ghana's National Communications Authority regulatory environment and national telecommunications infrastructure. vi. Prior experience implementing national-scale infrastructure, cloud, or government data-centre environments preferred.	
g) Integration/API Engineer Minimum Qualification and Experience: i. Bachelor's degree in computer science, Engineering, Information Systems, or related field. ii. Minimum six (6) years of API and enterprise integration engineering experience. iii. Demonstrated experience implementing integrations with government, financial-sector, or regulated systems. iv. Proficiency in: - o OpenAPI 3.x; - o OAuth 2.0 / OpenID Connect (OIDC); - o API gateways; - o event-driven integration patterns; - o secure integration frameworks. v. Experience integrating with Ghanaian or West African government systems preferred.	2pts
h) AML/CFT Compliance & Financial Crime Specialist Minimum Qualification and Experience: i. Bachelor's degree (Master's preferred) in Finance, Law, Compliance, Risk Management, Economics, or related field. ii. Minimum ten (10) years of AML/CFT compliance or financial crime risk-management experience. iii. Certified Anti-Money Laundering Specialist (CAMS) or equivalent certification (mandatory). iv. Demonstrated experience with: - o FATF compliance frameworks; - o STR/SAR reporting; - o transaction monitoring systems; - o sanctions and PEP screening; - o risk scoring methodologies; - o suspicious transaction analytics.	2pts

v. Prior experience in banking, fintech, extractives, or regulated sectors preferred. vi. Familiarity with Ghana's Anti-Money Laundering Act, FIC reporting requirements, and OECD/LBMA responsible sourcing frameworks preferred.	
i) ASM/Gold Sector Subject Matter Expert's Minimum Qualification and Experience: i. Master's degree in mining engineering, Mineral Economics, Development Studies, or related field. ii. Minimum ten (10) years of experience in the artisanal and small-scale gold mining sector. iii. Demonstrated field engagement with Ghanaian ASM communities (mandatory). iv. Working knowledge of: - o Minerals and Mining Act, 2006 (Act 703); - o Ghana Gold Board Act, 2025 (Act 1140); - o Minamata Convention; - o LBMA Responsible Gold Guidance; - o OECD Due Diligence Guidance. v. Prior advisory experience with government, Minerals Commission, GoldBod, donor programmes, or development partners preferred.	2pts
j) Mining Operations Expert's Minimum Qualification and Experience: i. Bachelor's degree (Master's preferred) in Mining Engineering, Geology, Mineral Processing, Metallurgy, or related field. ii. Professional registration with the Ghana Institution of Engineering, Mining Engineers Association of Ghana, or equivalent professional body. iii. Minimum ten (10) years of operational mining experience including at least five (5) years in ASM or small-scale gold mining environments. iv. Demonstrated field-level operational engagement in Ghanaian or West African mining environments. v. Experience with: - o assay systems; - o reconciliation; - o mineral processing; - o operational controls; - o mass-balance methodologies. vi. Familiarity with Ghana mining regulations and environmental compliance frameworks required.	2pts

k) Mobile Applications & Offline Systems Expert Minimum Qualification and Experience: i. Bachelor's degree (Master's preferred) in Computer Science, Mobile Computing, Software Engineering, or related field. ii. Minimum eight (8) years of enterprise mobile application development experience. iii. Demonstrated experience developing offline-capable mobile applications operating in low-connectivity environments. iv. Proven expertise in: - o Android/iOS enterprise applications; - o encrypted offline storage; - o secure synchronization; - o GPS/location services; - o device authentication and management; - o field-deployment optimization. v. Experience deploying mobile systems in rural or field-based operational environments preferred.	2pts
l) Legal & Regulatory Compliance Advisor (Ghanaian) Minimum Qualification and Experience: i. LL.B and LL.M (or equivalent) with valid practising certification from the General Legal Council of Ghana. ii. Minimum ten (10) years of legal and regulatory advisory experience in mining, financial services, data protection, AML/CFT, or public-sector governance. iii. Demonstrated familiarity with: - o Ghana Gold Board Act; - o Data Protection Act, 2012 (Act 843); - o Anti-Money Laundering Act, 2020 (Act 1044); - o Public Procurement Act 663 as amended; - o Minerals and Mining Act. Etc. iv. Experience advising on technology governance, data sharing, digital compliance, or regulated enterprise systems preferred.	2pts
m) Logistics and Supply Chain Expert Minimum Qualification and Experience: i. Bachelor's degree (Master's preferred) in Supply Chain Management, Logistics, Operations Management, or related field. ii. CILT (Chartered Institute of Logistics and Transport) Diploma, APICS CSCP, or CIPS qualification.	2pts

1. **Qualification of the Products and Related Services**

For the purposes of establishing the qualifications of the proposed Products and Related Services, Tenderers must provide the documentary evidence or certifications stipulated for each mandatory qualification requirement included in the following table.

Criteria	Requirement	Necessary Documentary Evidence or Certification
<u>Eligibility</u>		
<u>Market Penetration</u>		
<u>Similar Installation Sites</u>		

III.D. Post-Qualification Measures

The Purchaser may employ the following measures to determine at its own cost and to its satisfaction whether the Tenderer selected as having submitted the Lowest Evaluated Tender is qualified to perform the Contract satisfactorily, in accordance with ITT Clause 40:

- (1) Financial Capability/Viability and lines of Credit
- (2) Equipment holding
- (3) After Sale Service Support and Warranty

Section IV. Tender Forms

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General Instructions for the Tender Forms

1. Complete the Price Schedules and Tender Form in accordance with ITT Clause 13 and Section VI, the Schedule of Supply. Non-conformities, errors and omissions in the completed Schedules and Forms will be handled in accordance with ITT Clause 30.
2. Tenderers should provide the actual (trade) name of the offered items (instead of the generic descriptions shown in the Schedule of Supply), along with page or section number cross-reference(s) to the corresponding section(s) of the Technical Proposal.
3. Item codes used in the Price Schedules should correspond exactly to those specified in the Schedule of Supply, in accordance with ITT Clause 13.3. Item codes shall in all cases be unique (to facilitate contract administration, e.g., deliveries, invoice processing, change orders, etc.) and not repeated in any given Price Schedule. However, if the Tenderer's offerings do not correspond one-to-one with the items in the Schedule of Supply, proceed as follows:
 - When offering multiple Products or Services to fulfill a single requirement from the Required Products and Services Schedules, create unique item codes for each offered item by appending a hyphen and a sequential suffix to the Purchaser's original item code (e.g., Server-1, Server-2).
 - When offering a single Product or Service to fulfill multiple requirements, define a replacement item code and indicate the original item codes which it fulfills in either the Item Name field or in a footnote (e.g., RServ1, "Server and Operating System combination for Serv1 and OS1")
 - Always underline replacement item codes in the Price Schedules and clearly describe the rationale for the replacement and how the offered item fulfills all the technical requirements in the cross-referenced sections of the Technical Proposal.
4. Tally total product and service prices in the appropriate columns based on the currency of the unit prices. Use a separate line in the Price Schedule for each different currency component of the unit price of a Product or Service, using only one currency combination per Product or Service. Specify currencies in accordance with ITT Clause 14 using ISO (standard 4217) currency codes, which are available from most banks or on the Internet, as follows:
 - ☐ The URL for the official ISO 4217 currency codes at time of STD publication is:
<http://www.bsi.org.uk/bsi/products/standards/products/currency.xhtml>
 - ☐
 - Official ISO 4217 maintenance agency home page: <http://www.bsi.org.uk/>
 - ☐ International Standards Organization (ISO) home page: <http://www.iso.ch/>

General Instructions for the Tender Forms

- | | |
|----|---|
| 5. | <p>Specify countries of origin using official ISO (standard 3166) country codes, which are available from most banks or on the Internet, as follows:</p> <ul style="list-style-type: none">• ☐ The URL for the official ISO 3166 country codes at time of STD publication is:
http://www.din.de/gremien/nas/nabd/iso3166ma/codlstp1/index.html• ☐ ISO 3166 maintenance agency home page:
http://www.din.de/gremien/nas/nabd/iso3166ma/• ☐ International Standards Organization (ISO) home page: http://www.iso.ch/ |
|----|---|

Imported Products Price Schedule

Date: _____

Invitation for Tender No.: _____

ICT No.: _____

Tender & Lot Nos.: _____

Name of Tenderer: _____

Signature of Tenderer: _____

Required Products Schedule Item Code	Product Name	Technical Proposal Cross Ref.	Country of Origin	Product Quantity	CIP Unit Price ¹	Currency	Total CIP Product Price in Currency: (H = E x F)	Total CIP Product Price in Currency: (I = E x F)	Total CIP Product Price in Currency: (J = E x F)	Total CIP Product Price in Currency: (L = E x F)
(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H = E x F)	(I = E x F)	(J = E x F)	(L = E x F)
Imported Products Price Totals (To Tender Price Summary Schedule):										

1. Specify CIP Unit Prices in accordance with ITT Clauses 13.4(a) and 14.1(b) and the General Instructions for the Tender Forms.
2. Quote only one CIP price per Product, regardless of any incremental differences in transportation and insurance charges for different Project Sites.

Locally supplied Products Price Schedule

Date: _____

Invitation for Tender No.: _____

ICT No.: _____

Tender & Lot Nos.: _____

Name of Tenderer: _____

Signature of Tenderer: _____

Required Products Schedule Item Code	Product Name	Technical Proposal Cross Reference.	Country of Origin	Product Quantity	EXW Unit Product Price ¹ in Currency: _____	Total EXW Product Price
(A)	(B)	(C)	(D)	(E)	(F)	(G = E x F)
Locally supplied Products Price Total (To Bid Price Summary Schedule)						

3. Specify a single EXW Price for each locally supplied Product in accordance with ITT Clauses 13.4(b) and 14.1(a) and the General Instructions for the Tender Forms. Indicate costs for delivery and insurance of each Installation Batch separately on the Related Services Price Schedule.

Related Services Price Schedule

Date: _____

Invitation for Tender No.: _____

ICT No.: _____

Tender & Lot Nos.: _____

Name of Tenderer: _____

Signature of Tenderer: _____

Required Products Schedule Item Code	Service Description	Technical Proposal Cross Ref.	Country of Origin	Currency	Price ¹ per Year			Total Service Price			
								Currency:	Currency:	Currency:	Local Currency
					Year 1	Year 2	Year 3	_____	_____	_____	_____
(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I = F+G+H)	(J =F+G+H)	(K=F+G+H)	(L=F+G+H)
Related Services Price Totals (To Tender Price Summary Schedule):											

4. Specify Prices for services required under columns C and D of the Required Services Schedule in Section VI, Schedule of Supply, i.e. to be provided during Project Implementation and during the Warranty Period. Quote prices in accordance with ITT Clauses 13.4(c) and 14.1 and the General Instructions for the Tender Forms. If needed, add columns for additional years of Project Implementation and Warranty.

Maintenance Costs Schedule (Highly Required)

Date: _____

Invitation for Tender No.: _____

ICT No.: _____

Tender & Lot Nos.: _____

Name of Tenderer: _____

Signature of Tenderer: _____

Products/ Services Schedule Item Code	Maintenance Item ¹	Technical Proposal Cross Ref.	Country of Origin	Currency	Price ¹ per Year					Total Service Price
					Y1	Y2	Y3	Y4	Y5	Local Currency
(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)	(J)	(M = ∑F-J)
Related Services Price Totals (To Tender Price Summary Schedule):										

5. Specify Prices for services required under column E of the Required Services Schedule in Section VI, Schedule of Supply, and for Products to be provided during Maintenance Period (only). Quote prices in accordance with ITT Clauses 13.4(d) and 14.1 and the General Instructions for the Tender Forms.
6. Specify Maintenance Costs for the Products beginning on the date of expiration of the Warranty Period and running through the end of the Useful Life Period. Since the Warranty Period is usually at least one year long, there will generally be no Maintenance Costs for the first year (Y1) following the start Date of the Implementation Schedule (as defined in GCC Clause 3), even if some Products are installed immediately thereafter. Similarly, Products with one year Warranty installed and accepted at the end of a year-long Implementation Period may generate Maintenance Costs only for years Y3-Y6, given a 4 year Maintenance Period. The Schedule provides for up to 7 years of Maintenance Costs beyond the Start Date, and columns for additional years of Maintenance may be added as needed.

Tender Price Summary Schedule (Highly Required)

Date: _____

Invitation for Tender No.: _____

ICT No.: _____

Tender & Lot Nos.: _____

Name of Tenderer: _____

Signature of Tenderer: _____

Tender Price Component	Total in Local Currency:
Imported Products Price Totals	
Locally Supplied Products Price Totals	
Related Services Price Totals	
Total Supply Including Warranty Costs	

Note: Provide detailed price breakdown

Schedule of Key Subcontractors (Not Applicable)

Date: _____

Invitation for Tender No.: _____

ICT No.: _____

Tender & Lot Nos.: _____

Name of Tenderer: _____

Signature of Tenderer: _____

[illegible]

Tender Form

Date: _____

ICB No.: _____

Invitation for Tender No.: _____

Tender & Lot Nos.: _____

To: _____

We, the undersigned, declare that:

- (a) We have examined and have no reservations to the Tender Document, including Addenda Nos. _____, the receipt of which is hereby duly acknowledged;
- (b) We offer to supply in conformity with the Tendering Document and in accordance with the Implementation Schedule the following Products and Related Services: _____
_____;
- (c) The total amount and currency of our Tender is: _____
_____,

or such other sums as may be ascertained in accordance with the Price Schedules attached herewith and made part of this Tender;
- (d) The discounts offered and the methodology for their application are: _____
_____;
- (e) If our Tender is accepted, we will obtain the guarantee of a bank in a sum equivalent to _____ percent of the Contract Price for the due performance of the Contract, in the form prescribed by the Purchaser;
- (f) Our tender shall be valid for the Tender Validity Period specified in Clause 19 of the Tender Data Sheet and it shall remain binding upon us and may be accepted at any time before the expiration of that period;
- (g) The following commissions, gratuities, or fees have been paid or are to be paid with respect to the tendering process or execution of the Contract (If none, indicate "none"):

Name of Recipient	Address	Purpose	Amount & Currency
_____	_____	_____	_____
_____	_____	_____	_____
_____	_____	_____	_____

- (h) Until a formal Contract is prepared and executed, this Tender, together with your written acceptance thereof and your notification of award, shall constitute a binding Contract between us.
- (i) We understand that you are not bound to accept the lowest or any tender you may receive.
- (j) We certify/confirm that we and all subcontractors and producers for any part of the contract comply with the eligibility requirements as per ITT Clause 4 of the Tender Document.
- (k) We hereby certify that the Intellectual Property Rights in all the Products offered in this tender are either owned by us, or are offered under a legal and valid authorization and/or license from the owner.

Dated this _____ day of _____,

Name _____ In the capacity of _____

Signed _____

Duly authorized to sign the tender for and on behalf of _____

Tender Security Form

Date: _____

ICT No: _____

Invitation for Tender No.: _____

Tender & Lot Nos.: _____

To: _____

Whereas _____
(hereinafter "the Tenderer") has submitted its tender dated _____ for the
above-mentioned ICB No. for the supply of _____
hereinafter called "the Tender."

KNOW ALL PEOPLE by these presents that WE _____
of _____ having our registered office at
_____ (hereinafter "the Guarantor"), are bound unto
_____ (hereinafter
"the Purchaser") in the sum of _____
for which payment well and truly to be made to the aforementioned Purchaser, the Guarantor
binds itself, its successors, or assignees by these presents. Sealed with the Common Seal
of this Guarantor this _____ day of _____, _____.

THE CONDITIONS of this obligation are the following:

1. If the Tenderer withdraws any of its tenders during the period of tender validity specified by the Tenderer in the Tender Form, except as provided in ITT Clause 25.1; or
2. If the Tenderer, having been notified of the acceptance of its tender by the Purchaser, during the period of tender validity, fails or refuses to:
 - (a) execute the Contract Form; or
 - (b) furnish the Performance Security, in accordance with the ITT Clause 46; or
 - (c) accept the correction of its tender by the Purchaser, pursuant to ITT Clause 30.

We undertake to pay the Purchaser up to the above amount upon receipt of its first written demand, without the Purchaser having to substantiate its demand, provided that in its demand the Purchaser states that the amount claimed by it is due to it, owing to the occurrence of one or more of the above conditions, specifying the occurred conditions

This security shall remain in force up to and including twenty-eight (28) days after the period of tender validity, and any demand in respect thereof should be received by the Guarantor no later than the above date.

Name _____ In the capacity of _____

Signed _____

Duly authorized to sign the tender security for and on behalf of _____

Dated on _____ day of _____, _____.

Producer's Authorization Form

Date: _____

ICT No.: _____

Invitation for Tender No.: _____

Tender & Lot Nos.: _____

To: _____

WHEREAS _____ who are official producers of
_____ and having production facilities at
_____ do hereby authorize
_____ located at
_____ (hereinafter, the "Tenderer")
to submit a tender and subsequently negotiate and sign a Contract with you for resale of the
following Products produced by us, for the quantities, specifications, and delivery schedule
called for by the Supply Requirements associated with the above-mentioned Invitation for
Tenders: _____

_____.

We hereby extend to you a full guarantee and warranty in accordance with Clause 26,
Warranty, of the General Conditions of Contract and with our own standard product warranty,
and duly authorize the Tenderer to act on our behalf in fulfilling all warranty obligations with
respect to the above-listed products offered for resale by the Tenderer in relation to this
Invitation for Tenders.

We also certify that the Tenderer is qualified by us to provide the following maintenance,
support, software version upgrade or other required services related to the above-listed
Products in accordance with Clause 9, Scope of the Contract, of the General Conditions of
Contract: _____
_____.

Name _____ In the capacity of _____

Signed _____

Duly authorized to sign the authorization for and on behalf of: _____

Dated on _____ day of _____, _____.

Note: This letter of authority must be on the letterhead of the Producer, must be signed by a person competent and having the power of attorney to bind the Producer, and must be included by the Tenderer in its tender as specified in the Instructions to Tenderers

Section V. Eligible Countries

Public Procurement Authority of the Republic of Ghana

Eligibility for the Provision of Goods, Works and Services financed from the Public Funds of the Republic of Ghana

As of 20 .

For the information of Tenderers, and in accordance with ITT Clause 4, set forth below is a list of countries from which Tenderers, goods and services are not eligible to participate in procurement financed from the public funds of the Republic of Ghana:

-

PART 2 – Supply Requirements

Section VI. Schedule of Supply

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General Instructions for the Schedule of Supply

1. The Schedule of Supply contains several related schedules itemizing the Purchaser's Product and Service requirements, as follows:
 - a) The Products are to be installed and the Services are to be delivered at the Purchaser's Project Sites and according to the timetables specified in the Implementation Schedule. The Purchaser's Implementation Schedule defines the major milestones and required timetable for the project, which Tenderers shall prepare their version of in their tender.
 - b) The total quantities of Products and Related Services to be supplied under the Contract are listed in the Required Products and Services Schedules, which aggregate the requirements for all Project Sites and Installation Batches. All items in the Required Products and Services Schedules must be listed by Tenderers in the appropriate Price Schedules in their tenders.
 - c) The specific quantities of Products and Services comprising each Installation Batch at each Project Site are specified in the Deployment Schedule. Particulars about the delivery and installation timetables and associated tasks for each Installation Batch shall be provided by Tenderers in their detailed Project Plans.
2. An Installation Batch is a unique name given to specific quantities of Products to be Installed and Related Services to be delivered together at a particular Project Site at a specific time during project implementation. Installation Batches should generally be defined by the Purchaser to correspond to the Project Sites, except where a given Site will receive multiple Installations, e.g., in projects with several implementation phases or application sub-systems being Installed and/or Accepted at significantly different points in time. Installation Batches are defined as appropriate for each particular project, but should not span multiple delivery destinations (to facilitate tender pricing, comparisons and Contract administration, esp. of delivery and insurance costs for locally supplied (EXW) products quoted as a service in accordance with ITT Clause 13.4(b)(ii)).
3. The unique Item Codes specified for the Products and Services here in the Schedule of Supply shall correspond exactly to the Item Codes used by Tenderers in the Price Schedules furnished in Section IV. Tendering Forms, shall be included in the Firm's tender in accordance with ITT Clause 13 and the General Instructions for the Tendering Forms in Section IV.
4. Terms defined in Clause 1 of Section VIII, General Conditions of Contract (GCC) have the meanings defined therein, e.g., Acceptance is defined in GCC Clause 1.1(a) and the Project Plan in GCC Clause 1.1(t). Acceptance testing schedules and procedures are specified in GCC Clause 12, Inspections and Tests, and elaborated further in the Technical Specifications and the Project Plan. The schedule for payments based on deliveries (shipment), Installation and Acceptance are as specified in GCC and SCC Clauses 19.

Implementation Schedule

The Purchaser's Implementation Schedule defines the major milestones and required timetable for the Tenderer's tender. Tenderers shall prepare their response to the Purchaser's Implementation Schedule requirements as part of their tender. The Firm's Implementation Schedule shall specify the committed timetable for arrival and Installation of the Products at each of the Purchaser's Project Sites, and it shall be supported by a detailed Project Plan that fulfills the requirements of GCC Clause 1.1(t) and the Technical Specifications. Pursuant to GCC Clauses 1.1(m), 22, 32 and 34.1, the Firm's performance of the Contract shall be managed based on the Implementation Schedule provided in its tender and included in the Contract.

Site Code	Project Site Name	Site Location (Delivery Address)	Arrival Schedule	Installation Schedule ¹	Project Plan ² Cross Ref.	Installation Batch ³	
						Code ⁴	Description
(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)

1. Installation Schedule dates are measured in weeks from the Start Date for project implementation, as defined in GCC Clause 3.1. The Installation Schedule column indicates the acceptable ranges of dates for Installation (as defined in GCC Clause 1.1(o)) of each of the uniquely named Installation Batches at its Project Site. In defining dates in the Implementation Schedule, the time required for delivery to the Project Site, for any customs clearance procedures and for performance of all necessary services for Installation must be taken into account.
2. The Purchaser's input and requirements for the Project Plan are included in sub-section 5, Project Planning and Management, of the Technical Specifications, which should describe the rationale for how project Installations are broken down and scheduled. Tenderers shall provide a complete and detailed Project Plan in their tender for successful and timely project implementation in conformance with the Implementation Schedule, GCC Clause 1.1(t) and the Technical Specifications.
3. If there is only one Installation Batch for a given Project Site, then the Batch Code can equate to the Site Code. Logical associations between Installation Batches (e.g., based on project phases or stages, on deployment of a common application or sub-system, or to a particular department or building) may be indicated by naming conventions for the batches (e.g., HQ-Phase1, HQ-Accounting, Phase2-District3, as appropriate for the project), by grouping related batches in consecutive rows, and can be further elaborated in the Technical Specifications.
4. Installation Batch Codes must be unique, so that the Deployment Schedule can clearly and unambiguously identify the exact quantities of the specific Products and Services comprising each Installation Batch

Required Products and Services Schedules

The total quantities for each Product and Service item for the entire Contract (aggregated over all Installation Batches and Project Sites on the Deployment Schedule) are shown in the Required Products Schedule and the Required Services Schedule below. All items in the Required Products and Services Schedules must be listed and priced separately by Tenderers in the appropriate Price Schedules in accordance with ITT Clause 13.3 and the General Instructions for the Tendering Forms in Section IV.

Required Products Schedule

Site	Item Code	Total ¹ Quantity	Product Description	Key ² Item Yes/No	Technical Specifications Cross-reference
	(A)	(B)	(C)	(D)	(E)

1. Total Quantities for each Product should be carried forward from the Total Quantity column (H) from the Deployment Schedule, which sums the quantities required for each Product across all Installation Batches. Specify required Software License quantities and types according to the business requirements elaborated in the Technical Specifications (sub-section 2.3) and using the terminology from GCC Clause 1.1. Also indicate the required numbers of copies of the Software Distribution Media and associated documentation for each Project Site/Installation Batch in the Deployment Schedule.
2. Products flagged as “Key Items” that are not produced by the Firm require the inclusion of Producer’s Authorization Forms with the tender, in accordance with ITT Clause 18.2(b), and also require that the subcontractor or producer be specifically identified in the Schedule of Key Subcontractors in accordance with ITT Clause 12.1 (f)

Required Services Schedule

Item Code	Service Description	Required ¹ (With Separate Pricing ³) During:			Key ² Item Yes/No	Technical Specifications Cross-Reference
		Implementation Period	Warranty Period	Maintenance Period		

1. Requirements for each Service should be carried forward from the Total Quantity column (H) from the Deployment Schedule, which aggregates the requirements for each Service across all Installation Batches, broken down by project Period, as appropriate.
2. Supply items (including “subscription” services, e.g., for supply of new software releases) flagged as “Key Items” that are not produced by the Firm require the inclusion of Producer’s Authorization Forms with the tender, in accordance with ITT Clause 18.2(b), and also require that the subcontractor or producer be specifically identified in the Schedule of Key Subcontractors in accordance with ITT Clause 12.1(f).
3. Pursuant to ITT Clause 13.4, prices for Services during the Implementation and Warranty Periods for the Products should be specified on the Related Services Price Schedule, and Maintenance Period Costs should be specified on the Maintenance Costs Schedule

Deployment Schedule

The Deployment Schedule specifies the detailed Product and Service quantities required for each of the Installation Batches indicated in the Implementation Schedule, which may logically correspond to Project Sites, implementation phases, application sub-systems, or some combination thereof. Each row in the Implementation Schedule identifies and specifies the Installation schedule for one Installation Batch.

Product/ Service Item Code ¹	Installation Batch Code ²						Total Quantity ³	Notes
	<i>Batch 1</i>	<i>Batch 2</i>	<i>Batch 3</i>	<i>Batch 4</i>	<i>Batch 5</i>	<i>Batch 6</i>		
(A)	(B)	(C)	(D)	(E)	(F)	(G)	(H)	(I)

1. Each Item Code from the Required Products and Services Schedules should have a corresponding entry (row) in the Deployment Schedule, indicating the detailed deployment (delivery & installation) requirements for each and every Product and Service for each Installation Batch.
2. Installation Batch Codes are uniquely defined in column (G) of the Implementation Schedule. Separate columns are used here in the Deployment Schedule to specify the Product and Service quantities comprising each Installation Batch. Columns should be added as necessary to accommodate more Installation Batches.
3. The Total Quantity of each Product and Service (aggregated across all Installation Batches) in the Deployment Schedule shall correspond to the Total Quantities specified for the corresponding item in the Required Products and Services Schedules.



GOLDBOD
GHANA GOLD BOARD

Section VII. Technical Specification

TERMS OF REFERENCE (TOR)

PROCUREMENT OF TECHNICAL SERVICES FOR THE DESIGN, DEVELOPMENT, IMPLEMENTATION AND SUPPORT OF A TRACEABILITY AND ASSURANCE SYSTEMS (TAS)

This Terms of Reference (ToR) is issued pursuant to the Ghana Gold Board Act, 2025 (Act 1140) and the Public Procurement Act, 2003 (Act 663), as amended. The ToR forms an integral part of the Request for Proposals (RFP) and shall be read together with the Final Master RFP.

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ANNEX 6: CHANGE CONTROL PROCEDURE 1**ABBREVIATIONS/ACRONYMS**

Abbreviation and Acronyms	Meaning
AML/CFT	Anti-Money Laundering / Countering the Financing of Terrorism
ASM	Artisanal and Small-Scale Mining
API	Application Programming Interface
BO	Beneficial Ownership
BoG	Bank of Ghana
CTR	Currency Transaction Report
DPIA	Data Protection Impact Assessment
DR	Disaster Recovery
FIC	Financial Intelligence Centre
FATF	Financial Action Task Force
FX	Foreign Exchange
KRI	Key Risk Indicator
KPI	Key Performance Indicator
KYC/CDD	Know Your Customer / Customer Due Diligence
LBMA	London Bullion Market Association
MinCom	Minerals Commission
OECD	Organisation for Economic Co-operation and Development
PEP	Politically Exposed Person
PPA	Public Procurement Authority
QCBS	Quality and Cost-Based Selection
QBS	Quality-Based Selection
RFP	Request for Proposals
RPO	Recovery Point Objective
RTO	Recovery Time Objective
SAR / STR	Suspicious Activity / Transaction Report
SLA	Service Level Agreement
TAS	Traceability & Assurance System
TP	Traceability Point

BACKGROUND

The Ghana Gold Board (GoldBod) is established under the Ghana Gold Board Act, 2025 (Act 1140) and operates under the oversight of the Ministry of Finance. GoldBod is the statutory authority mandated to buy, sell, weigh, grade, assay, value, and export gold and other precious minerals in Ghana. In addition to its commercial mandate, GoldBod is responsible for promoting responsible mining practices, supporting the formalisation of artisanal and small-scale mining (ASM), and strengthening transparency, traceability, and regulatory compliance across the gold value chain.

Ghana's ASM gold sector continues to face structural and systemic challenges, including informality, inconsistent licensing compliance, reliance on manual and paper-based transaction records, and limited end-to-end visibility of gold movements from extraction to export. These challenges heighten exposure to gold smuggling, revenue leakage, money laundering, and terrorist financing risks. Existing practices at community and processing levels rely heavily on physical seals and handwritten receipts, resulting in fragmented, non-standardised, and difficult-to-audit chains of custody.

To address these challenges, GoldBod seeks to establish a national Traceability & Assurance System (TAS) underpinned by a secure, permissioned blockchain-based architecture. The TAS will digitise, standardise, and secure the chain of custody for gold by recording custody events, transformations, and compliance actions on an immutable, append-only distributed ledger, providing a single, tamper-evident source of truth from point of extraction through processing, aggregation, and final GoldBod intake.

The blockchain layer of TAS is intended to ensure data integrity, non-repudiation, and auditability across the gold value chain, while supporting controlled access for authorised actors and regulators. On-chain smart contracts (Hyperledger Fabric chaincode in Go, Solidity for Besu, or Kotlin for Corda) shall enforce business rules relating to custody transfer, batch amalgamation, assay reconciliation, and licence/compliance checks. Smart contracts handling annual throughput exceeding USD 1 million shall undergo formal verification (Certora, K Framework, or equivalent) prior to production deployment. Smart-contract upgrades shall require multi-signature governance approval as defined in Section 3.4. Off-chain data repositories may be used for scalability and performance, but all critical events, hashes, and state changes shall be anchored to the blockchain ledger.

The TAS will support Ghana's domestic legal obligations and international commitments, including compliance with the Anti-Money Laundering Act, 2020 (Act 1044), the Data Protection Act, 2012 (Act 843), the FATF Recommendations, the OECD Due Diligence Guidance for Responsible Supply Chains of Minerals (Gold Supplement), and the LBMA Responsible Gold Guidance. The system will also contribute to Ghana's broader objectives on revenue assurance, responsible sourcing, sustainability, and international market access.

The TAS is also intended to support conformance with the LBMA Responsible Gold Guidance (RGG), including Chain of Custody (CoC) requirements applicable to LBMA Good Delivery refiners and downstream market participants.

ASM Trade Realities and Critical Processing Nodes

Ghana's artisanal and small-scale gold trade does not operate as a strictly linear progression between licensing tiers. In practice, gold moves through interconnected and overlapping trading relationships driven by access to working capital, trust-based financing arrangements, and aggregation efficiency. Licensed Tier 1 and Tier 2 buyers frequently trade laterally among themselves, consolidating gold within communities and township markets prior to onward sale to aggregators.

At community level, gold is predominantly traded in sponge form. Purity is commonly assumed based on geographic origin rather than formally tested, as sponge gold cannot be density-tested without consolidation. Formal purity determination typically occurs only after smelting at community or township "kitchens." These kitchens represent the first physical point at which multiple small purchases are consolidated into a testable quantity and therefore constitute critical control points for traceability, assay capture, and AML/CFT monitoring.

Following kitchen processing, refined gold may be re-tested at Tier 2 buyer facilities, aggregation centres, and ultimately at GoldBod intake. Variances between purity and weight measurements across these stages are recognised commercial and regulatory risks. The TAS blockchain ledger shall therefore preserve end-to-end lineage through consolidation and re-assay, enabling tolerance-based reconciliation, exception handling, and forensic audit without disrupting legitimate trade flows.

The Traceability & Assurance System shall be designed with these operational realities in mind, embedding blockchain-backed traceability controls at natural transaction and processing points, aligned with established ASM practices, and capable of supporting real-time regulatory oversight and international market confidence in Ghanaian gold.

AIM AND OBJECTIVES

Aim

To design, develop, pilot, and nationally deploy a secure, scalable Traceability and Assurance System (TAS) that captures movement and transformation of gold from point of extraction to GoldBod intake. The system will record transactions across the full trade chain, including community-level purchases, Tier 1 and Tier 2 buyer transactions, kitchen processing, aggregator amalgamation, and final assay, generating analytics and audit trails that satisfy Ghanaian regulatory requirements and international responsible-sourcing standards.

Outcomes and Functional Objectives

The specific objectives are organised into five functional areas: origin and custody, processing and reconciliation, identity and due diligence, compliance and reporting, and system sustainability.

2.2.1 Origin and Custody

- **Establish verifiable origin at source.** Register actors and sites to bind each gold lot to a geographic zone, and where possible to a licensed concession, capturing geolocation, extraction method, and seller identity. This creates the foundational data layer that prevents unverified production from entering the formal chain.
- **Record custody events across the trade chain.** Model and capture the documented pathways through which gold moves: from miner or gleaner to community buyer, through Tier 1 and Tier 2 transactions, to aggregator consolidation and GoldBod delivery. The system must accommodate the multiple routes observed in practice, including direct sales that bypass intermediate tiers, while maintaining unbroken custody records.
- **Preserve lineage through amalgamation.** When aggregators merge lots from multiple sources into consolidated batches, maintain a composition record linking each contributor to their weight, estimated purity, and price. Batch-level attributes must be recomputed to preserve origin visibility even after physical consolidation.

2.2.2 Progressive Compliance Framework

The TAS shall be designed to support a phased and progressive compliance model aligned with ASM operational realities. The system shall accommodate varying levels of formality across actors and enable gradual transition into full traceability compliance.

At a minimum, the system shall support:

- **Level 1 (Basic):** Transaction recording with actor identification and location.
- **Level 2 (Intermediate):** Tagged custody tracking and partial lineage preservation.
- **Level 3 (Advanced):** Full chain-of-custody traceability with AML/CFT integration and assay reconciliation.

The system shall allow actors to transition between compliance levels over time without exclusion from the formal market.

2.2.3 Processing and Reconciliation

- **Capture assay data at multiple points.** Record purity and weight measurements at Tier 2 kitchens, aggregator processing facilities, and GoldBod intake. The system must support both density-based field testing and formal assay methods.

- **Reconcile variance across the chain.** Compare declared weights and purity values at successive touchpoints, applying defined tolerances to identify discrepancies. Where variance exceeds thresholds, generate exception cases for investigation while allowing compliant transactions to proceed.

2.2.4 Identity and Due Diligence

- **Establish a verified actor registry.** Create and maintain a digital identity database covering all licensed participants—miners, Tier 1 buyers, Tier 2 buyers, kitchen operators, aggregators, and transporters. Integrate Ghana Card verification where available. Where Ghana Card is unavailable, the fallback procedure shall comprise: (a) voter identification card; (b) biometric capture (fingerprint conformant to ISO/IEC 19794-2); (c) supervised onboarding co-signed by a registered Tier 2 sponsor accepting liability; and (d) a 90-day grace window during which the actor may transact up to a capped value (default GHS 50,000) pending Ghana Card issuance.
- **Embed KYC/CDD and beneficial ownership capture.** Screen registered actors against sanctions lists and politically exposed person (PEP) databases. Capture beneficial ownership information (threshold: $\geq 10\%$ direct or indirect ownership; PEP and adverse-media screening required for each beneficial owner above the threshold) for corporate entities in accordance with GoldBod licensing requirements and Anti-Money Laundering Act (Act 1044) obligations.

2.2.5 Compliance and Reporting

- **Enable AML/CFT monitoring.** Embed rule-based detection for transaction patterns indicative of money laundering or terrorist financing, including structuring, third-party payments, unusual volume spikes in border-proximate zones, capacity-versus-output mismatches, and grade anomalies.
- **Support regulatory reporting.** Provide electronic preparation and secure transmission of Suspicious Transaction Reports (STRs) and Suspicious Activity Reports (SARs) to the Financial Intelligence Centre. Generate standard compliance outputs aligned to FATF requirements.
- **Deliver standards-ready audit trails.** Produce due-diligence documentation and chain-of-custody reports compatible with OECD, LBMA, and Responsible Minerals Initiative (RMI) frameworks, supporting market access for Ghanaian gold with international buyers and refiners.
- **Ensure interoperability with regulatory systems.** Provide bi-directional data exchange with the Minerals Commission (licensing and cadastre), Ghana Revenue Authority Customs Division (export verification), Bank of Ghana (payments and foreign exchange),

and the Financial Intelligence Centre (financial intelligence sharing).

2.2.6 System Sustainability

- **Deploy physical and electronic traceability infrastructure.** Implement serialised tagging (QR codes, with NFC or RFID where appropriate) and mobile scanning capability. Capture timestamp, GPS coordinates, actor identification, device identifiers, lot references, weight, purity, method, photographic evidence, and digital signatures at each touchpoint. The system must function offline with secure synchronisation when connectivity is restored.
- **Build institutional capacity for sustained operation.** Deliver training programmes, change management communications, and formal handover procedures to enable GoldBod to operate TAS nationally without ongoing vendor dependency. Establish mechanisms for continuous improvement based on operational feedback and evolving regulatory requirements.
- **Establish and maintain a verifiable Chain of Custody (CoC)** for gold in accordance with LBMA Responsible Gold Guidance, ensuring continuous, auditable linkage between physical gold movements, ownership transfers, processing events, and compliance records.

The TAS shall support interim provenance mechanisms, including geographic zoning where mine-level concession identification is not immediately available, preserve lineage through commingling and batch amalgamation, and apply tolerance-based variance reconciliation with escalation workflows for material discrepancies.

SCOPE OF WORK

The firm shall deliver an end-to-end Traceability & Assurance System (TAS) underpinned by a permissioned blockchain platform (including but not limited to Hyperledger Fabric 2.5+, Hyperledger Besu, R3 Corda 5, Quorum, or equivalent enterprise-grade permissioned DLT platforms) using a Byzantine fault-tolerant consensus algorithm (IBFT2, QBFT, or Raft) that serves as the authoritative, immutable record of gold custody, transformation, and compliance events for regulatory, audit, and compliance purposes. Append-only relational journals with hash-chaining alone do not satisfy this requirement. The blockchain layer is a core and mandatory component of the TAS and shall not be treated as optional, experimental, or ancillary. All scope components described in this section shall be designed, implemented, and integrated to operate in conjunction with this blockchain-based ledger.

The TAS shall be implemented through a phased delivery approach that enables GoldBod to establish foundational traceability, progressively digitise priority touchpoints, and ultimately deploy a full end-to-end national system. These phases represent logical groupings of deliverables and do not limit the firm's obligation to deliver the complete scope of services within the contract period. Interim measures implemented during earlier phases shall be fully compatible with, and transition seamlessly into, the final TAS platform.

Blockchain Architecture and Responsibilities

The firm shall be fully responsible for the design, configuration, deployment, and operation of the blockchain layer of TAS during implementation and stabilisation. The blockchain shall function as an append-only, tamper-evident ledger that records and time-stamps all critical system events, including but not limited to:

- Actor onboarding and identity state changes
- Licence and authorisation status references
- Custody transfers between actors
- Kitchen processing, smelting, and batch amalgamation events
- Assay results, re-assay outcomes, and variance resolutions
- AML/CFT alerts, escalations, and case dispositions
- Administrative overrides and approvals

Off-chain databases may be used for performance, analytics, and storage of large files; however, all critical events, state changes, and integrity hashes shall be anchored to the blockchain ledger to ensure immutability, non-repudiation, and auditability.

For the purposes of LBMA Chain of Custody assurance, the permissioned blockchain ledger shall function as the authoritative system of record for custody events, transformations, and reconciliations, supported by off-chain documentation where required.

3.1.1 Enterprise Architecture Layers

The proposed TAS solution shall conform to the principles of the Ghana Government Enterprise Architecture (GGEA) and the e-Government Interoperability Framework (eGIF).

The firm shall provide a clearly defined layered enterprise architecture comprising, at a minimum, the following layers:

- User Interface Layer supporting mobile applications, web portals, dashboards, and reporting interfaces for miners, buyers, aggregators, transporters, regulators, inspectors, and GoldBod personnel.
- Security Layer incorporating identity and access management, multi-factor authentication, role-based and attribute-based access controls, encryption, transport security, monitoring, and audit controls.

- Core Application Layer covering identity and KYC management, chain-of-custody and traceability, assay and reconciliation, AML/CFT analytics, case management, reporting, and workflow services.
- Integration Layer supporting API gateways, middleware, interoperability services, event orchestration, and integration with government and external systems.
- Data Layer comprising the permissioned blockchain ledger, relational databases, object storage, audit repositories, archival systems, backup, and disaster recovery components.
- Hosting and Infrastructure Layer including cloud and/or on-premise infrastructure, network topology, node configuration, load balancing, failover, monitoring, and observability controls.

The architecture documentation submitted by the firm shall clearly define interactions, dependencies, interfaces, and security boundaries across all architecture layers.

3.1.2 Technology Justification and Performance Requirements

The firm shall provide a clear justification for the selected blockchain platform and consensus algorithm, demonstrating that the proposed solution meets the data integrity, immutability, governance, and performance requirements set out in this ToR. Proposals advocating centralised databases, hashed-RDBMS journals, or hybrid non-DLT alternatives in place of a permissioned blockchain will be considered non-responsive. The justification shall demonstrate suitability for Ghana's ASM operating environment.

The proposed solution shall meet the following minimum performance requirements:

- Demonstrate scalable architecture capable of supporting projected national transaction volumes under peak operational conditions.
- Provide benchmark testing results for sustained and peak throughput under representative workloads.
- Support a minimum sustained throughput of 100 transactions per second and peak throughput of 300 transactions per second under representative national operating conditions.
- Ensure transaction confirmation latency and synchronisation performance are sufficient to support near-real-time operational workflows and regulatory oversight.
- Demonstrate reliable offline-to-online synchronisation without data loss or duplication.

The firm shall also define:

- Node distribution architecture (including number, ownership, and roles of nodes).
- Governance model for node participation, validation rights, and consensus mechanisms.
- Data access controls for regulators and authorised stakeholders.

Mandatory National Go-Live Capabilities

For national go-live, the firm shall, at a minimum, deliver the following fully operational capabilities:

- Digital identity and actor registry covering miners, buyers, kitchens, aggregators, and transporters
- End-to-end chain-of-custody capture, including regulated kitchens and aggregation points
- Assay data capture, variance reconciliation, and mass-balance reporting
- AML/CFT monitoring, case management, and STR/SAR generation
- Core dashboards and regulator-ready reports for GoldBod and authorised agencies

Advanced analytics enhancements and optimisation features may continue post-pilot provided that the above capabilities are operational and accepted.

A live, production-ready permissioned blockchain network operational within Ghana, recording all mandatory TAS events and accessible to authorised GoldBod users for audit, verification, and reporting purposes.

Component	Summary	Detailed Service
A. System Design & Architecture	Develop a full technical, operational, and security architecture for the TAS covering the entire gold value chain.	1. Conduct stakeholder mapping and business process analysis (miners, Tier 1, Tier 2, aggregators, GoldBod and other relevant regulators). 2. Define TAS data model across TP0–TP7 touchpoints. 3. Develop logical architecture covering mobile apps, APIs, data platform, analytics, and integrations. 4. Clearly define on-chain vs off-chain responsibilities and data

		flows. - Design and implement a Sealed Processing Protocol for kitchens and aggregation centres, including: - Seal application, validation, and breakage controls - Assignment of responsible personnel - Capture of inputs, outputs, timestamps, GPS, and photo evidence - Prepare Data Protection Impact Assessment (DPIA) in line with Act 843. - Define performance benchmarks (latency, throughput, offline sync reliability). - Specify disaster recovery (RPO ≤ 15 min / RTO ≤ 4 hrs). - Produce Architecture & Design Report for GoldBod approval.
B. Digital Identity & KYC Module	Establish a comprehensive registry for all actors and facilities with integrated identity verification and due diligence	- Create a centralised actor registry (miners, Tier 1, Tier 2/kitchens, aggregators, transporters). - Integrate with existing National Identification databases (such as Ghana Card/NIA APIs) with fallback manual verification. - Capture licence and concession numbers, beneficial ownership, and sanctions/PEP screening results. - Link actors to registered devices and digital signatures/identities.

		5. Enforce periodic KYC refresh cycles 6. Maintain immutable audit trails of identity lifecycle events (onboarding, suspension, reactivation).
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C. Tagging & Traceability Infrastructure	Deploy hybrid physical and digital system for tracking all gold movements and transformations	1. Issue serialised QR/NFC tags or seals to lots and batches at extraction. 2. Require scans at every custody/processing event (TP1–TP7) capturing timestamp, GPS, actor, device ID, weight, purity, photo, and signature. 3. Implement anti-reuse logic and exception workflows for damaged or missing seals. 4. Enable merge/split/commingle with lineage matrix preserving miner-level source data. 5. Store photo and document evidence for each tagged event. 6. Detect and prevent tag reuse, duplication, or unauthorised reassignment across transactions. 7. Implement cross-verification mechanisms using weight patterns, actor transaction history, and geolocation data. 8. Require mandatory photographic evidence linked to each tagging and scanning event. 9. Establish exception workflows for lost, damaged, or missing tags, including supervisory validation and audit trails. 10. Define fraud scenarios and corresponding system controls to mitigate manipulation of traceability data.
D. Mobile & Web Applications	Build and deploy secure, user-friendly, offline-capable applications for all user roles	1. Develop Android and iOS mobile applications for all actors

		(including miners, Tier 1, Tier 2/kitchens, aggregators, and transporters). - Design web portals for supervisors, GoldBod HQ, and regulators. - Provide offline data capture with encrypted local storage. - Implement secure synchronisation upon connectivity restoration. - Implement device trust and endpoint-security controls including: - device registration and binding - rooted/jailbroken device detection - remote device disablement and wipe capability - encrypted local credential storage - device attestation and authentication - session timeout and forced re-authentication - offline transaction signing protection - Restrict TAS access to registered and approved devices with traceable device identifiers.
E. Assay & Mass-Balance	Enable capture, verification, and reconciliation of purity and weight readings across the chain. Assay readings, re-assay outcomes, and variance reconciliations shall be anchored	- Integrate assay data from Tier 2 and aggregator kitchens. - Capture intermediate and final assay results. - Define allowable tolerance thresholds for variance detection: $\pm 0.5\%$ on weight per touchpoint,

	to the blockchain ledger to preserve an immutable measurement history.	$\pm 1.5\%$ on purity per touchpoint, and $\pm 2\%$ cumulative mine-to-intake. Variances exceeding tier thresholds shall auto-open a case for investigation. 4. Conduct final re-assay at GoldBod to validate purity and weight. 5. Produce variance and mass-balance reports. 6. Define: • acceptable tolerance (auto-approve) • moderate variance (flag + review) • high variance (block / escalate) 7. Automatically trigger alerts and investigation workflows. 8. Maintain audit trails of variance resolution decisions. 9. Enable configurable enforcement rules by GoldBod. The TAS shall support LBMA-compliant Chain of Custody through commingling and aggregation using mass-balance principles and contributor-level lineage preservation. Where gold from multiple sources is consolidated, the system shall maintain auditable attribution and reconciliation records sufficient to demonstrate continuity of custody, traceability, and responsible sourcing in accordance with LBMA Responsible Gold Guidance.
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F. Central Data Platform	Establish a secure data backbone ensuring immutability and auditability.	1. Deploy enterprise-grade relational database (e.g., PostgreSQL or equivalent) that supports ACID compliance, cryptographic integrity controls, and high-availability configurations. 2. Maintain append-only permissioned blockchain ledger. 3. Configure API gateway for integrations. 4. Implement object storage for photos and documents. 5. Enforce encryption (in transit and at rest). 6. Provide full audit logging and access controls.
G. AML/CFT Analytics & Case Management	Integrate AML/CFT analytics to identify, flag, and report suspicious activities. AML/CFT alerts, investigations, and case dispositions shall be recorded or hash-anchored on the blockchain ledger to ensure evidentiary integrity.	1. Develop rules and anomaly detection models for red-flag behaviors (e.g., structuring, under-reporting, excessive purity jumps, third-party payments). 2. Assign dynamic risk scores to actors based on activity. 3. Create a case management module for investigations. 4. Generate and export STR/SAR files electronically to the FIC. 5. Maintain audit logs for all compliance actions. 6. Phased AML Implementation • Basic: transaction logging + simple rules • Intermediate: structured alerts + risk scoring

		• Advanced: anomaly detection + full case management • Minimize false positives and avoid disruption to ASM trade
H. Integration & Interoperability	Enable data exchange with national systems.	1. Implement bi-directional APIs with Minerals Commission, Bank of Ghana, FIC, and GoldBod licensing/assay systems. 2. Establish secure authentication and data-sharing protocols (OpenAPI 3.x). 3. All TAS APIs shall implement OAuth 2.0 for authorization and OpenID Connect (OIDC) for identity federation in compliance with Government API Standards and interoperability requirements published by NITA. The API framework shall additionally support: • API versioning and lifecycle management • standardized response and error-handling structures • rate limiting and throttling controls • API monitoring and audit logging • secure token management • service authentication and authorization • API documentation aligned with OpenAPI 3.x standards

		The firm shall ensure interoperability alignment with the Ghana e-Government Interoperability Framework (eGIF) and applicable government API governance policies. 4. Map data fields for licences, export permits, FX settlements, and AML data fields. 5. Provide sandbox and production environments for testing. 6. Provide integration documentation and API test scripts. Integration Readiness Controls: 7. GoldBod shall secure signed Memoranda of Understanding (MoUs) with the Minerals Commission, Ghana Revenue Authority (Customs Division), Bank of Ghana, and Financial Intelligence Centre prior to bid issue. Each MoU shall confirm API readiness, name technical contacts, and commit sandbox access dates. The firm shall conduct an integration readiness assessment against each MoU. 8. Identify technical and governance constraints. 9. Provide fallback mechanisms: • batch uploads • manual data entry
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		workflows • delayed synchronisation 10. Ensure system functionality even without full real-time integration.
I. Pilot Implementation	Test and validate system functionality in selected districts	1. Select at least five ASM-dense districts representing varied geography and supply routes. 2. Register and onboard $\geq 1,500$ miners, ≥ 50 aggregators, and ≥ 3 domestic refiners. 3. Operate the system under live field conditions for a minimum of 3 months. 4. Achieve $\geq 95\%$ lineage integrity of aggregated batches during pilot (rising to $\geq 99\%$ prior to national go-live). 5. Validate integrations and usability. 6. Conduct user satisfaction surveys and evaluation
J. Training, Communications & Adoption	Build national capacity for sustainable use of TAS and promote awareness in ASM communities.	1. Develop a detailed training curriculum and ToT program. 2. Deliver role-specific training to miners, Tier 1/2 operators, aggregators, and inspectors. 3. Produce user manuals, quick guides, and video modules. 4. Conduct community outreach and sensitisation on traceability and AML/CFT relevance. 5. Establish helpdesk and support channels

		6. Design incentives for participation: • better pricing • faster transactions • access to formal markets 7. Link TAS use to regulatory enforcement 8. Monitor adoption metrics, behavioral change and feedback loops.
K. National Rollout & Support	Scale TAS nationwide and ensure system sustainability through continuous support and maintenance.	1. Update system based on pilot feedback and finalize for full rollout. 2. Implement phased onboarding aligned with licence renewal cycles. 3. Provide 24-month stabilisation and O&M support with SLAs. 4. Conduct quarterly performance and compliance reviews. 5. Deliver annual assurance and penetration tests. 6. Prepare the final handover package with documentation and capacity transfer.

Operational Readiness Boundary

The firm's responsibility extends to system readiness, workflow configuration, SOP development, training delivery, and initial stabilisation support. Statutory enforcement, staffing, and regulatory actions remain the responsibility of GoldBod. For avoidance of doubt, configuration and tuning of system-based AML/CFT rules and alerts constitute technical support functions and do not transfer statutory enforcement authority from GoldBod.

Blockchain governance & ownership

The blockchain network deployed for TAS shall operate under the ownership, control, and governance of the Ghana Gold Board. Node participation, access rights, and administrative privileges shall be defined in accordance with GoldBod policy and regulatory requirements. The firm shall not retain proprietary control over the blockchain network, smart contract logic, or ledger data beyond the contract term, subject to agreed support and transition arrangements.

All smart contract deployments, upgrades, emergency patches, and version changes shall be governed through formal change-control procedures approved by GoldBod.

The firm shall implement:

- smart contract version control
- upgrade approval workflows
- rollback procedures
- emergency pause mechanisms
- multi-signature authorisation for production deployment

All production smart contract changes shall maintain immutable audit records and require documented testing and approval prior to deployment.

The TAS shall be implemented using open and well-documented standards, APIs, schemas, and interoperability protocols sufficient to permit future maintenance, enhancement, or migration by GoldBod or third-party vendors without unreasonable dependency on proprietary tooling or undocumented vendor-specific components.

GoldBod shall establish a TAS Governance and Oversight Structure comprising technical, operational, regulatory, and compliance stakeholders.

The governance framework shall define:

- governance committees and reporting structures
- decision-making authority and approval rights
- blockchain node governance and participation rules
- smart-contract governance procedures
- change management authority
- inter-agency coordination mechanisms
- regulatory oversight responsibilities

- operational escalation procedures
- data governance and access-control responsibilities

The firm shall support GoldBod in developing governance charters, operational procedures, escalation matrices, and decision frameworks required for sustained national operation of TAS.

IMPLEMENTATION METHODOLOGY, DELIVERABLES AND TIMELINE

Methodology

Bidders shall propose a comprehensive, risk-aware implementation methodology demonstrating how functional, technical, security, and compliance requirements will be delivered. The methodology shall explicitly address system design, development, testing, deployment, integration, pilot implementation, national rollout, and acceptance. The proposed methodology shall form part of the technical evaluation.

Transition and Migration Strategy

The firm shall define a clear transition plan from existing manual and semi-digital systems to TAS, including:

- Data migration strategy (where applicable).
- Parallel run arrangements during pilot and early rollout phases.
- Change management processes for stakeholders transitioning from current practices.
- Risk mitigation measures to prevent disruption to ongoing gold trade activities.

The transition approach shall ensure continuity of operations and gradual system adoption.

The implementation methodology shall explicitly demonstrate how risks identified at system, process, and actor levels are decomposed into executable controls, assigned ownership, and enforced through operational workflows. Bidders shall provide evidence of how field-level realities, including shared devices, informal trading behaviours, and connectivity limitations, are addressed within the system design and deployment approach.

Change Management & Adoption Framework

The firm shall develop and implement a comprehensive Change Management and Change Adoption Plan covering both technical change governance and organisational adoption requirements.

The Change Management Plan shall, at a minimum, include:

Technical Change Management Requirements

- Detailed change-control procedures and governance processes
- Defined approval and escalation matrix for system changes
- Audit trails and change traceability requirements
- Rollback and recovery procedures for failed changes
- Operational readiness assessments prior to production deployment
- Emergency change handling and incident-response procedures

Organisational Change & Adoption Requirements

- Stakeholder impact assessment framework
- Structured communications and user-awareness strategy
- Resistance management and adoption-support measures
- Clearly defined adoption success indicators and performance metrics
- User feedback and continuous-improvement mechanisms
- Capacity-building and institutional-transition planning

The Change Management Plan shall define:

- roles and responsibilities;
- governance authorities;
- escalation procedures;
- phased adoption strategy;
- implementation-readiness criteria and ;
- sustainability mechanisms to support long-term national operation of TAS.

Deliverables and Timeline

The total implementation period shall be eighteen (18) months from the date of contract signature and shall be implemented through a phased delivery model:

- Phase 1 (0–6 months): Core system development, pilot deployment, and validation of key functionalities.

- Phase 2 (6–12 months): System scaling, integration with external systems, and enhancement of AML/CFT capabilities.
- Phase 3 (12–18 months): Optimisation, national rollout completion, and system stabilisation.

National rollout shall be contingent upon successful pilot validation and defined readiness criteria.

Phase	Deliverable	Timeline	Key Milestones
1. Mobilisation & Inception	1. Mobilisation and Inception Pack (validated requirements, RACI, TP0–TP7 data dictionary v1, detailed work plan). 2. Risk and dependency register (integrations, tagging supply, identity/KYC). 3. Stakeholder workshops & inception sign-offs.	Month 1-2	1. Stakeholder & scope sign-off completed. 2. Pilot districts selected and approved. 3. Tagging/seal samples approved for field use. 4. DPIA plan initiated and data classifications baselined.
2. Design & Alpha	1. Design Dossier (process maps incl. kitchens/amalgamation, canonical data model, OpenAPI specs, security & privacy, DR plan, sealed-processing control design). 2. Alpha MVP prototype in sandbox (mine→Tier 2/kitchen→aggregator→GoldBod).	Month 2-3	1. Design Dossier approved by the GoldBod. 2. Alpha demo passed (end-to-end flow incl. lineage & assays). 3. Data mappings v1 issued for MinCom, GRA/Customs, BoG (and FIC where accessible).
3. Build & Integrations (Gate 1)	1. Core modules: Identity/KYC, Tagging and Chain-of-Custody, Assay capture, Event ledger/DB, Web portal, Offline sync. 2. Integration Gate 1:	Month 3- 6	1. API gateway live and rate limits verified. 2. Offline→online sync proven in field conditions. 3. Security test plan approved (pen test booked).

	MinCom live + ≥ 1 of GRA/Customs or BoG. 3. Test harness, logs, and Gate-1 acceptance		4. Gate-1 acceptance signed.
4. Pilot Implementation	1. Pilot deployment in 5 districts; ToT delivered. 2. M&E framework, dashboards, and adoption tracking live. 3. Pilot Evaluation Report (draft) with lineage, variance, uptime, and UX findings. The pilot shall be conducted in no fewer than five (5) ASM-dense districts and shall include Tier 1 buyers, Tier 2 buyers, kitchens, aggregators, and GoldBod intake points. A minimum lineage integrity rate of ninety-five percent (95%) for aggregated batches during pilot, rising to ninety-nine percent (99%) prior to national go-live, shall be achieved as a condition of acceptance.	Month 6-8	1. $\geq 1,500$ miners, ≥ 50 aggregators onboarded; Tier 2 kitchens registered. 2. Lineage preserved in $\geq 95\%$ of aggregated batches. 3. Uptime $\geq 99.0\%$; variance within defined tolerances. 4. User satisfaction baseline captured.
5. Hardening & Integrations (Gate 2)	1. Remediation from pilot (performance, UX, data quality). 2. Integration Gate 2 to reach ≥ 3 counterpart systems (MinCom, GRA/Customs, BoG; FIC where accessible). 3. Security & DR tests (RPO ≤ 15 min; RTO ≤ 4 hrs). 4. Go-Live readiness	Month 8-9	1. P95 API latency ≤ 1s achieved. 2. DR test passed; backup/restore verified. 3. Gate-2 acceptance signed with data-mapping sign-offs. 4. Go-Live readiness approved by the Steering Committee.

	checklist & cutover plan.		
6. National roll-out	1. Production deployment & cutover. 2. Aggregator-first national onboarding (phased with licence cycles). 3. Go-Live Report and Operations Pack (SOPs/runbooks/guide)	Month 12-15 (limited production from Month 10; full national rollout commences after Gate-3 acceptance at Month 12)	1. Go-Live sign-off achieved. 2. $\geq 60\%$ of aggregators active and transacting. 3. First compliant export run completed: ≥ 50 kg cumulative shipped to a named LBMA Good Delivery refiner, with a full RGG-aligned due-diligence dossier accepted by the refiner's compliance team. 4. L1–L3 support model implemented
7. Training, Support and Handover	1. Develop a detailed training curriculum and ToT program. 2. Deliver role-specific training to miners, Tier 1/2 operators, aggregators, and inspectors. 3. Produce user manuals, quick guides, and video modules. 4. Conduct community outreach and sensitisation on traceability and AML/CFT relevance. 5. Establish helpdesk, Ghana-hosted ticketing system and USSD-based field support (WhatsApp is not permitted as a support channel due to in-	Month 15-18	Training and handover completed for GoldBod staff, miners, Tier 1/2 operators, aggregators, and inspectors.

	country data residency requirements), and field clinics. 6. Monitor adoption metrics and feedback loops.		
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Acceptance & Sign-off Criteria

Each deliverable shall be subject to formal acceptance by GoldBod based on pre-agreed functional, performance, security, and documentation criteria. Acceptance shall be a prerequisite for milestone payments. GoldBod reserves the right to request remediation where delivery does not materially meet agreed specifications.

GoldBod reserves the right to appoint an independent technical assurance advisor to review architecture, cybersecurity, performance, smart-contract integrity, interoperability, and implementation quality throughout the project lifecycle. The firm shall provide reasonable cooperation, technical access, documentation, and demonstration support required for such independent assurance activities.

Acceptance shall not be granted where:

- Lineage integrity across defined touchpoints is materially incomplete
- AML/CFT rules, alerting, or reporting functions are non-operational
- Pilot KPIs, including a minimum of ninety-five percent (95%) batch lineage integrity, are not met
- Security, disaster recovery, or audit requirements are unmet
- Operational risk controls at actor level, including enforcement workflows, data integrity validation, and behavioural risk monitoring, are not fully implemented and demonstrated under field conditions

AML/CFT COMPLIANCE AND FUNCTIONAL SPECIFICATIONS

The TAS shall be designed to support the detection, prevention, investigation, and reporting of money-laundering and terrorist-financing risks across the gold supply chain. All AML/CFT rules, thresholds, and analytical models shall be explainable, auditable, and configurable by authorised GoldBod personnel without vendor intervention.

Phased AML/CFT Implementation Approach

The TAS shall implement AML/CFT controls using a risk-based, tiered approach aligned with actor categories and operational realities:

- Basic Level: Transaction recording, actor identification, and basic rule-based flagging.
- Intermediate Level: Enhanced monitoring including structured rule-based alerts and risk scoring for Tier 1 and Tier 2 actors.
- Advanced Level: Full anomaly detection, case management, and regulatory reporting applied to aggregators, exporters, and high-risk actors.

The system shall minimize false positives and ensure that AML/CFT controls do not unduly disrupt legitimate ASM trade flows.

Actor Scale, Cooperatives, and Device Use

The AML/CFT controls described in this section shall be implemented in accordance with the phased approach defined in Section 5.1.

The TAS shall be designed for national-scale deployment. Without limiting scalability requirements, the system is expected to accommodate tens of thousands of miners and community sellers, several thousand Tier 1 buyers, several hundred Tier 2 buyers and registered kitchens, and a limited number of aggregators and transporters.

Cooperative mining entities shall be registered as legal actors within TAS. Where required for AML/CFT, beneficial ownership, or due diligence purposes, individual members of cooperatives shall be captured as associated records linked to the cooperative entity. For AML monitoring tractability, cooperatives shall be capped at 50 members per registered entity. Member-level KYC and beneficial-ownership capture shall be required for any cooperative whose cumulative monthly throughput exceeds 5 kg, and payment-splitting logic among members shall be documented and auditable.

Shared mobile devices are permitted in ASM operating environments. However, all transactions recorded in TAS must be attributable to authenticated individual users, with role-based access controls, device binding where applicable, and full audit trails to ensure non-repudiation.

AML Risk Typologies

AML/CFT monitoring shall explicitly include capacity-versus-output mismatches by geographic zone, transaction clustering in border-proximate districts, and abnormal purity or grade shifts occurring immediately after kitchen processing.

Feature	Specification	Category
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Actor KYC & Beneficial Ownership Registry	1. Register miners, Tier 1, Tier 2 (and kitchens), aggregators, transporters. 2. Capture IDs (Ghana Card or fallback), licences, concession links, BO details. 3. Bind actors to facilities/devices; roles & permissions. 4. Enforce periodic KYC refresh with expiry alerts.	Detection/Prevention
Sanctions/PEP Screening	1. Screen actors and BOs at onboarding and periodically against OFAC SDN, UN Consolidated, EU Consolidated, and UK HMT sanctions lists, plus a commercial PEP/adverse-media provider (e.g., World-Check or Dow Jones). Re-screen daily and within 24 hours of any list update. 2. Fuzzy matching, overrides with justification. 3. Full audit of hits, clears, and escalations.	Detection/Prevention
Risk Scoring (Actor/Facility)	1. Dynamic scores using geography, role, volumes, cash intensity, history, alerts. 2. Tag high-risk kitchens/aggregators and border-proximate sites. 3. Drive EDD triggers and workflow routing.	Detection/Prevention
Rules-Based Monitoring	1. Configurable rules (structuring, rapid turnover, purity jumps, third-party payments). 2. Evaluate all custody/processing/payment events in near-real time. 3. Thresholds, whitelists, false-positive suppression.	Detection/Prevention
Anomaly Detection	1. Outliers on weight/purity, capacity vs output, route deviations. 2. Peer baselines by district/actor type. 3. Model monitoring, back-testing, drift checks; target false-positive rate $\leq 70\%$ post-tuning with a 24-month roadmap to $\leq 50\%$; monthly FP measurement reporting.	Detection/Prevention
Kitchen Registration & Sealed-Processing Control	1. Register Tier 2 and aggregator kitchens as regulated processing points. 2. Digital seals with officer/inspector reference at sensitive steps. 3. Record operator, method, inputs/outputs, time/GPS, photos. 4. Mercury balance: capture	Detection/Prevention

	mercury purchase, consumption, and stockpile per kitchen; reconcile against gold output (gravimetric ratio); flag implausible deviations for environmental and Minamata Convention compliance.	
Lot/Batch Lineage & Commingling Controls	1. Preserve per-contributor lineage when merging blades/lots to 1kg+ batches. 2. Maintain split/merge histories with weighted-purity math. 3. Block batch closure if lineage/seal checks fail.	Detection/Prevention
Assay Variance Reconciliation & Mass-Balance	1. Capture Tier 2/aggregator tests; re-assay at GoldBod. 2. Define tolerances; auto-reconcile or raise exceptions. 3. Periodic mass-balance by actor/facility with explanations for losses/gains.	Detection/Prevention
Payments & Beneficiary Monitoring	1. Record method, payer/payee, bank/wallet, FX refs. 2. Detect third-party payments and beneficiary mismatch. 3. Reconcile value vs assay/weight and market price.	Detection/Prevention
Cash Threshold & CTR Controls	1. Configurable cash thresholds per role/event with a statutory floor of USD 10,000 equivalent (or GHS at Bank of Ghana mid-rate) per the Anti-Money Laundering Act, 2020 (Act 1044). 2. Auto-flag CTR-eligible events and compile CTR summaries; aggregate detection across 24-hour, 7-day, and 30-day windows. 3. Detect repeated near-threshold behavior.	Detection/Prevention
Route & Geo-Risk Monitoring	1. GPS at each event. Detect route deviations/custody gaps. 2. Geo-fence borders (≤ 25 km from official crossings, ≤ 10 km from informal crossings per MinCom/Customs records), national parks, and restricted zones. 3. District risk weighting at 1.5–2$\times$ for border-proximate transactions. 4. Cross-reference with Ghana Immigration Service movement data and Ghana Customs ASYCUDA export declarations; auto-generate	Detection/Prevention

	STR where a registered actor's gold is intercepted at a crossing.	
Capacity vs Output & Grade Outliers	1. Maintain expected capacity per licence/facility. 2. Alert on implausible output or abnormal grade jumps. 3. Require supervisor sign-off for repeats.	Detection/Prevention
Case Management & Investigations	1. Unified alert queue with severity, assignee, SLAs. 2. Attach evidence (photos, seals, receipts, chat). 3. Track actions, dispositions, regulator escalations.	Reporting/Assurance
STR/SAR e-Filing to FIC	1. Prepare STR/SAR with structured fields, narrative, attachments. 2. Export in regulator-compatible format/API. 3. Track submissions, acknowledgements, queries.	Reporting/Assurance
Watchlist & Adverse Media Management	1. Maintain internal banned lists and external feeds. 2. Version/timestamp updates; bulk re-screening. 3. Overrides with approvals and audit.	Detection/Prevention
Access Control, SoD & Dual Control	1. RBAC/ABAC with least privilege. 2. Segregation of duties for sensitive actions. 3. Dual approval for high-risk events/whitelist changes.	Detection/Prevention
Tamper-Evidence, Device Security & Offline Integrity	1. QR/NFC seals with anti-reuse; photo at scan. 2. Device binding; jailbreak/root detection; signed payloads. 3. Offline queues with secure sync and tamper-evidence.	Detection/Prevention
Audit Trails & Evidence Locker	1. Immutable event ledger with hash-chaining. 2. Non-repudiation on user/action/time/IP/device. 3. Evidence locker with retention controls.	Reporting/Assurance
Data Governance & Retention	1. Data classification; retention (≥ 7 years) and legal holds. 2. Encryption in transit/at rest; field-level access.	Reporting/Assurance

	3. DPIA and privacy notices in app flows.	
Regulatory & Management Reporting (KRI/KPI)	1. Dashboards for lineage integrity, variance, alert volume, STR/SAR throughput, CTR counts, adoption. 2. Scheduled regulator packs (MinCom/GRA/BoG/FIC). 3. CSV/PDF/API exports. 4. Public-facing transparency dashboard publishing aggregate volumes by district, declared-vs-assayed variance by aggregator, and STR/SAR counts (anonymised, no case content), with periodic publication of Merkle root hashes enabling third-party verification of ledger integrity.	Reporting/Assurance
Model & Rule Governance	1. Versioning and changing logs. 2. Back-testing/validation before promotion. 3. Periodic tuning with FIC feedback/pilot learnings.	Reporting/Assurance
Training, Red-Flag Library & Comms	1. In-app guidance and red-flag catalog by role. 2. Micro-learning and refresher prompts. 3. Community comms templates.	Detection/Prevention
Whistleblowing & Grievance Channel	1. Confidential reporting channel linked to cases. 2. Multi-language; offline capture option. 3. Anti-retaliation tracking & escalation paths.	Reporting/Assurance

AML Responsibilities

Initial AML/CFT rule configuration shall be proposed by the firm and approved by GoldBod. Ongoing rule tuning and false-positive optimisation post-go-live are within scope. For avoidance of doubt, system rule configuration and optimisation do not transfer statutory enforcement authority from GoldBod.

TAS provides detection, alerting, and reporting support. Statutory responsibility for STR/SAR filing remains with GoldBod in accordance with applicable law.

These controls support GoldBod's risk-based due diligence obligations under the OECD Due Diligence Guidance for Responsible Supply Chains of Minerals (Gold Supplement).

Operational Risk Decomposition, Field Controls, and Enforcement Framework

The TAS shall incorporate a comprehensive Operational Risk Decomposition and Control Framework to ensure that identified risks are addressed at the lowest executable level across all actors and transaction points within the ASM gold value chain.

5.5.1 Actor-Level Risk Mapping and Control Assignment

The firm shall design and implement a structured risk decomposition model that maps risks to specific actor categories, including but not limited to:

- Miners and community sellers
- Tier 1 buyers
- Tier 2 buyers and kitchen operators
- Aggregators
- Transporters and logistics actors
- GoldBod inspectors and authorised officers

For each actor category, the system shall:

- Identify key operational, behavioural, and fraud-related risks
- Define preventive, detective, and corrective controls
- Assign clear ownership of each control (system-enforced vs human-enforced)
- Establish accountability mechanisms linked to user identity, device, and transaction records

5.5.2 Data Integrity at Source Controls

In addition to ensuring immutability of recorded data, the TAS shall implement controls to ensure the accuracy and authenticity of data at the point of capture.

At a minimum, the system shall support:

- Mandatory multi-factor data capture (e.g., weight, photo, GPS, timestamp, actor identity, device ID)
- Cross-validation rules across actors (e.g., buyer vs seller confirmations, weight and price consistency checks)
- Randomised and risk-based field verification workflows by GoldBod inspectors
- Detection of inconsistent or implausible entries prior to ledger confirmation
- Supervisory validation requirements for high-risk transactions

No transaction shall be considered fully valid where minimum data integrity requirements are not met.

5.5.2.1 Minimum Field Verification and Audit Requirements

To ensure the effectiveness of data integrity controls and prevent systemic manipulation, the TAS shall support structured and measurable field verification and audit processes.

At a minimum, the following audit and inspection thresholds shall be supported and configurable within the system:

- **Routine Transaction Verification:** A minimum of **5–10% of all transactions per month** shall be subject to randomised field verification by authorised GoldBod inspectors.
- **High-Risk Actor Verification:** A minimum of **20–30% of transactions involving high-risk actors** (as determined by AML/CFT risk scoring) shall be subject to verification.
- **High-Variance or Flagged Transactions:** 100% of transactions exceeding defined variance thresholds or triggering high-risk alerts shall be subject to mandatory review and/or physical inspection.
- **Kitchen and Aggregation Point Inspections:** All registered kitchens and aggregation centres shall undergo at least one physical inspection per quarter, with additional risk-based inspections as required.
- **Ad-hoc and Intelligence-Led Audits:** The system shall support targeted inspections triggered by:
 - AML/CFT alerts
 - Whistleblower reports
 - Pattern anomalies
 - Regulatory directives

The TAS shall:

- Generate automated inspection schedules and assignment lists
- Track inspection coverage, outcomes, and compliance rates
- Provide dashboards showing audit coverage vs required thresholds
- Escalate non-compliance with audit requirements

Failure to meet minimum audit thresholds shall be flagged as a compliance breach and reported to GoldBod management.

5.5.3 Field Enforcement Workflow and Escalation Protocols

The TAS shall define and operationalise clear enforcement workflows to ensure that detected risks, alerts, and exceptions are acted upon within defined timelines.

The system shall:

- Trigger alerts linked to predefined AML/CFT, traceability, and operational risk conditions
- Assign alerts to designated GoldBod officers or authorised personnel
- Define response time thresholds based on severity levels
- Provide escalation pathways for unresolved or repeated violations

- Record all enforcement actions, decisions, and outcomes in an auditable manner

For avoidance of doubt, TAS shall not only detect and report risks but shall support enforceable operational response workflows.

5.5.3.1 Enforcement Roles, Responsibilities, and Chain of Command

To ensure effective execution of enforcement workflows, the TAS shall incorporate a clearly defined operational enforcement structure with assigned roles and responsibilities across GoldBod personnel.

At a minimum, the following roles and responsibilities shall be supported within the system:

- **Field Inspectors / Authorised Officers:** Responsible for initial verification, field validation, and first-level response to alerts, including physical inspection and transaction confirmation where required.
- **Regional Supervisors / Coordinators:** Responsible for oversight of field operations, validation of escalated cases, approval of enforcement actions, and coordination across districts.
- **AML/CFT Compliance Unit (GoldBod HQ):** Responsible for reviewing high-risk alerts, managing investigations, approving Suspicious Transaction Reports (STRs) and Suspicious Activity Reports (SARs), and coordinating with the Financial Intelligence Centre.
- **Central Operations and Regulatory Oversight Unit:** Responsible for system-wide monitoring, enforcement policy application, exception approvals, and coordination with national regulatory bodies.

The TAS shall:

- Route alerts and cases automatically to the appropriate role based on severity and geographic jurisdiction
- Enforce role-based access and decision rights within the system
- Maintain a complete audit trail of all actions taken by each role
- Support escalation across roles where response timelines or thresholds are exceeded

For avoidance of doubt, all enforcement actions shall be attributable to specific authorised personnel, and no enforcement process shall remain unassigned or untracked within the system.

5.5.3.2 Enforcement Response Time Standards (SLAs)

The TAS shall enforce defined Service Level Agreements (SLAs) for response to alerts, exceptions, and enforcement actions. These SLAs shall be configurable by GoldBod but shall, at a minimum, meet the following baseline thresholds:

- **High-Risk Alerts (e.g., suspected fraud, major variance, AML/CFT triggers):**
Initial response within 24 hours, with investigation initiated within 48 hours
- **Medium-Risk Alerts (e.g., moderate variance, unusual transaction patterns):**
Initial review within **72 hours**
- **Low-Risk Alerts (e.g., minor inconsistencies, data anomalies):**
Review within 5 working days

The system shall:

- Automatically track and monitor compliance with SLAs
- Generate escalation alerts where response timelines are breached
- Reassign or escalate cases to higher authority levels where necessary
- Provide dashboards showing response times, backlog, and compliance rates

Failure to meet defined SLA thresholds shall be recorded as a compliance exception and included in performance monitoring reports.

5.5.4 Human Behaviour and Collusion Risk Controls

Recognising the operational realities of ASM environments, the TAS shall incorporate controls to mitigate risks arising from human behaviour, including collusion, proxy transactions, and intentional misreporting.

The system shall include:

- Device-to-user binding and authentication controls to prevent anonymous transactions
- Detection of abnormal interaction patterns between actors (e.g., repeated exclusive trading pairs, circular transactions)
- Controls for shared device usage ensuring individual accountability for each transaction
- Behavioural risk indicators integrated into actor risk scoring
- Audit trails capturing user actions, edits, overrides, and approvals

5.5.5 Incentive, Restriction, and Enforcement-Linked Compliance Mechanisms

To ensure sustained adoption and enforce compliance, the TAS shall implement mandatory linkage between actor behaviour, compliance status, and access to market participation, services, and regulatory approvals.

The system shall:

A. Compliance-Based Incentives

- Assign compliance levels (e.g., Level 1–3) to all actors based on traceability, reporting accuracy, and AML/CFT adherence
- Enable preferential treatment for compliant actors, including:
 - priority access to GoldBod purchasing channels
 - faster transaction validation and processing
 - eligibility for formal financing, aggregation, and export participation

B. Mandatory Restrictions for Non-Compliance

- Automatically flag non-compliant actors based on predefined thresholds
- Enforce restrictions including:
 - temporary suspension from TAS transactions
 - restriction from participating in aggregation or export chains
 - mandatory review or re-validation before re-entry into the system

C. Regulatory Linkages

The TAS shall support integration of compliance status with:

- Licensing and permit validation processes
- Export approval workflows
- Payment and settlement systems

Non-compliant actors shall not be eligible for full participation in formal gold trade channels unless corrective actions are completed and validated.

D. Behavioural Monitoring and Progression

- Track compliance trends over time
- Enable progression or regression between compliance levels
- Provide audit-ready records linking behaviour to system actions

For avoidance of doubt, compliance status shall have direct operational and economic consequences and shall not be treated as advisory or informational only.

5.5.6 Operational Risk Register and Continuous Monitoring

The firm shall develop and maintain a detailed operational risk register that:

- Breaks down high-level risks into actionable sub-risks at transaction and actor level
- Links each risk to specific system controls, ownership, and monitoring indicators
- Defines Key Risk Indicators (KRIs) for continuous monitoring
- Supports periodic review and update based on pilot findings and operational data

The operational risk framework shall be fully integrated into TAS dashboards and reporting modules to enable proactive risk management.

TECHNICAL AND SECURITY SPECIFICATIONS

The solution shall conform to the minimum technical and security specifications outlined below. Bidders shall respond to each requirement with a clear explanation and, where appropriate, references to standards, documentation, or prior implementations that demonstrate compliance.

The TAS shall be built upon a permissioned blockchain platform (Hyperledger Fabric 2.5+, Hyperledger Besu, R3 Corda 5, or Quorum) using a Byzantine fault-tolerant consensus algorithm (IBFT2, QBFT, or Raft) that functions as a core production component of the system. The blockchain layer shall provide immutability, non-repudiation, and verifiable audit trails for all critical TAS events and shall be designed to operate continuously as part of the national production environment.

Feature	Specification
System Architecture (Minimum Requirements)	The system shall employ coherent, maintainable architecture that minimizes operational complexity, supports offline-first field operations, ensures data integrity, and can be hosted fully in-country under GoldBod control. Vendors may propose monolithic or modular architectures, provided functional, security, and performance requirements are met On-Chain and Off-Chain Data Responsibilities The firm shall clearly define and document the separation of on-chain and off-chain data within TAS. On-chain records shall include transaction metadata, state changes, event timestamps, cryptographic hashes, digital signatures, and references required to establish an immutable chain of custody and compliance history. Off-chain repositories may be used for performance, analytics, reporting, and storage of large files such as documents, images, and assay certificates. All off-chain records that relate to traceability, compliance, or enforcement shall be cryptographically anchored to the blockchain ledger to ensure integrity and auditability.
Blockchain Node Architecture & Hosting	The system shall be hosted entirely within Ghana under GoldBod's control. Bidders may propose physical servers, virtualised infrastructure, or private cloud deployments, provided all data residency, security, and performance requirements are met. 1. Hardened Linux host in Ghana DC under GoldBod control. 2. Multi-node deployment: minimum of four (4) validator nodes distributed across at least two (2) physical sites within Ghana (primary and disaster-recovery), each provisioned with no less than 32 vCPU, 128 GB ECC RAM, and NVMe RAID-10 storage. Application, database, and object-store services shall not be co-resident

	on a single host. 3. ECC RAM, RAID-10 storage, redundant power. The blockchain network shall be deployed as a permissioned node-based architecture hosted in-country. Disaster recovery arrangements shall include node redundancy and ledger replication to ensure continuity.
Data Residency & Network Egress Control	1. Firewall egress allow-list 2. Optional Admin via VPN only.
Database & Storage	1. SQL database for transactions (actors, licences, facilities, lots/batches, assays, events). 2. Row-level audit cols + soft delete.
Event & Audit Ledger (Immutable Trail)	1. Append-only Immutable permissioned blockchain ledger 2. Auditor export tools.
Transport Layer Security (TLS) & Encryption at Rest	1. TLS 1.3 for all external traffic; mTLS for all service-to-service communication (SPIFFE/SPIRE or equivalent), with short-lived (≤ 24h) workload identities. 2. Data-at-rest AES-256. 3. Keys in HSM/KMS vault, FIPS 140-2 Level 3 minimum (Level 4 for blockchain validator root keys), with documented key ceremony and M-of-N (e.g., 3-of-5) quorum for production root key custody.
Authentication & Authorisation	1. MFA for admins, SSO optional. 2. RBAC + ABAC for fine-grained access. 3. JWT or short-lived sessions with rotation.
Offline-First Mobile Sync APIs	1. Android 12+ (API 31), iOS 16+, with Google Play Integrity API and Apple App Attest device attestation 2. Offline data capture 3. Local encrypted store 4. Conflict resolution on sync
API Governance & Rate Limiting	1. Publish OpenAPI + RFC 7807 errors. 2. Global/per-client rate limits. 3. Scoped API keys for MinCom/FIC/GRA/BoG.
Tagging & Device Integration	1. Tamper-evident polymer seals with embedded NFC chips supporting cryptographic mutual authentication and anti-cloning (e.g., NTAG 424 DNA AES-128 or equivalent), plus printed QR for visual fallback. 2. Bluetooth scale ingest with signed readings. 3. Mandatory photos at critical steps.

Assay Device & Lab Interfaces	1. CSV/API ingest from portable XRF devices (Olympus Vanta or equivalent) for field assay and fire-assay laboratory instruments for GoldBod final assay, with periodic round-robin proficiency testing across participating labs. 2. Store method/calibration/operator. 3. Auto reconcile vs GoldBod final assay.
Security Operations (Monitoring & Alerts)	1. On-prem SIEM stack. 2. On-call alerts with runbooks.
Backups, Recovery & Continuity	1. Daily encrypted backups (hot+cold). 2. RPO $\leq 15m$, RTO $\leq 4h$, with a disaster-recovery site in a separate Ghana region (e.g., primary Accra, DR Kumasi), active-passive replication with $\leq 30s$ replication lag, and an annual full-cutover DR drill in addition to quarterly restore drills. 3. Quarterly restore drills.
Secrets & Configuration Management	1. Vaulted secrets; regular rotation. 2. Configuration managed as environment variables, signed and checksummed. 3. Pre-commit secret scans.
Secure Development Lifecycle	1. Peer review; signed builds. 2. SBOM published per release in CycloneDX or SPDX 2.3+ format, signed with vendor release key, with vulnerability scan output (Trivy or Gype) attached.
Change & Release Management	1. Versioned releases (semver). 2. Auto DB migrations with rollback.
Privacy & Data Protection (Act 843)	1. DPIA; in-app privacy notices. 2. Field-level masking for sensitive data. 3. DSAR/erasure workflows: all personally identifying information shall be stored off-chain only; the blockchain ledger shall reference identities solely by opaque salted-hash identifiers, with salt rotation supporting effective erasure under Act 843 while preserving ledger integrity.
Operations & Observability	1. Metrics/logs/traces; health endpoints. 2. Dashboards for latency, errors, CPU/RAM, I/O, queues. 3. Capacity alerts with runbooks.
Penetration Testing & Vulnerability Management	1. Continuous testing: SAST on every commit, DAST nightly, quarterly external penetration test by CREST or OSCP-certified testers, annual red-team exercise; track remediation against agreed SLAs.

	2. Monthly CVE scans (host/app deps). 3. Patch windows with comms.
Source Code Escrow & IP Terms	1. Source-code escrow with independent Ghana-based escrow agent, quarterly updates, with release triggers: vendor insolvency, breach uncured within 30 days, change-of-control, contract end, or documented and audited GoldBod request. 2. Deliver build scripts/IaC/admin manuals. 3. License allows internal ops/mods.
Localisation, Accessibility & UX	1. Offline error/retry guidance. Mobile UI shall support English plus Twi and at least two additional mining-community languages (e.g., Dagbani, Hausa, Ewe, Ga, or Frafra), with voice-prompt support for low-literacy users. WCAG 2.1 AA accessibility compliance required. 2. High-contrast/large targets for field.
Retention & Archival	1. ≥ 7 -year retention. 2. Tiered retention: 10 years for AML records (per Act 1044 section 23), 7 years for general transaction data, 12 months for raw scan photos with permanent retention of their cryptographic hashes, and indefinite retention for security-incident logs; policy-driven purge after expiry except where subject to legal hold.

Any interim or transitional reporting or data-capture tools used during pilot or early implementation phases shall not constitute production systems and shall be fully replaced by in-country hosted TAS infrastructure before national rollout.

6.1 Security Standards Compliance

The TAS shall be designed, implemented, tested, and maintained in accordance with internationally recognised cybersecurity standards and secure-development practices, including:

- ISO/IEC 27001 (Information Security Management)
- ISO/IEC 27701 (Privacy Information Management)
- NIST Cybersecurity Framework
- OWASP ASVS and OWASP Mobile Security Guidelines
- CIS Security Benchmarks
- Secure Software Development Lifecycle (SSDLC) principles

The firm shall implement:

- vulnerability management procedures
- secure coding practices
- penetration testing
- dependency and patch management
- audit logging and monitoring
- privileged-access management
- endpoint protection controls

All critical vulnerabilities identified during testing or operations shall be remediated within agreed SLA timelines.

6.2 Data Residency, Sovereignty and Hosting Requirements

All primary TAS production data, blockchain validator nodes, cryptographic key-management infrastructure, and disaster recovery replicas shall be hosted within the jurisdiction of Ghana unless otherwise approved in writing by GoldBod.

GoldBod shall retain sovereign ownership and administrative control over:

- blockchain ledger data
- node governance
- cryptographic root keys
- identity registries
- audit records
- smart contract repositories

Cross-border replication, offshore backup storage, or foreign administrative access shall require prior written approval from GoldBod and must comply with the Data Protection Act, 2012 (Act 843).

The firm shall provide a hosting and sovereignty architecture demonstrating:

- node hosting locations
- jurisdictional control
- key custody arrangements
- failover locations
- privileged-access governance

EVALUATION CRITERIA

Bids shall be evaluated according to a two-stage Quality- and Cost-based selection (QCBS) approach as provided under the Public Procurement Act, 2003 (Act 663), as amended, and applicable Public Procurement Authority guidelines. Only firms achieving a minimum technical score of 75% shall proceed to financial evaluation.

GoldBod reserves the right to verify claims through demonstrations, reference checks, and technical clarifications prior to final scoring.

Upon handover, GoldBod shall retain full administrative control over the blockchain network, including node credentials and smart contract repositories, subject to agreed support arrangements.

Criterion	Weight (%)	Scoring Guide	Evidence Required
Technical Approach & Methodology (TP0–TP7 fit)	15	1. Clearly maps to ASM flow including kitchens and amalgamation lineage. 2. Shows offline-first field operations and exception handling (seal reuse, variance). 3. Demonstrates end-to-end testing approach and acceptance criteria.	1. Process maps/user journeys covering miner → Tier 1/Tier 2/kitchen → aggregator → GoldBod. 2. Sample forms/screens and exception workflows. 3. Acceptance test list aligned to ToR touchpoints.
System Architecture & Hosting Strategy (Monolith or modular architectures with APIs; in-country)	15	1. Monolith or modular modular architectures, (provided functional, security, and performance requirements are met) with well-documented APIs for mobile and third parties. 2. In-country, self-hosted design with DR targets. 3. Performance plan (latency, throughput)	1. Architecture diagram + API catalogue (OpenAPI). 2. Hosting/runbook showing Ghana-resident setup and DR (RPO/RTO). 3. Performance test plan.

		for one-box deployment and scale-up.	
AML/CFT Functional Capability	15	1. Complete KYC/BO, screening, rules + anomaly detection, STR/SAR, and mass-balance. 2. Evidence of case management and regulator-ready outputs. 3. Clear governance for rules/model updates.	1. Feature matrix mapped to ToR /AML items. 2. Sample alerts/cases and SAR templates. 3. Rule/model governance procedure.
Interoperability & Priority Integrations	10	1. Bi-directional APIs to Minerals Commission, GRA/Customs, Bank of Ghana, FIC. 2. Data mappings for licensing, export, FX, STR/SAR status. 3. Credible test strategy (sandboxes, harnesses).	1. Integration plan and mappings. 2. Letters/track record for similar gov't integrations. 3. Test evidence plan for Gate 1/2.
Security & Privacy (Act 843; auditability)	10	1. TLS, encryption at rest, MFA, RBAC/ABAC, secure SDLC. 2. Immutable event/audit ledger with export. 3. DPIA and data-minimisation practices.	1. Security design + policies. 2. Audit/logging architecture and sample exports. 3. DPIA outline and retention schedule.
Implementation Plan & Schedule (18-month)	10	1. Feasible phased implementation plan aligned with the 18-month delivery schedule 2. Clear risks/mitigations (integrations, hardware, adoption).	1. Gantt and milestone plan. 2. Risk register with owners/mitigations. 3. Resourcing matrix by sprint/phase.

		3. Resource loading aligned to milestones.	
Pilot Design & Adoption/Change	5	1. Realistic pilot in ≥ 5 diverse districts, lineage and variance KPIs. 2. Training-of-trainers, community comms, helpdesk. 3. Go/No-Go criteria tied to data completeness and uptime.	1. Pilot plan and M&E framework. 2. Training curriculum and comms assets. 3. Go/No-Go checklist.
Team & Key Experts	10	1. Proven experience in traceability, AML/CFT and mining systems. 2. Ghana/West Africa field delivery experience. 3. Time commitments and backfill plans.	1. CVs and role matrix. 2. Evidence of regional deployments. 3. Signed availability letters.
Past Performance & References	8	1. At least two comparable implementations (mining/commodity). 2. Demonstrated uptime/throughput and audit success. 3. Positive client references.	1. Case studies with contacts. 2. KPIs from live systems. 3. Reference letters.
Value-Add & Sustainability	2	1. Source-code escrow and knowledge transfer for a small in-house team. 2. Tools/templates that reduce TCO. 3. Post-go-live optimisation plan.	1. Escrow and O&M documentation. 2. Tooling lists (monitoring, automation). 3. 12-month optimisation roadmap.

FIRM SUBMISSION REQUIREMENTS

Proposals must include the components listed below. Submissions that omit any mandatory items will be considered non-responsive. Provide concise, evidence-based content aligned to the ToR objectives and Ghana's ASM context.

Section	Requirements	Page Limits & Format	Mandatory (Y/N)
A. Technical Proposal	1. Executive Summary and Understanding of the Assignment. 2. Methodology and Work Plan mapped to TP0–TP7 touchpoints (miner → Tier 1 → Tier 2/kitchens → aggregator → GoldBod). 3. System Architecture: monolith or modular architecture with APIs (mobile + third-party), in-country hosting and DR targets. 4. Detailed Technology Stack (DB, ledger/audit, security). 5. Compliance Matrix (AML/CFT, OECD, LBMA, Fairtrade and ISO 27001) feature-by-feature response. 6. Governance, Risk & Security Framework (privacy/Act 843, auditability). 7. Team Composition & CVs of Key Experts (roles, FTEs, time allocations)**. 8. Implementation Schedule & Milestones (18-month phased implementation) 9. Training, Change & Adoption Plan for ASM/Tier 1/Tier 2/aggregators/GoldBod team. 10. Interoperability Plan (Minerals Commission, GRA/Customs, Bank of Ghana, FIC).	1. Max 30 pages for items 1–6 (A4, 11pt min). 2. CVs of key experts 3. Schedules/diagrams may be annexed. 4. Submit PDF plus editable source (DOCX, XLSX, PPTX) for matrices and plans.	Y

B. Financial Proposal	1. Detailed cost breakdown by deliverable and phase (hardware, software, services, training, tagging supplies). 2. Milestone-linked payment schedule consistent with Section 4. 3. Assumptions & Exclusions (taxes/duties, FX, warranties). 4. Operating & Maintenance (Months 7–12) with SLAs. 5. Price validity ≥ 180 days.	1. Max 12 pages (tables allowed). 2. Provide XLSX for cost sheets + PDF.	Y
C. Administrative & Compliance Attachments	1. Certificate of Incorporation/Registration; valid Tax Clearance (or equivalent). 2. Signed Conflict of Interest and Anti-Corruption declarations. 3. Signed acknowledgment of data residency in Ghana and in-country hosting. 4. Acceptance of source-code escrow and knowledge-transfer obligations. 5. At least two client references for comparable work (traceability/AML/mining). 6. Proof of National Partnership for foreign firms/consortia (National firms should play the leading role) 7. Consulting Firms/Consortia must hold current ISO 27001 certification (with certificate number and issuing auditor name); ISO 27017 (cloud) and ISO 27701 (privacy) are strongly preferred and may be obtained within 12 months of contract signature where not already held 8. Must be certified by the Data protection commission of Ghana. 9. Must indicate how they intend to satisfy OWASP Web Top 10 (2021), OWASP API Security Top 10 (2023), OWASP MASVS L2+R for mobile, and OWASP DSOMM for the development pipeline; annual re-	1. No page limit (attachments). 2. Submit as a single Annex (PDF).	Y

	assessment required.		
D. Annexes (Technical Evidence)	1. Compliance Matrix (filled) mapped to Section 5 table (features/specs/rationale). 2. Architecture diagrams (monolith/modular + APIs), data model excerpts. 3. Sample screens/forms (offline flows, tagging, assay, case management). 4. Prototype links or screenshots (if available).	1. No page limit for diagrams/screens. 2. Provide editable files where applicable.	Y

***** For the programming capabilities, the experience of the team should be a minimum of five (5) years of relevant experience in the industry.***

8.1 Minimum Key Personnel Requirements

The Bidder shall provide a qualified, multi-disciplinary implementation team with demonstrated experience in enterprise blockchain systems, national-scale digital transformation, AML/CFT compliance systems, ASM operations, cybersecurity, mobile/offline systems deployment, and responsible mineral supply chains.

Bidders shall submit:

- detailed curriculum vitae (CVs) for all proposed Mandatory Key Personnel;
- academic and professional certifications;
- project references demonstrating relevant experience;
- proposed role allocation and reporting structure;

- **written availability commitments;**
- implementation organogram;
- Ghana-based staffing and support plan.

GoldBod reserves the right to interview proposed Mandatory Key Personnel during technical evaluation to validate competence, experience, and project availability.

Replacement of Mandatory Key Personnel during implementation shall require prior written approval from GoldBod. Proposed replacements shall possess equal or superior qualifications and experience.

The Supplier shall maintain a Ghana-based implementation and support presence throughout the project lifecycle, including deployment, training, technical support, integration coordination, and stakeholder engagement personnel located within Ghana.

MANDATORY KEY PERSONNEL

1. Project Director / Team Leader

Qualifications & Experience

- vii Master's degree in computer science, Engineering, Information Technology, Project Management, Business Administration, or related discipline.
- viii Minimum fifteen (15) years of senior programme delivery experience in large-scale public-sector or enterprise technology programmes.
- ix Demonstrated leadership of at least three (3) comparable national-scale digital transformation, traceability, enterprise systems, or projects of similar scale and complexity.
- x PMP, PRINCE2 Practitioner, or equivalent senior project/programme management certification (mandatory).
- xi Demonstrated experience leading multi-disciplinary teams of at least twenty (20) personnel across multiple operational locations.
- xii Africa or Ghana implementation experience strongly preferred.

2. Project Manager

Qualifications & Experience

- vii. Bachelor's degree (Master's preferred) in Information Technology, Engineering, Computer Science, or Project Management.
- viii. Minimum ten (10) years of project management experience, including at least five (5) years managing complex enterprise ICT implementations.
- ix. PMP, PRINCE2 Practitioner, PMI-ACP, or equivalent certification (mandatory).
- x. Demonstrated experience managing phased or gated implementation methodologies and milestone-linked delivery frameworks.
- xi. Experience coordinating multi-vendor and multi-stakeholder programmes preferred.
- xii. Africa or Ghana delivery experience strongly preferred.

3. Enterprise Blockchain Architect

Qualifications & Experience

- viii. Master's degree in Computer Science, Software Engineering, Distributed Systems, or related field.
- ix. Minimum eight (8) years of enterprise systems architecture experience with at least five (5) years in permissioned blockchain implementation.
- x. Demonstrated experience designing and deploying at least two (2) enterprise-grade blockchain or distributed ledger platforms.
- xi. Proven expertise with one or more enterprise blockchain platforms including Hyperledger Fabric, Hyperledger Besu, R3 Corda, Quorum, or equivalent enterprise-grade permissioned DLT platforms.
- xii. Demonstrated experience with:
 - smart contracts / chaincode;
 - consensus architecture;
 - node governance;
 - blockchain security;
 - cryptographic integrity controls;
 - on-chain/off-chain architecture.

- xiii. Experience integrating blockchain platforms with enterprise APIs, mobile systems, and regulatory systems.
- xiv. Relevant blockchain certifications strongly preferred.

4. Senior Software Developer & Enterprise Architect

Qualifications & Experience

- vi. Master's degree in Computer Science, Software Engineering, Information Systems, or related field.
- vii. Minimum ten (10) years of experience in software development, enterprise architecture, systems design, and solution delivery.
- viii. Demonstrated experience delivering at least three (3) large-scale enterprise platforms involving multiple stakeholders, system integrations, or high transaction volumes.
- ix. Proven experience in secure solution design aligned with ISO/IEC 27001 and OWASP Application Security Verification Standard (ASVS).
- x. Strong experience in:
 - o API integrations;
 - o microservices;
 - o event-driven architecture;
 - o cloud-native and distributed systems;
 - o enterprise integration patterns.

5. Cybersecurity / Information Security Lead

Qualifications & Experience

- vii. Master's degree in Cybersecurity, Information Security, Computer Science, or related field.
- viii. Minimum ten (10) years of information security experience with at least five (5) years leading cybersecurity for regulated or critical systems.

- ix. CISSP, CISM, or CISA certification (mandatory).
- x. ISO/IEC 27001 Lead Implementer or Lead Auditor certification (mandatory).
- xi. Demonstrated experience with:
 - enterprise security architecture;
 - identity and access management;
 - blockchain security;
 - mobile application security;
 - incident response;
 - penetration testing oversight.
- xii. Experience implementing cybersecurity controls aligned with NIST and OWASP frameworks preferred.

6. Technology, Network & Infrastructure Lead

Qualifications & Experience

- vii. Bachelor's degree (Master's preferred) in Computer Engineering, Telecommunications, Network Engineering, Information Systems, or related field.
- viii. Minimum ten (10) years of enterprise infrastructure, network, and data-centre operations experience, including at least five (5) years in leadership roles.
- ix. CCNP, CCNA, or equivalent enterprise network certification (mandatory).
- x. ITIL v4 Foundation or higher certification preferred.
- xi. Familiarity with Ghana's National Communications Authority regulatory environment and national telecommunications infrastructure.
- xii. Prior experience implementing national-scale infrastructure, cloud, or government data-centre environments preferred.

7. Integration / API Engineer

Qualifications & Experience

- vi. Bachelor's degree in Computer science, Engineering, Information Systems, or related field.
- vii. Minimum six (6) years of API and enterprise integration engineering experience.
- viii. Demonstrated experience implementing integrations with government, financial-sector, or regulated systems.
- ix. Proficiency in:
 - OpenAPI 3.x;
 - OAuth 2.0 / OpenID Connect (OIDC);
 - API gateways;
 - event-driven integration patterns;
 - secure integration frameworks.
- x. Experience integrating with Ghanaian or West African government systems preferred.

8. AML/CFT Compliance & Financial Crime Specialist

Qualifications & Experience

- vii. Bachelor's degree (Master's preferred) in Finance, Law, Compliance, Risk Management, Economics, or related field.
- viii. Minimum ten (10) years of AML/CFT compliance or financial crime risk-management experience.
- ix. Certified Anti-Money Laundering Specialist (CAMS) or equivalent certification (mandatory).
- x. Demonstrated experience with:
 - FATF compliance frameworks;
 - STR/SAR reporting;
 - transaction monitoring systems;
 - sanctions and PEP screening;
 - risk scoring methodologies;

- suspicious transaction analytics.
- xi. Prior experience in banking, fintech, extractives, or regulated sectors preferred.
- xii. Familiarity with Ghana's Anti-Money Laundering Act, FIC reporting requirements, and OECD/LBMA responsible sourcing frameworks preferred.

9. ASM / Gold Sector Subject Matter Expert

Qualifications & Experience

- vi. Master's degree in Mining Engineering, Mineral Economics, Development Studies, or related field.
- vii. Minimum ten (10) years of experience in the artisanal and small-scale gold mining sector.
- viii. Demonstrated field engagement with Ghanaian ASM communities (mandatory).
- ix. Working knowledge of:
 - Minerals and Mining Act, 2006 (Act 703);
 - Ghana Gold Board Act, 2025 (Act 1140);
 - Minamata Convention;
 - LBMA Responsible Gold Guidance;
 - OECD Due Diligence Guidance.
- x. Prior advisory experience with government, Minerals Commission, GoldBod, donor programmes, or development partners preferred.

10. Mining Operations Expert

Qualifications & Experience

- vii. Bachelor's degree (Master's preferred) in Mining Engineering, Geology, Mineral Processing, Metallurgy, or related field.
- viii. Professional registration with the Ghana Institution of Engineering, Mining Engineers Association of Ghana, or equivalent professional body.

- ix. Minimum ten (10) years of operational mining experience including at least five (5) years in ASM or small-scale gold mining environments.
- x. Demonstrated field-level operational engagement in Ghanaian or West African mining environments.
- xi. Experience with:
 - assay systems;
 - reconciliation;
 - mineral processing;
 - operational controls;
 - mass-balance methodologies.
- xii. Familiarity with Ghana mining regulations and environmental compliance frameworks required.

11. Mobile Applications & Offline Systems Lead

Qualifications & Experience

- vi. Bachelor's degree (Master's preferred) in Computer Science, Mobile Computing, Software Engineering, or related field.
- vii. Minimum eight (8) years of enterprise mobile application development experience.
- viii. Demonstrated experience developing offline-capable mobile applications operating in low-connectivity environments.
- ix. Proven expertise in:
 - Android/iOS enterprise applications;
 - encrypted offline storage;
 - secure synchronization;
 - GPS/location services;
 - device authentication and management;
 - field-deployment optimization.

- x. Experience deploying mobile systems in rural or field-based operational environments preferred.

12. Senior Change Management & Adoption Lead

Qualifications & Experience

1. Master's degree in Organisational Development, Behavioural Science, Communications, Public Administration, or related field.
2. Minimum ten (10) years of change-management experience with at least five (5) years on national or large-scale transformation programmes.
3. PROSCI ADKAR, CCMP, APMG Change Management Practitioner, or equivalent certification preferred.
4. Demonstrated experience implementing behavioural-change and stakeholder-adoption programmes in emerging markets.
5. Experience in informal-sector formalisation programmes strongly preferred.
6. Ghana or West Africa implementation experience preferred.

13. Legal & Regulatory Compliance Advisor – Ghana

Qualifications & Experience

- v. LL.B and LL.M (or equivalent) with valid practising certification from the General Legal Council of Ghana.
- vi. Minimum ten (10) years of legal and regulatory advisory experience in mining, financial services, data protection, AML/CFT, or public-sector governance.
- vii. Demonstrated familiarity with:
 - Ghana Gold Board Act;
 - Data Protection Act, 2012 (Act 843);
 - Anti-Money Laundering Act, 2020 (Act 1044);
 - Public Procurement Act;
 - Minerals and Mining Act.

- viii. Experience advising on technology governance, data sharing, digital compliance, or regulated enterprise systems preferred.

14. Logistics & Supply Chain Expert

Qualifications & experience

- vi. Bachelor's degree (Master's preferred) in Supply Chain Management, Logistics, Operations Management, or related field.
- vii. CILT (Chartered Institute of Logistics and Transport) Diploma, APICS CSCP, or CIPS qualification.
- viii. Minimum 10 years in logistics and supply chain, with 5+ years on national-scale field-deployed equipment programmes.
- ix. Demonstrated experience with security-sensitive supply chains (tamper-evident materials, biometric devices, secure-tag distribution).
- x. Prior West Africa logistics delivery experience preferred.

PREFERRED / SUPPORTING PERSONNEL

The Bidder is additionally encouraged to provide the following supporting personnel where relevant to the proposed implementation methodology:

- Data Protection & Privacy Specialist
- GIS / Geospatial Specialist
- Data Scientist / Fraud Analytics Specialist
- Logistics & Supply Chain Expert
- Training & Capacity Building Specialists
- DevSecOps / Cloud Operations Engineer
- QA / Testing Lead
- Business Process Analyst

LEGAL, ETHICAL, AND REGULATORY COMPLIANCE

The Traceability & Assurance System (TAS) and all related services must comply with applicable Ghanaian laws and regulations, recognised international standards, and ethical norms for responsible sourcing. The firm shall implement controls that ensure legal conformity, data sovereignty (in-country processing and storage), human rights protections, environmental safeguards, and verifiable audit trails from mine to market.

Domain	Key instruments / standards
A. Mining & Licensing Compliance	1. Minerals and Mining Act, 2006 (Act 703) and amendments. 2. Minerals Commission directives and licence conditions.
B. Anti-Money Laundering / Countering the Financing of Terrorism (AML/CFT)	1. Anti-Money Laundering Act, 2020 (Act 1044). 2. Anti-Terrorism Act, 2008 (Act 762) and amendments. 3. Financial Intelligence Centre (FIC) directives. 4. Financial Action Task Force (FATF) Recommendations.
C. Data Protection & Privacy	1. Data Protection Act, 2012 (Act 843). 2. Electronic Transactions Act, 2008 (Act 772).
D. Data Sovereignty & Hosting	1. Bank of Ghana and government data-residency policies (as applicable). 2. GoldBod data-sovereignty policy.
E. Environmental & Mercury Controls	1. Environmental Protection Agency Act, 1994 (Act 490). 2. Minamata Convention on Mercury (Ghana implementation).
F. Trade, Customs & Foreign Exchange	1. Ghana Revenue Authority (Customs Division) rules. 2. Foreign Exchange Act, 2006 (Act 723). 3. Bank of Ghana export/FX circulars.
G. Guidelines/standards/Policies for Responsible Sourcing & Traceability	1. Bank of Ghana/GoldBod Responsible Sourcing and Traceability policy 2. London Bullion Market Association Responsible Gold Guidance (LBMA RGG v9 or later) and Chain of Custody (CoC). The firm shall produce a clause-level traceability matrix mapping each RGG process step (Establish CoC, Identify Risks, Risk Management, Third-party Audit, Reporting, Recordkeeping, Grievance) to specific TAS modules, controls, and reports. 3. Organisation for Economic Co-operation and Development Due Diligence Guidance (OECD DDG).

	4. Responsible Minerals Initiative / Responsible Minerals Assurance Process (RMI/RMAP).
H. Anti-Corruption & Conflicts	1. Public Procurement Act and anti-corruption codes (as applicable). 2. United Nations Convention against Corruption (UNCAC) principles. 3. Firms shall disclose any existing, prior, or potential conflicts of interest involving gold trading entities, financial institutions, assay laboratories, logistics providers, or regulated actors participating within the TAS ecosystem. GoldBod reserves the right to assess, mitigate, or reject any bid where such conflicts may compromise independence, impartiality, regulatory integrity, or system trustworthiness.
I. Human Rights & Labour	1. National labour laws and child/forced labour prohibitions. 2. International Labour Organisation (ILO) core conventions.
J. Information Security	1. Transport Layer Security (TLS) 1.3: Advanced Encryption Standard (AES-256) at rest. 2. Security best practices aligned to Open Web Application Security Project (OWASP) guidance.
K. Records, Retention & Disclosure	1. Record-keeping provisions under Anti-Money Laundering Act and sector laws. 2. Data retention minima per GoldBod policy ($\geq$ seven years unless otherwise mandated).
L. Community & Ethics Oversight	1. GoldBod community engagement and grievance standards. 2. National guidelines for stakeholder engagement (where applicable).

Licensing and concession data remain authoritative with the Minerals Commission. TAS shall flag discrepancies but shall not silently override regulatory records. Automated transactions blocking on licence expiry is not required; such events shall be flagged, restricted, and escalated for GoldBod decision-making. Any overrides or grace periods require authorised approval, justification, and full audit logging.

9.1 Government ICT Compliance and NITA Technical Clearance

The TAS solution shall comply with applicable Government of Ghana ICT policies, standards, interoperability frameworks, cybersecurity directives, and technical-clearance requirements issued by the National Information Technology Agency (NITA).

The firm shall ensure alignment with:

- Ghana Government Enterprise Architecture (GGEA)
- e-Government Interoperability Framework (eGIF)
- Government API Standards
- applicable national cybersecurity and data-governance requirements
- NITA technical-clearance and conformity requirements

The firm shall cooperate with GoldBod and NITA throughout technical review, architecture validation, interoperability assessment, security assurance, implementation verification, and certification processes.

Where required, the firm shall provide technical documentation, architecture diagrams, API specifications, security assessment reports, integration artefacts, and implementation evidence necessary to support NITA review and certification activities.

9.2 Intellectual Property, Escrow & Exit Management

All system configurations, data models, documentation, and source code developed under this contract shall vest in GoldBod. A source-code escrow arrangement shall be maintained throughout the contract. The firm shall provide reasonable transition support in the event of contract termination or system migration.

All AML/CFT rules, risk models, scoring logic, training datasets generated under the project, anomaly-detection configurations, and model outputs developed, configured, or generated under TAS shall remain the intellectual property of GoldBod. The firm shall provide full documentation, export capability, and administrative access necessary for GoldBod to maintain, modify, or transfer such models and rules without vendor dependency.

REPORTING

Reporting should provide timely, accurate, and regulator-ready visibility over TAS operations and compliance. Output must be generated in-country, from a single source of truth, and aligned to operational Service Level Agreements (SLAs), and Anti-Money Laundering / Countering the Financing of Terrorism (AML/CFT) obligations.

Report name	Purpose	Reporting Period	Core contents
-------------	---------	------------------	---------------

Monthly Progress Report (activities, risks, mitigation)	Provide a concise status for decision-making and unblock risks.	Monthly (Months 1–18)	1. Activities completed/next. 2. Risks/issues & mitigations. 3. Key Performance Indicators (KPI): lineage %, assay variance, adoption, uptime/latency vs SLA. 4. Decisions/escalations required.
Gate Readiness Pack (Integration Gate 1 & Gate 2)	Prove gate exit criteria are met before proceeding.	1. Gate-1 window - End of Month 6 2. Gate-2 window - End of Month 9	1. Government integrations uptime & data-quality (Minerals Commission / Customs / Bank of Ghana / Financial Intelligence Centre). 2. Security/DR tests (Recovery Point/Time Objectives). 3. Known defects & work-offs. 4. Sign-off checklist.
Pilot Evaluation Report	Demonstrate that the pilot meets functional, data, and adoption targets.	Submit end of Month 8	1. Lineage integrity & commingling exceptions. 2. Assay variance vs tolerance; root causes. 3. Adoption & training outcomes; helpdesk patterns. 4. Recommendations for hardening.
Go-Live Readiness & Launch Report	Confirm production readiness and record launch outcomes.	1. Pre-launch readiness pack - End of Month 9. 2. Launch post-cutover results - Mid-Month 10.	1. Readiness checklist (performance, security, integrations). 2. Cutover plan/results; incident log. 3. Initial production KPIs and stabilisation actions.
Quarterly Financial & Technical Report	Provide consolidated	Quarterly through Month 12	1. Budget vs actuals; forecast & variances.

	programmes and spend oversight.		2. Technical KPIs and availability. 3. Compliance: Suspicious Transaction Report / Suspicious Activity Report (STR/SAR) register; Currency Transaction Report (CTR) summary; data-residency attestation. 4. Procurement/tagging stock status.
Final Project Completion and commission Report	Close the project and hand over for steady-state operations.	End of Month 18.	1. Objectives vs results; KPI attainment. 2. Benefits realisation; lessons learned. 3. Handover: SOPs, runbooks, training, source-code escrow. 4. Sustainability & optimisation roadmap.

CONTRACT VALUE AND TYPE

The selected firm will be awarded a fixed-price contract covering all deliverables, equipment, licenses, and services. Payments shall be made against approved milestones upon verification and acceptance by GoldBod.

Performance-Based Penalties and Incentives

The contract shall include performance-linked provisions, including:

- Penalties for delays in milestone delivery.
- Financial deductions for failure to meet agreed SLAs (uptime, performance, security).
- Withholding of milestone payments pending acceptance criteria fulfillment.
- Incentives for early delivery or exceeding performance benchmarks.

These provisions ensure accountability and alignment with project objectives.

11.1 Payment Schedules

Phase	Deliverable	Payment Schedule	Remarks
1	Mobilisation & Inception	5% of the contract price shall be paid after submission and acceptance of the inception report	5 hard-bound copies and a soft copy on a drive
2.	Design & Alpha	5% of the contract price shall be paid after submission and acceptance of certified solution designs as well as an agreed user acceptance test content	
3.	Build & Integrations (Gate 1)	10% of the contract price shall be paid after submission and acceptance of Gate-1 deliverables (core modules, MinCom integration live, offline-to-online sync proven, Gate-1 acceptance signed).	
4.	Pilot Implementation	10% of the contract price shall be paid after the submission and acceptance a fully functional system with a User Acceptance Test signed off	

5.	Hardening & Integrations (Gate 2)	20% of the contract price shall be paid after the submission and acceptance a fully functional system with a User Acceptance Test signed off	
6.	National roll-out	10% of the contract price shall be paid after the submission and acceptance a fully functional system with a User Acceptance Test signed off	
7.	Training, Support and Handover	40% of the contract price shall be paid after assignment completion report*	Training Schedule and Content and final project handover report

**GoldBod may retain up to 20% of the total contract value as performance retention, releaseable upon successful commissioning of the stabilisation period.*

SERVICES AND FACILITIES TO BE PROVIDED BY THE CLIENT

The Client will assist the firm to facilitate interactions with other stakeholders, especially the GoldBod staff (both National and field staff), small-scale miners, cooperative mining schemes, licensed gold buyers and all those that may be relevant to the execution of the assignment to ensure the smooth execution of the contract. The client will provide the following facilities to the consultant:

- a. Letter(s) of introduction
- b. Relevant reports and documents related to the assignment.
- c. Preliminary data (if available).

PROJECT COMPLETION & HANDING OVER

The Firm shall:

- i. On substantial completion of the works, perform checkouts and verify the completeness and compliance of facilities with the contract requirements. Prepare

- a deficiency list of outstanding items for attachment to the provisional take-over certificates and expedite the clearing of the deficiency list after the Client's acceptance of the facilities for use.
- ii. witness all the final handing over to the Client and prepare a report on any system defects identified in the process.
 - iii. review and approve the Contractor's procedures for correction of defects; and
 - iv. Witness the acceptance (after corrections) and report on all acceptability with respect to contract requirements.

Project Completion

The Consultant shall ensure the provision of:

- i. review and file permanent records within two weeks of project completion; and
- ii. Certificates – Prepare certificates of substantial completion, Take-over, and any other Final certificates.

Source-code escrow shall be updated at least quarterly. Exit support and knowledge transfer are mandatory and subject to formal sign-off.

GoldBod may require a parallel run with a successor firm where continuity risk exists, of up to six (6) months at no additional licence cost, with reasonable time-and-materials support hours capped at 20% of the annual O&M fee.

SUBMISSION AND CONTACT INFORMATION

- Interested firms may obtain further information or clarifications by contacting:
Procurement Directorate [Email-frank.ademan@goldbod.gov.gh]
- Proposals must be submitted electronically to the above address no later than [INSERT DEADLINE DATE AND TIME, GHANA LOCAL TIME — TO BE COMPLETED BEFORE ISSUE; PROPOSALS SHALL ALSO BE SUBMITTED VIA THE GHANEPS PORTAL WHERE APPLICABLE], clearly marked:

“Tender –Traceability and Assurance System (TAS) for the Ghana Gold Board.”

Late submissions will not be considered.

ANNEXES

ANNEX 1: ASM → GOLDBOD PROCESS MAP

ANNEX 2: TAS TOUCHPOINTS (END -TO-END FLOW)

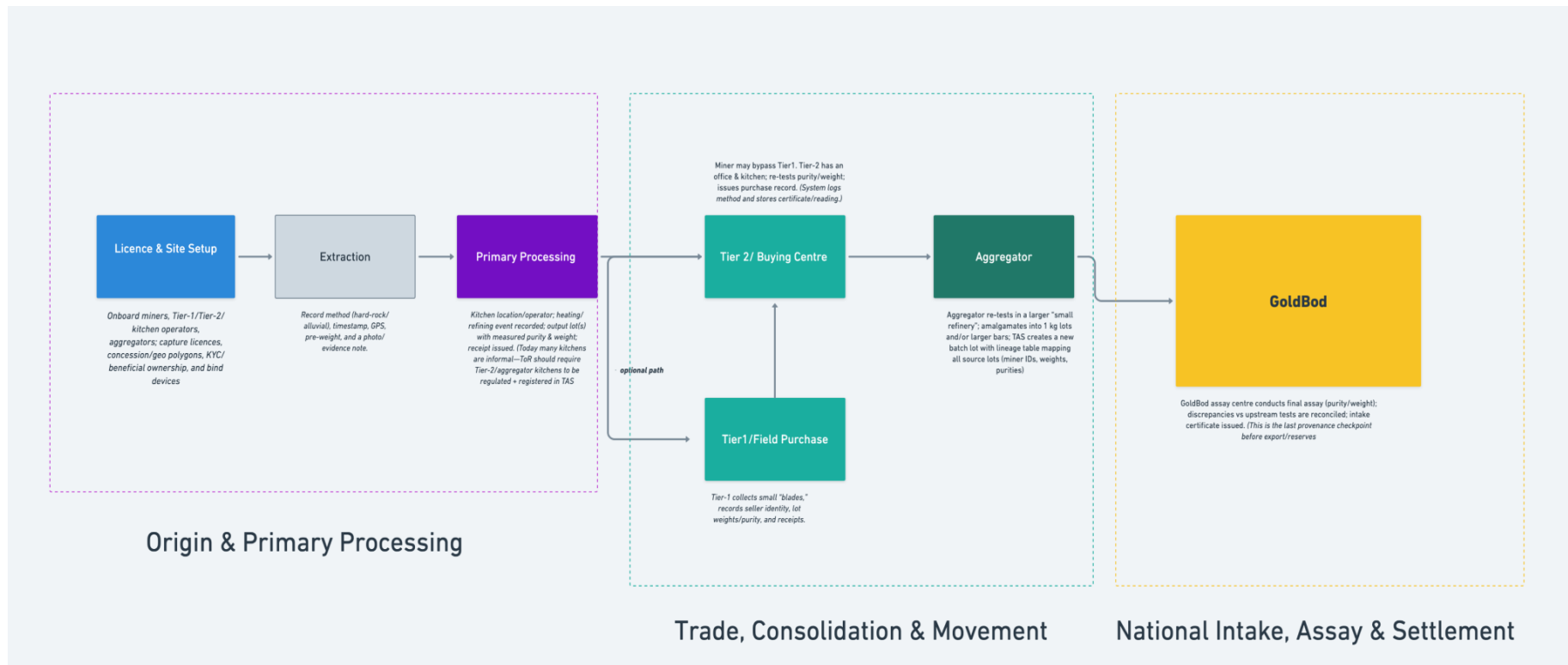
ANNEX 3: RISK ALLOCATION AND MANAGEMENT MATRIX

ANNEX 4: DATA OWNERSHIP, CONTROL, AND ACCESS MODEL

ANNEX 5: EXIT, TRANSITION, AND CONTINUITY PLAN

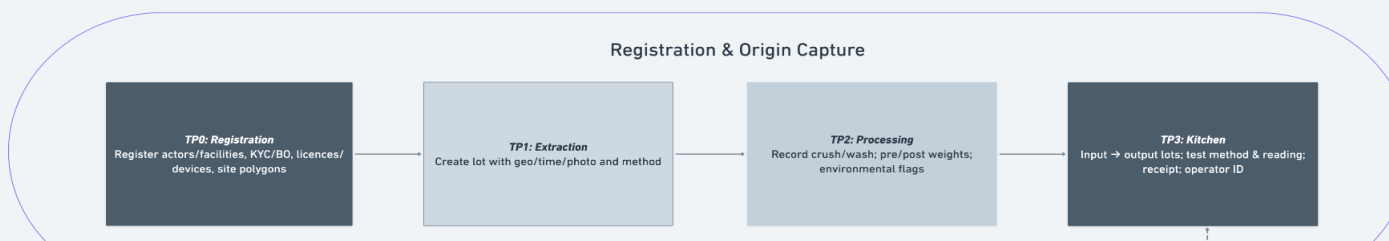
ANNEX 6: CHANGE CONTROL PROCEDURE

ANNEX 1 : ASM → GOLDBOD PROCESS MAP



ANNEX 2 : TAS TOUCHPOINTS (END -TO-END FLOW)

Registration & Origin Capture



ANNEX 3: RISK ALLOCATION AND MANAGEMENT MATRIX

The Risk Allocation and Management Matrix shall comprehensively cover operational, behavioural, financial, and regulatory risks across all actors and transaction points within the ASM gold value chain. The firm shall implement, configure, and maintain controls aligned to the matrix below.

Risk Category	Sub-Risk Description	Actor Level	Control Type	Control Mechanism	Control Owner	System/Manual	Escalation Path	KRI
Data Integrity	False weight entry at source	Miner / Tier 1	Preventive	Photo + device-linked scale + dual confirmation	System + Buyer	Hybrid	Supervisor	% variance anomalies
	False purity declaration	Tier 1 / Tier 2	Detective	Assay comparison across stages	System	System	AML Unit	Purity variance rate
	Missing transaction data	All actors	Preventive	Mandatory fields validation			System block	% incomplete records
Identity Fraud	Proxy transactions	All actors	Preventive	Device-user binding + authentication			Supervisor	% shared-device anomalies
	Fake or duplicate identities	All actors	Preventive	KYC + Ghana Card verification			Compliance Unit	Duplicate ID rate

Traceability & Assurance System (TAS)-ToR

Collusion	Closed-loop trading networks	Tier 1 / Tier 2	Detective	Network pattern analysis			AML Unit	Repeated pair frequency
	Buyer–seller price manipulation	Tier 1 / Aggregator	Detective	Price deviation analytics			Supervisor	Price variance index
AML/CFT	Structuring transactions	All actors	Detective	Threshold-based rules				
	Third-party payments	Aggregator		Beneficiary mismatch checks	% mismatched payments			
	Suspicious volume spikes			Volume anomaly detection	Volume deviation			
Operational	Non-response to alerts	All actors	Corrective	SLA-based escalation	GoldBod	Manual	Regional Office	Response time
	Delayed data synchronisation		Preventive	Offline sync controls	System	System	Tech team	Sync delay rate
	Device tampering		Preventive	Root/jailbreak detection			IT Security	Device risk score

Behavioural	Intentional misreporting		Detective	Cross-validation + audit	System + Inspector	Hybrid	Supervisor	Audit discrepancy rate
	Shared device misuse		Preventive	Individual login tracking	System	System	Supervisor	Multi-user device rate
Logistics	Route diversion	Transporters	Detective	GPS tracking & geofencing			Regional Office	Route deviation rate
	Custody gaps	All actors	Detective	Chain-of-custody validation			Supervisor	Missing custody links
Processing	Unregistered kitchen operations	Tier 2	Preventive	Kitchen registration enforcement			Compliance Unit	% unregistered kitchens
	Incorrect batch amalgamation	Aggregator	Detective	Lineage validation rules			AML Unit	Lineage integrity %
Assay	Excessive variance in assay	All actors	Detective	Tolerance thresholds			Supervisor	Variance exceedance
Financial	Underpricing to avoid reporting	Tier 1	Detective	Price benchmarking			AML Unit	Price anomaly rate

Traceability & Assurance System (TAS)-ToR

	Payment outside system	All actors	Preventive	Mandatory payment recording			Compliance Unit	Off-system payments
Compliance	Expired licence usage	All actors	Detective	Licence validation API			Compliance Unit	Expired licence usage
	Non-compliant actor participation	All actors	Preventive	Access restriction rules			System block	% blocked transactions
Export	Untraceable gold entering export chain	Aggregator	Preventive	Full lineage requirement			Export Unit	% unverified exports
	Smuggling/diversion risk	Transporters	Detective	Geo-risk alerts			AML Unit	Border proximity alerts
Governance	Unauthorised overrides	Supervisors	Detective	Audit logs + dual approval			HQ Oversight	Override frequency
	Weak enforcement follow-through	GoldBod	Corrective	SLA monitoring dashboards		Hybrid	HQ	SLA compliance rate
Cybersecurity	Data breach	System-wide	Preventive	Encryption + SIEM		System	IT Security	Incident count
	Unauthorised access	System-wide	Preventive	RBAC + MFA				Access violations

Risk Category	Sub-Risk Description	Actor Level	Control Type	Control Mechanism	Control Owner	System/Manual	Escalation Path	KRI
Data Integrity	False weight entry at source	Miner / Tier 1	Preventive	Photo + device-linked scale + dual confirmation	System + Buyer	Hybrid	Supervisor review	% variance anomalies
Collusion	Repeated closed-loop trading	Tier 1 / Tier 2	Detective	Pattern analysis and network alerts	System	System	AML officer	Repeated pair frequency
Compliance	Non-response to alerts	Aggregator	Corrective	SLA-based alert escalation	GoldBod	Manual	Regional office	Alert response time
Identity Fraud	Proxy transactions	All actors	Preventive	Device-user binding + authentication	System	System	Supervisor	% shared-device anomalies

Traceability & Assurance System-ToR

ANNEX 4: DATA OWNERSHIP, CONTROL AND ACCESS MODEL

- All data generated under the TAS shall be the exclusive property of the GoldBod
- GoldBod is the Data Controller under the Data Protection Act, 2012 (Act 843) and the firm acts solely as a Data Processor.

Principles:

1. Role-based access control
2. Full audit logging
3. No data export without GoldBod authorisation
4. Minimum data retention of seven (7) years

ANNEX 5: EXIT, TRANSITION AND CONTINUITY PLAN

This Annex applies upon contract expiry, termination, or system transition.

The firm shall:

- Provide full source code through escrow
- Transfer all documentation and configurations
- Deliver data exports in open formats
- Provide up to six (6) months transition support

Exit shall be certified complete only upon written confirmation by GoldBod.

ANNEX 6: CHANGE CONTROL PROCEDURE

All changes to scope, cost, or schedule shall follow a formal Change Request process.

1. Submission of Change Request
2. Impact assessment by firm
3. Review by GoldBod Change Review Committee
4. Written approval or rejection

No change shall take effect without written approval.

Software Licensing (ref. GCC Clause 24), Media and Documentation Requirements

[illegible]

PART 3 - CONTRACT

Section VIII. General Conditions of Contract

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Section VIII. General Conditions of Contract

A. Contract and Interpretation

1. Definitions

1.1 The following words and expressions shall have the meanings hereby assigned to them:

- (a) “Acceptance” means the Purchaser’s written certification that the System (or a subset thereof) is confirmed to have met the functional and performance requirements and passed the inspections and Acceptance tests specified in GCC Clause 12 and the Technical Specifications.
- (b) “Contract” means the agreement entered into between the Purchaser and the Firm, as recorded in the Contract Form signed by the parties, including all Contract Documents, all attachments and appendices thereto, and all documents incorporated by reference therein.
- (c) “Contract Documents” means the documents specified in GCC Clause 2.
- (d) “The Contract Price” means the price payable to the Firm under the Contract for the full and proper performance of its contractual obligations, subject to such additions and adjustments thereto or deductions therefrom, as may be made pursuant to the Contract. Maintenance Costs for Products and Services to be provided during the Maintenance Period and which are the subject of Supplemental Agreements are not included in the Contract Price.
- (e) “Country of Origin” or “Origin” means the place where the products are produced or from where the Related Services are supplied. Products are produced when, through manufacturing, processing, software development or substantial and major assembly or integration of components, a commercially recognized product results that is substantially different in basic characteristics or in purpose or utility from its components. The origin of products and services is distinct from the nationality of the Tenderer and may be different.
- (f) “Coverage Period” means the time periods during which the maintenance, operational and technical support services required by the Contract must be available, which, unless

otherwise specified in SCC, shall be 8:00 AM to 5:00 PM local time at the Project Site, Monday through Friday, exclusive of official national holidays.

- (g) “Day” means calendar day of the Gregorian Calendar.
- (h) “Delivery” means the transfer of the Products from the Firm to the Purchaser in accordance with the applicable Incoterm under the terms and conditions of the Contract.
- (i) “Eligible Countries” means the countries and territories listed as eligible in Section V.
- (j) “GCC” means the General Conditions of Contract.
- (k) “Implementation Period” means the period from the Start Date until all the Products and the entire System have achieved full and complete Acceptance by the Purchaser and all Project Implementation Services have been delivered.
- (l) “Implementation Schedule” is the part of the Firm’s tender (provided in response to the Purchaser’s requirements therefor specified in the Schedule of Supply) specifying the committed schedule for arrival and Installation of Batches of Products and delivery of related Services at the Project Sites, including any agreed amendments thereto. The Implementation Schedule defines the major milestones and timetable for the detailed Project Plan and is used for managing the Firm’s performance under the Contract, including for the assessment of liquidated damages pursuant to GCC Clause 22.
- (m) “Information System,” also called “System,” means the complete set of Products and Related Services that the Firm is required to provide under the Contract.
- (n) “Installation” means the Firm’s written notification that a specific set of Products and associated Services identified as an “Installation Batch” in the Schedule of Supply have been delivered, unpacked, set up, connected, configured and tested, and are ready for Acceptance testing and subsequent production use by the Purchaser in accordance with Contract requirements.
- (o) “Intellectual Property Rights” means any and all copyright, moral rights, trademark, patent, and other intellectual and proprietary rights, title and interests, world-wide, whether

vested, contingent or future, including without limitation all economic rights and all exclusive rights to reproduce, fix, adapt, modify, translate, create derivative works from, extract or re-utilize data from, manufacture, introduce into circulation, publish, distribute, sell, license, sub-license, transfer, rent, lease, transmit or provide access electronically, broadcast, display, enter into computer memory, or otherwise use any portion or copy, in whole or in part, in any form, directly or indirectly, or to authorize or assign others to do so.

- (p) “Maintenance Period” means the remainder of the Useful Life Period following the expiration of the Warranty Period for each Installation Batch of the Products during which the Firm is obligated to provide recurrent supply of Products, Software Releases and License renewals as specified in the Supply Requirements, and also specific Maintenance, Support and other Services for the System in accordance with GCC Clause 8, for costs not exceeding those specified in the Maintenance Costs Schedule.
- (q) “Products” or “Goods,” means all equipment and Software, Licenses, documentation, and any other materials, supplies and tangible items that the Firm is required to supply under the Contract.
- (r) “Project Manager” means the person named in SCC and appointed by the Purchaser to manage the project and represent the interests of the Purchaser.
- (s) “Project Plan” is the part of the Firm’s Technical Proposal providing detailed information about the timetable, duration, sequence, dependencies and required resources for all activities needed to complete the Contract. The Project Plan shall conform to the timelines specified in the Contract Implementation Schedule and shall include a complete description of each activity, resource and project milestone. The Project Plan shall clearly indicate the requirements, timetable and resource loading for all activities and resources that are the responsibility of the Purchaser to perform, provide or otherwise arrange for. The Project Plan shall also meet the requirements specified therefore in the Technical Specifications.
- (t) “Project Site(s)” or simply “Site” means the place(s) specified in the Supply Requirements for Installation of the Products and delivery of the Services.

- (u) “Purchaser” means the entity purchasing the Products and Related Services, as specified in the SCC.
- (v) “Purchaser’s Country” is the country named in the SCC.
- (w) “Related Services,” also called “Services,” means all technical, logistical, managerial, fiduciary, training, maintenance, new software release supply and any other activities and obligations incidental to the supply of the Products that must be carried out by the Firm under the Contract.
- (x) “SCC” means the Special Conditions of Contract.
- (y) “Software” means the set of executable instructions and any related control and reference data and procedures that cause computer systems to perform specific business or technical operations. Contract terminology for classification or identifications of forms of Software is as follows:
 - (i) “Application Software” means Software that performs or facilitates key business, governmental and/or technical functions for end-users of the System or for equipment being monitored or controlled by the System, including Software for general purpose data processing such as word processing, spreadsheet, electronic mail, database management, etc.
 - (ii) “Custom Software” means Software developed by the Firm at the Purchaser’s expense under the Contract, including customizations made to Off-the-shelf Software.
 - (iii) “Off-the-shelf Software” means Software that is provided as a pre-packaged Product for multiple clients.
 - (iv) “Source Code” means the program source files, database structures, dictionaries, definitions, scripts, libraries and procedures for compiling and linking all source elements into the executable code, design and user documentation, including precise identification of all prerequisite Software (compilers, libraries, System Software, linkers, software tools, etc.) and versions, and any other files, documents, keys or symbolic representations necessary for the generation, compilation, execution, verification testing and subsequent maintenance of the Software.

- (v) “Surreptitious Code” means any virus, Trojan horse, worm, or other routine or component designed to permit unauthorized access to the System, or to disable, erase, or otherwise harm any component of the System, or any back door, time bomb, drop dead device, or other routine designed to disable Software automatically with the passage of time or under the positive control of a party other than the Purchaser, or to perform any other such malicious, mischievous or disruptive actions. Surreptitious Code excludes routines designed to allow remote maintenance, upgrades or technical support under the positive control of the Purchaser.
- (vi) “System Software” means Software that provides the operating, management and control instructions for the hardware, or protocol and middle-ware support for communications functions. System Software includes but is not limited to firmware, micro-code, operating system, network protocol, system and network management, file system, backup, clustering, disaster recovery, security, system integration middle-ware, compilers, linkers, virtual machines and system utility Software.
- (aa) “Software License” means the Purchaser’s right to install and use the Software and related documentation under the terms and conditions of a Software License Agreement pursuant to GCC Clause 24, which can be classified according to the scope of authorized usage, as follows:
 - (i) “Computer-Based Software License” means a Software License that is granted for use on a specific computer or set of computers and which may also include some User-Based Licensing restrictions.
 - (ii) “Perpetual Use License” means a Software License that is fully paid for, irrevocable, and granted in perpetuity for all or specified versions of the Software.
 - (iii) “Renewable License” means an irrevocable and fully paid for Software License for all or specified version(s) of the Software that is valid only for a specified amount of time and which will be renewed at specified intervals requiring the payment of additional License fees as shall be indicated in the Maintenance Costs Form in the Firm’s tender.

- (iv) “Site or Enterprise License” means a Software License authorizing simultaneous use of the Software by all or a specified set of the Purchaser’s users on all or a specified set of computers at all or a specified set of the Purchaser’s Project Sites.
- (v) “User-Based Software License” means a Software License which is granted for a specified maximum number of registered (authorized) or concurrent (simultaneously active) users of the Software.
- (bb) “Software Producer” means the legal entity that owns the Intellectual Property Rights in the Software and the Source Code therefor.
- (cc) “Software Release” means a licensed copy of a particular version of the Software, which may be a complete new version, or only an incremental update to a prior Release. Contract terminology for Software Releases is as follows:
 - (i) “Maintenance Release” means an incremental Release of the Software which primarily fixes problems in the current version and which may also include some functional enhancements.
 - (ii) “Media” (also called the “Distribution Media”) means the physical media or the files comprising an electronic distribution that tangibly embodies a Software Release, often including associated on-line documentation.
 - (iii) “Version Release” means a complete Release of the Software that incorporates significant additional functionality over previous Releases, which may not be fully compatible with previous versions and should therefore generally be evaluated for potential system, business and user training impact and then be thoroughly tested by the Purchaser prior to deployment. Version Releases generally change the primary Software version identifier, such as the integer part of a version number, or the entire Product name.
- (dd) “Start Date” means the date following contract signing that the Implementation Schedule commences, as defined in **GCC Clause 3**.
- (ee) “Subcontractor” means any natural person, private or government entity, or a combination of the above, including

its legal successors or permitted assigns, to whom any part of the Products to be supplied or execution of any part of the Related Services is directly or indirectly subcontracted by the Firm.

(ff)“Firm” means the natural person, private or government entity, or a combination of the above, whose tender to perform the Contract has been accepted by the Purchaser and is named as such in the Contract Form, and includes the legal successors or permitted assigns of the Firm.

(gg)“Useful Life Period” means the time that the Products are required to be in production use, which is five (5) years from the date of Acceptance of the Products, unless otherwise specified in SCC. The Useful Life Period equals the Warranty Period plus the Maintenance Period. If the Implementation Schedule specifies multiple Installation Batches with Installation and Acceptance of the Products staggered over time, then the Warranty, Maintenance and Useful Life Periods shall be calculated separately for each Installation Batch.

(hh) “Warranty Period” means the period following Acceptance of the Products during which the Firm’s warranty obligations in respect of the Products are in force in accordance with GCC Clause 26. The duration of the Warranty Period shall be:

- 1 year for Software;
- 1 year for servers and communications equipment; and
- 1 year for workstations, printers and all other Products.

unless otherwise specified in SCC.

(ii) “In writing” means communicated in written form with proof of receipt.

2. Contract Documents

- 2.1 The documents listed below shall constitute the Contract between the Purchaser and the Firm. In the event of ambiguity or conflict between the documents, the order of precedence shall be the order in which they are listed below. Subject to this order of precedence, all documents forming the Contract (and all parts

thereof) are intended to be correlative, complementary, and mutually explanatory.

- (a) Contract Form furnished in Section X, Contract Forms
- (b) Section IX, Special Conditions of Contract
- (c) Section VIII, General Conditions of Contract
- (d) Section VI, Schedule of Supply
- (e) Section VII, Technical Specifications
- (f) The Firm's tender, including the Price Schedules and the project Implementation Schedule
- (g) Any other documents identified in the Contract Form.

3. Start Date

3.1 The beginning of the Implementation Period of the Contract and the effective commencement date for the Implementation Schedule and the Project Plan shall be the date, hereinafter called the "Start Date," when all of the following conditions have been fulfilled, generally in the order specified herein:

- (a) The Contract Form has been duly executed for and on behalf of the Purchaser and the Firm;
- (b) All necessary Supplemental Agreements for the required Maintenance Period Products and Services have been duly executed for and on behalf of all the parties thereto in accordance with GCC Clause 8;
- (c) The Firm has submitted the performance and advance payment securities to the Purchaser in accordance with GCC Clause 21;
- (d) The Purchaser has paid the Firm the advance payment, in accordance with GCC Clause 19; and
- (e) Any other conditions specified in the SCC.

4. Language and Interpretation

4.1 **Language**—The Contract and all related correspondence and documents exchanged by the Firm and the Purchaser, shall be written in the language specified in the SCC and shall be construed and interpreted in accordance with that language. Supporting documents and printed literature that are part of the Contract may be in another language provided they are accompanied by an accurate translation of the relevant passages in

the language specified in the SCC, in which case, for purposes of interpretation of the Contract, this translation shall govern. The Firm shall bear all costs of translation to the governing language and all risks of the accuracy of such translation.

- 4.2 **Incoterms**—Unless inconsistent with any provision of the Contract, the meaning of any trade term (EXW, CIP, etc.) and the rights and obligations of parties thereunder shall be as prescribed by Incoterms 2000. “Incoterms 2000” means the international rules for interpreting trade terms published by the International Chamber of Commerce, 38 Cours Albert 1er, 75008 Paris, France.
- 4.3 **Entire Agreement**—The Contract constitutes the entire agreement between the Purchaser and Firm with respect to the subject matter of Contract and supersedes all communications, negotiations and agreements (whether written or oral) of parties with respect to the subject matter of the Contract made prior to the date of Contract signing.
- 4.4 **Amendment**—No variation in or modification of the terms of the Contract shall be made except by written amendment signed by the parties.
- 4.5 **Independent Firm**—The Firm shall be solely responsible for the manner in which the Contract is performed. All employees, representatives or Subcontractors engaged by the Firm in connection with the performance of the Contract shall be under the complete control of the Firm and shall not be deemed to be employees of the Purchaser.
- 4.6 **Joint Venture, Consortium or Association**—If the Firm is a joint venture, consortium, or association, all of the parties shall be jointly and severally liable to the Purchaser for the fulfillment of the provisions of the Contract and shall designate one party to act as a leader with authority to bind the joint venture, consortium, or association, and to incur liabilities and receive instructions for and on behalf of all the parties. The entire execution of the Contract, including payment, shall be done exclusively with the leader. The composition or the constitution of the joint venture, consortium, or association shall not be altered without the prior consent of the Purchaser.

5. Notices

- 5.1 Any notice given by one party to the other pursuant to this Contract shall be sent to the other party in writing and confirmed in writing to the other party's address specified in SCC, within two days.

- 5.2 A notice shall be effective when delivered or on the notice's effective date, whichever is later. Any notice sent by airmail post or special courier shall be deemed (in the absence of evidence of earlier receipt) to have been delivered ten (10) days after proper dispatch.

6. Governing Law

6.1 The Contract shall be governed by and interpreted in accordance with the laws of the Purchaser's country, unless otherwise specified in the SCC.

7. Resolution of Disputes

7.1 If any dispute or difference of any kind whatsoever shall arise between the Purchaser and the Firm in connection with or arising out of the Contract, the parties shall make every effort to resolve amicably such dispute or difference by mutual consultation.

7.2 If, after thirty (30) days, the parties have failed to resolve their dispute or difference by such mutual consultation, then either the Purchaser or the Firm may give notice to the other party of its intention to commence arbitration, as hereinafter provided, as to the matter in dispute, and no arbitration in respect of this matter may be commenced unless such notice is given.

(a) Any dispute or difference in respect of which a notice of intention to commence arbitration has been given in accordance with this Clause shall be finally settled by arbitration. Arbitration may be commenced prior to or after Installation of the Products under the Contract.

(b) Arbitration proceedings shall be conducted in accordance with the rules of procedure specified in the SCC.

7.3 Notwithstanding any reference to arbitration herein, the parties shall continue to perform their respective obligations under the Contract unless they otherwise agree, and the Purchaser shall pay the Firm any monies due the Firm.

8. Supplementary Agreements

8.1 The following services required under the Contract shall be the subject of supplemental agreements (hereinafter the "Supplemental Agreements") covering the entire Maintenance Period and, where appropriate, the Warranty Period, for the affected Installation Batches of the Products:

(a) Any Software Maintenance and Technical Support Services contracts required pursuant to GCC Clause 16,

(b) Any Hardware and Maintenance and Operational Support Services contracts required pursuant to GCC Clause 17, and

(c) Any Source Code Escrow agreements required pursuant to GCC Clause 24.7

- 8.2 The Supplementary Agreements shall include prices no higher than those specified in the Maintenance Costs Schedule, and terms and conditions substantially equivalent to those specified in the Contract and offered in the Firm's tender, including the clauses for Force Majeure, Termination for Default, Termination for the Purchaser's Convenience, and any additional provisions designated in the SCC, which shall also be substantially equivalent to those contained in the Contract. If there is any conflict between the Contract and any Supplemental Agreement, the provisions of the Contract shall prevail.
- 8.3 Should the Purchaser terminate a Supplemental Agreement or cause it to be terminated during the Maintenance Period, then the Maintenance Period for the affected Product or Service shall terminate on the date that the corresponding Supplemental Agreement is terminated

B. Subject Matter of Contract

9. Scope of the Contract

- 9.1 As further elaborated throughout the Contract, the Firm shall deliver, Install and obtain the Purchaser's Acceptance of all the Products specified in the Supply Requirements and deliver the Related Services indicated in GCC Clause 9.2 below in accordance with the Implementation Schedule, the Project Plan and GCC Clauses 10 and 13. The Firm shall also fulfill all Warranty obligations pursuant to GCC Clause 26 and shall execute the Supplemental Agreements for the Maintenance Period in accordance with GCC Clause 8.
- 9.2 The Related Services which the Firm shall provide under the Contract are as follows, even if not clearly identified and separately priced in the Related Services Price Schedule, in which case their price shall be deemed to be included in the Contract Price in accordance with GCC Clause 18.4:
- (a) The Project Implementation Services specified in GCC Clause 15.2;
 - (b) The Software Maintenance and Technical Support Services specified in GCC Clause 16.2; and
 - (c) The Hardware and Maintenance and Operational Support Services specified in GCC Clause 17.4.

10. Upgrades and Cost Reductions

- 10.1 For Products and Services not yet delivered, the Firm shall offer the Purchaser any cost reductions, promotional programs, ancillary or extended services or warranties, or other options offered to the Firm's clients in the Purchaser's country.
- 10.2 The Firm shall offer to the Purchaser to replace Products not yet delivered during Project Implementation by new or upgraded models, new Software Releases and replacement Products of the latest technology and equal or better performance or functionality, which the Firm offers its clients in the Purchaser's country at the same or lower relative prices.
- 10.3 Changes to the Contract necessitated by the Purchaser's acceptance of an offer made in accordance with GCC Clause 10.1 or 10.2 shall be implemented by a Change Order pursuant to GCC Clause 33

11. Transport and Delivery

- 11.1 The Firm shall provide such packing of the Products as is required to prevent their damage or deterioration during shipment. The Firm shall promptly repair or replace any Products that are damaged in transit due to inadequate packing or related causes, where such damage is not otherwise fully covered by insurance. The packing, marking, and documentation within and outside the packages shall also comply strictly with the Purchaser's instructions to the Firm.
- 11.2 Transportation of the Products, including insurance and storage, to the named place of destination in the Purchaser's country, as specified in the Implementation Schedule, shall be arranged and paid for by the Firm, and the cost thereof shall be included in the Contract Price in accordance with GCC Clause 18. The Firm shall be free to use transportation through carriers registered in any eligible country and to obtain insurance from any eligible source country.
- 11.3 Transportation of any Products from their original delivery destination to a different Project Site for final Installation shall be the responsibility of the Purchaser, unless otherwise specified in the SCC.
- 11.4 Early or partial deliveries require the explicit written consent of the Purchaser, which consent shall not be unreasonably withheld.

11.5 For Products supplied from abroad (CIP Terms): Upon shipment, the Firm shall notify the Purchaser and the Insurance Company by cable with the full details of the shipment. The Firm shall promptly send the following documents to the Purchaser by mail or courier, as appropriate, with a copy to the Insurance Company:

- (a) two copies of the Firm's invoice showing the Products' description, quantity, unit price, and total amount;
- (b) usual transportation documents;
- (c) insurance certificate for CIP only; and
- (d) certificate(s) of origin.

11.6 For Products supplied from within the Purchaser's country (EXW Terms): Upon shipment, the Firm shall notify the Purchaser and shall promptly send the following documents to the Purchaser by mail or courier, as appropriate:

- (a) two copies of the Firm's invoice showing the Products' description, quantity, unit price, and total amount
- (b) delivery note, railway receipt, or truck receipt; and
- (c) certificate(s) of origin.

11.7 The documents listed in GCC Clauses 11.5 and 11.6 shall be received by the Purchaser at least one week before arrival of the Products and, if not received on time, the Firm will be responsible for any resulting additional expenses.

11.8 At the request of the Purchaser, the Firm will make available a representative or agent during the process of customs clearance in the Purchaser's country for Products supplied from abroad. In the event of delays in customs clearance that are not the fault of the Firm, the Contract Price shall be adjusted in accordance with GCC Clause 33 to compensate for any additional storage charges that the Firm may incur as a result of the delay.

12. Inspection and Tests

12.1 The Purchaser or its representative shall have the right to inspect and test the Products to confirm their conformity to the Supply Requirements at the Project Sites or another location specified in the SCC at no extra cost to the Purchaser.

- 12.2 The Purchaser or its authorized representative shall conduct Acceptance Tests, in the presence and with the support of the Firm, in accordance with the test procedures set out in the SCC and Project Plan.
- 12.3 When the Products have successfully passed the Acceptance Tests, the Purchaser shall notify the Firm in writing within 21 days of the completion of the tests, certifying that the Products supplied and Installed under the Contract are Accepted.
- 12.4 Should any Product fail to conform to the Supply Requirements or to pass the Acceptance tests, the Purchaser shall notify the Firm in writing, indicating the nature of the failure or nonconformity, and the Firm shall promptly either replace the rejected Products or make the necessary repairs or corrections as required for Acceptance at no additional cost to the Purchaser and in accordance with the procedures and response times specified in the SCC.

12.1 12.5 Nothing in GCC Clause 12 shall in any way release the Firm from any warranty or other obligations under this Contract or limit the Purchaser's ability to seek other remedies as specified in the Contract.

13. Firm's Obligations

- 13.1 The Firm shall complete delivery and Installation of the Products and Related Services, obtain the Purchaser's Acceptance therefor, and shall manage and perform all its other obligations under the Contract in accordance with the following and any additional requirements specified in the SCC.
- 13.2 The Firm shall appoint an experienced Representative to manage its performance of the Contract. The Representative shall be authorized to accept orders and notices on behalf of the Firm, and to generate notices and commit the Firm to specific courses of action within the scope of the Contract. The Representative may be replaced only with the prior written consent of the Purchaser.
- 13.3 Supply, Installation, Acceptance, Warranty and Maintenance of the Products shall include provision of the following, at each Project Site where appropriate:
- (a) any special or proprietary tools required for assembly, use or maintenance of the Products;

- (b) detailed operations, maintenance and usage manuals for each appropriate unit of the Products; and
- (c) original Media, license certificates and, where necessary, license authorization key codes for all Software Releases.

13.4 The Firm and all Subcontractors will abide by the job safety, insurance, customs and immigration measures prevalent and laws in force in the Purchaser's country, and will indemnify the Purchaser from all demands or responsibilities arising from accidents or loss of life, the cause of which is the Firm's negligence. The Firm will pay all indemnities arising from such incidents and will not hold the Purchaser responsible or obligated.

13.5 The Firm is responsible for, and obligated to conduct all contracted activities with due care and diligence, in accordance with the Contract and using state-of-the-art methods and economic principles, and exercising all reasonable means to achieve the performance specified in the Contract.

13.6 The Firm shall permit the Public Procurement Authority to inspect the Firm's accounts and records relating to the performance of the Firm and to have them audited by auditors appointed by the Authority, if so required by the Authority.

14. Purchaser's Obligations

14.1 The Purchaser will appoint a Project Manager with the authority to accept or reject all deliverables and to be the primary contact for the Firm's Representative.

14.2 The Purchaser shall be responsible for timely provision of all resources, facilities, information, equipment access, and other deliverables necessary for project implementation, as identified in the Project Plan. Pursuant to GCC Clause 34, delay by the Purchaser shall result in an extension of the Implementation Schedule. The Purchaser shall also be responsible for any additional obligations specified in the SCC.

14.3 The Purchaser will designate appropriate staff for any training to be imparted by the Firm and shall make all appropriate logistical arrangements therefor in accordance with the Project Plan.

**15. Project
Implementation
Services**

15.1 The specific Installation, Acceptance Testing and other services that the Firm shall provide during the Implementation Period are specified in GCC Clause 15.2, subject to the following service definitions and any requirements therefor specified in the SCC:

- (a) “Installation Service” includes all unpacking, assembling, wiring, installation, cabling between equipment units and components, connections to power supplies and networks, Software Installation and system configuration and integration of all the Products comprising each Installation Batch at the Project Site for no less than one full day per Batch and sufficient to ensure that all Products are Installed as required to fulfill the Supply Requirements, and suitably configured and tested to ensure readiness for Acceptance testing.
- (b) “Operational Orientation Service” consists of instructing the Purchaser’s technical staff at the Project Site in the proper operation, management, routine care and maintenance of the Products.
- (c) “Acceptance Test Support Service” includes provision by the Firm of on-site technical and operational support for each Installation Batch at each Project Site for the time periods specified in the Supply Requirements, with at least one expert systems engineer knowledgeable in the overall systems design and the major Products comprising the Systems, with backup support and expertise available off-site and/or at other Project Sites as necessary to provide coverage for the full suite of Products included in the Installation Batches being tested together, as specified in the Supply Requirements.

15.2 Except as otherwise specified in the SCC, the Firm shall provide the following Project Implementation Services in accordance with the service definitions specified in GCC Clause 15.1

- (a) Installation and Orientation Services for all Installation Batches at all Project Sites;
- (b) Provision of all project and technical documents and diagrams specified in the Supply Requirements;

- (c) Acceptance Test Support Services for all the Purchaser's Acceptance Testing activities as indicated in GCC Clause 12 and the Project Plan;
- (d) Technical and end-user training for the Purchaser's staff, as specified in the Supply Requirements;
- (e) Any additional Project Implementation Services specified in the Supply Requirements.

**16. Software
Maintenance
and Technical
Support
Services**

16.1 The specific Software Maintenance and Technical Support Services that the Firm shall provide under the Contract and which may be the subject of Supplemental Agreements pursuant to GCC Clause 8 are specified in GCC Clause 16.2, subject to the following service definitions and any requirements therefor specified in the SCC:

- (a) "Software Maintenance Service" is a subscription service whereby the Purchaser automatically receives or is granted Internet access to the Media and associated documentation for all Maintenance Releases of the Software, and is also provided with access to a (preferably Internet) knowledge-base of information and/or a periodic publication about known problems with the Software and solutions therefor.
- (b) "Version Upgrade Service" is a subscription service whereby the Purchaser automatically receives or is granted Internet access to the Media and associated documentation for all new Version Releases of the Software.
- (c) "End-User Help Desk Support" is a service whereby the Purchaser's end-users of the system may report problems and submit questions related to the use of the Software (or the entire System) during the specified Coverage Period and promptly receive courteous, helpful, solution-oriented replies from technical experts in the subject matter of the service. The forms of communication and the required response times and any necessary escalation procedures are specified in the SCC.
- (d) "Remote Technical Support" is a service whereby designated members of the Purchaser's technical staff may report system problems and submit technical questions related to the use, operation and proper maintenance of the Systems or specific Products during the specified Coverage

Period and promptly receive courteous, helpful, solution-oriented replies from technical experts in the subject matter of the service. The forms of communication and the required response times and any necessary escalation procedures, e.g. to On-Call Technical Support, are specified in the SCC.

- (e) “On-Call Technical Support” is a service for either a fixed, periodic fee or a usage-based fee structure whereby during the Coverage Period designated members of the Purchaser’s technical staff may request and receive on-site technical assistance within a reasonable response time with the diagnosis, repair, operation, modification or proper maintenance of the Systems, the network or specific Products, including with the installation and testing of new Software Releases. The methods of communication, the required response times, service levels and escalation procedures are specified in the SCC.
- (f) “On-Site Technical Support” is a service whereby the Firm employs a specified number of systems engineers to work full time during the Coverage Period at designated Project Sites to provide technical support services identical to those provided under the On-Call Technical Support service, but with a more immediate response time from dedicated, on-site experts. The management procedures, required service levels and escalation procedures are specified in the SCC.

16.2 Except as otherwise specified in the SCC, the Firm shall provide the following Services for all Software provided under the Contract in accordance with the service definitions specified in GCC Clause 16.1, provided that provision of these Services shall not relieve the Firm of any warranty or other obligations under this Contract:

- (a) Software Maintenance Services for all Software for the entire Useful Life Period;
- (b) On-Call Technical Support during Acceptance testing for all Project Sites, Products and Acceptance tests;
- (c) Remote Technical Support Services for the System and individual Products from the date of Installation

through Acceptance testing and the Warranty Period;

(d) Version Upgrade Services for all Software for the Implementation Period; and

(e) Any additional Services specified in the SCC.

16.3 Unless otherwise specified in SCC, all End-user Help Desk and Technical Support Services that the Firm is required to provide under the Contract shall conform to the following:

- (a) The means of communicating questions, problems and responses between the Firm and the Purchaser shall include telephone, facsimile and electronic mail, and the Purchaser shall receive responses within one business day;
- (b) The Supplemental Agreement shall include provisions for escalation to more senior experts within 24 hours and, if so specified in the Supply Requirements, On-site Technical Support within two (2) business days, for persistent problems affecting the Purchaser's production operations.

16.4 Unless otherwise specified in the SCC, all Software Maintenance Release and Version Upgrade Services required under the Contract shall include to the following provisions:

- (a) All new Releases of the Software and associated documentation to be provided under the Contract shall be delivered to the Purchaser within 30 days of their availability in the Purchaser's country, and no later than 12 months after they are released in the Country of Origin of the Software.
- (b) If the License for the Software is version-restricted, the Purchaser's License is upgraded automatically to include new versions upon receipt thereof by the Purchaser.

17. Hardware and Maintenance and Operational Support Services

- 17.1 The specific Hardware and Maintenance and Operational Support Services which the Firm shall provide under the Contract and which may be the subject of Supplemental Agreements pursuant to GCC Clause 8 are specified in GCC Clause 17.4, subject to the service definitions contained in GCC Clauses 17.2 and 17.3.
- 17.2 Except as otherwise or as further specified in the **SCC**, the following fixed price services should include provision of all labor, spare parts, supplies, diagnostic equipment, escalation procedures, continuous effort until repair and periodic preventative maintenance activities, for fixed maximum prices indicated in the Maintenance Costs Form, sufficient to ensure that all equipment operates without defects or interruption for at least 99.5% of the Coverage Period (exclusive of scheduled down-time), and if any critical component is out of service for more than 48 hours, the Firm shall either immediately replace the defective component or the entire unit at its own cost:
- (a) “On-site System Maintenance Service,” whereby the Firm shall provide a specified number of full-time resident engineers on site for the Coverage Period at designated Project Sites.
 - (b) “On-Call System Maintenance Service,” whereby the Firm shall provide local equipment maintenance services during the Coverage Period for all or specific Products at designated Project Sites, whereby an engineer shall arrive on site within 4 hours from the time the Purchaser logs a service call.
 - (c) “Return-to-Service-Depot Maintenance” is a service suitable for smaller and less critical items (such as workstations, printers and peripheral equipment) that is made available for specific Products by the Firm on a carry-in or send-in basis during the Coverage Period. Equipment returned for service shall be repaired or replaced within five (5) days. The Purchaser shall be responsible for all shipping and insurance charges to the Service Depot and the Firm shall be responsible for return shipping and insurance charges. If so specified in the **SCC**, the Return-to-Service-Depot may also be priced as an on-demand service under the provisions of GCC Clause 17.3.
- 17.3 Except as otherwise or as further specified in the **SCC**, the following on-demand or other special services should be

available on a time-and-materials fee basis in accordance with Contract prices for both spare parts and labor indicated on the Maintenance Costs Form:

- (a) “Per-Call Hardware Maintenance Service,” whereby the Firm shall have an engineer on site within 4 hours from when the Purchaser logs a service call during the Coverage Period. This service includes provision for continuous effort until repair, escalation procedures, a local inventory of essential spare parts, maintenance supplies and diagnostic equipment. If specified in the SCC, the Per-Call service may provide backup for a self-maintenance program and/or offer additional coverage outside the regular Coverage Period by defining an “Extended Coverage Period.”
- (b) “Self-Maintenance” is a service whereby the Firm shall provide spare parts, diagnostic equipment and software, and technical support as necessary to facilitate the diagnosis, maintenance and repair of the Products by the Purchaser’s technical staff, as specified in the SCC.

17.4 Except as otherwise specified in the SCC, the Firm shall provide the following Services for the Products provided under the Contract in accordance with the service definitions specified in GCC Clause 17.1, provided that provision of these Services shall not relieve the Firm of any warranty or other obligations under this Contract:

- (a) Unless otherwise and as may be further specified in the Supply Requirements, the Firm shall provide On-call System Maintenance Services for the Useful Life Period for all Products and Project Sites;

C. Payment

18. Contract Price 18.1 The Contract Price shall be the amount specified in the Contract Form.

18.2 Except for Maintenance Costs (which are quoted as maximum costs with appropriate price escalation factored in and which may vary downwards from the prices quoted in the Maintenance Costs Schedule under Supplemental Agreements in accordance with GCC Clause 8), the Contract Price shall be a firm lump sum not subject to any alteration, except as a result of a Change Order pursuant to GCC Clause 33, or in accordance with any price adjustment formula specified in the SCC.

18.3 The Contract Prices of individual Products and Services are as listed in the Price Schedules, as may be amended during Contract execution, and adhere to the following terms:

- (a) Products supplied from outside the Purchaser's country are priced on a CIP (named place of destination) basis to the Project Site for the Installation Batch specified in the Implementation Schedule, unless otherwise specified in the SCC. Requirements for the contract of carriage are as specified in the SCC.
- (b) Products supplied from within the Purchaser's country, unless otherwise specified in the SCC, are priced on an EXW (ex factory, ex works, ex warehouse, ex showroom or off-the-shelf, as applicable) basis, including all customs duties and sales and other taxes already paid or payable on the components and materials used in the production of the Products, or on the previously imported Products, but excluding sales or other taxes assessed on the Products in the Purchaser's country. Transportation, insurance and other local costs incidental to delivery of EXW Products to the Project Site shall be priced separately as a Service.
- (c) Prices for Services required by the Contract are quoted inclusive of all expenses and incidental costs associated with the provision thereof, and include all taxes, duties, levies and fees paid or payable by the Tenderer, its Subcontractors, or their employees on account of the Services in any country.
- (d) Taxes and duties associated with the supply of Products and Services under the Contract are treated in accordance with GCC Clause 20.

18.4 The Firm shall be deemed to have satisfied itself as to the correctness and sufficiency of the Contract Price, which shall, except as otherwise provided for in the Contract, cover all its obligations under the Contract.

19. Payment

19.1 The method and conditions of payment to be made to the Firm under this Contract shall be specified in the SCC. The Firm's requests for payment shall be made to the Purchaser in writing, accompanied by an invoice describing, as appropriate, the Products Installed or Accepted and Services delivered, and by

documents submitted pursuant to GCC Clauses 11.5 and 11.6, and upon fulfillment of other obligations stipulated in the Contract.

19.2 Payments shall be made promptly by the Purchaser, but in no case later than sixty (60) days after submission of an invoice or claim by the Firm.

19.3 Unless otherwise specified in the SCC, all payments shall be made in the currency specified in the applicable Price Schedules. If both the specified currency and, because of a membership in the European Monetary Union (EMU), the Euro are valid in the country of the specified currency, payments may also be made in Euro. In case the specified currency has ceased to be valid in because of the country's membership in the EMU, payments shall be made in the corresponding amount of the Euro. The conversion of the amount of the specified currency into Euro shall in both cases be made in accordance with the statutory rate of conversion applicable in the country of the specified currency.

19.4 Payment of the foreign currency portion of the Contract Price for imported Products and Services shall be made to the Firm through an irrevocable letter of credit opened by an authorized bank in the Firm's country, and will be payable on presentation to it of the appropriate documents. It is agreed that the letter of credit will be subject to Article 10 of the last revision of *Uniform Customs and Practice for Documentary Credits*, published by the International Chamber of Commerce, Paris

19.5 No payment made by the Purchaser herein shall be deemed to constitute Acceptance of the Products or any part thereof by the Purchaser.

20. Taxes and Duties

20.1 Except as otherwise specifically provided in the Contract, the Firm shall bear and pay all taxes, duties, levies, and charges assessed on the Firm, its Subcontractors, or their employees by all municipal, state, or national government authorities, both within and outside the Purchaser's Country, in connection with the Products and Services supplied under the Contract.

20.2 Notwithstanding GCC Clause 20.1 above and unless otherwise specified in the SCC, the Purchaser shall bear and promptly pay all customs duties, import taxes, sales or other indirect taxes imposed by law in the Purchaser's Country on Products supplied from outside the Purchaser's country under the Contract.

- 20.3 If any tax exemptions, reductions, allowances, or privileges may be available to the Firm in the Purchaser's Country, the Purchaser shall use its best efforts to enable the Firm to benefit from any such tax savings to the maximum allowable extent.
- 20.4 For the purpose of the Contract, it is agreed that Contract Prices are based on the taxes, duties, levies and charges prevailing on the date twenty-eight (28) days prior to the date of tender submission in the Purchaser's country (hereinafter called "Tax" in this GCC Clause 20.4). If any rates of Tax are increased or decreased, a new Tax is introduced, an existing Tax is abolished, or any change in interpretation or application of any Tax occurs in the course of the performance of Contract, which was or will be assessed on the Firm, Subcontractors, or their employees in connection with performance of the Contract, an equitable adjustment to the affected Contract Prices shall be made to fully take into account any such change.

21. Securities

- 21.1 ***Advance Payment Security:*** Within twenty-eight (28) days of notification of Contract Award the Firm shall provide a security in the currency and in an amount equal to that of the advance payment to be received under the Contract.
- 21.2 The Advance Payment Security shall be in the form provided in the Tendering Documents or in another form acceptable to the Purchaser. The amount of the security shall be reduced in proportion to the value of payments becoming due to the Firm, and shall automatically become null and void and shall be immediately returned to the Firm when the full amount of the advance payment has been recovered in kind by the Purchaser.
- 21.3 ***Performance Security:*** Within twenty-eight (28) days of the notification of Contract award the Firm shall provide a security denominated in the currency of the Contract, or in a freely convertible currency acceptable to the Purchaser, and in the amount specified in the SCC for the due performance of the Contract.
- 21.4 The proceeds of the Performance Security shall be payable to the Purchaser as compensation for any loss resulting from the Firm's failure to complete its obligations under the Contract. The Performance Security shall be in one of the following forms:
- (a) a bank guarantee or an irrevocable letter of credit issued by a reputable bank acceptable to the Purchaser

located in the Purchaser's country or abroad, in the form provided in the tendering documents or another form acceptable to the Purchaser; or

(b) a cashier's or certified check.

21.5 Unless otherwise specified in the SCC, the Performance Security shall be reduced by half six (6) months after completion of the Implementation Period, when the Firm has demonstrated the ability to fulfill its Warranty obligations.

21.6 Unless otherwise specified in the SCC, the Performance Security shall be discharged in full by the Purchaser and returned to the Firm not later than thirty (30) days following expiration of the Warranty Period for all the Products, provided, however, that if the Firm has obligations for the Maintenance Period under Supplemental Agreements pursuant to GCC Clause 8, the following conditions must also be met:

- (a) the Firm shall have fulfilled Maintenance Period obligations for six (6) months and shall not be in default on any of the Supplemental Agreements; and
- (b) the quoted annual Maintenance Costs are not lower than 2.5% of the total supply cost of the Products (indicating performance risk with costs substantially below a reasonable industry average percentage);
- (c) there shall be no operating or usage restrictions or Firm or Producer controls present in any key Software Products that would introduce a business continuity risk for the Purchaser, e.g., by means of Firm-controlled passwords, keys or codes or other Firm-controlled mechanisms of any kind.

21.7 Unless otherwise specified in the SCC, if any of the conditions specified in GCC Clauses 21.6(a), 21.6(b) or 21.6(c) are not met, the Performance Security shall remain in force for the entire Maintenance Period and decreased from its amount at the end of the Warranty Period by a prorated percentage the end of each year in the Maintenance Period (equal to one divided by the number of years in the Maintenance Period).

22. Liquidated Damages

22.1 Subject to the provisions of GCC Clauses 31 and 34, if the Firm fails to install the Products or to perform the Related Services within the period(s) specified in the Implementation Schedule,

the Purchaser shall, without prejudice to its other remedies under the Contract, deduct from the Contract Price, as liquidated damages, a sum equivalent to the percentage specified in SCC of the price of the delayed Products or unperformed Services for each week or part thereof of delay until actual Installation or performance, up to a maximum deduction of the percentage specified in SCC. Once the maximum is reached, the Purchaser may consider termination of the Contract pursuant to GCC Clause 35.

D. Intellectual Property

23. Copyright

23.1 The Intellectual Property Rights in all Off-the-shelf Software and associated documentation shall remain vested in the owner of such rights.

23.2 The Purchaser agrees to use, copy, assign, install and duplicate Off-the-shelf Software and its documentation in accordance with the associated Software License Agreement, pursuant to GCC Clause 24. However, the Purchaser may make additional copies of documentation for use within the scope of the Project in the event that the Firm is unable to deliver new copies within 30 days from receipt of an order.

24. Software License Agreements

24.1 The Firm hereby grants to the Purchaser a non-exclusive License to install, configure, use and integrate the Software and related documentation provided under the Contract, including all inventions, designs, and marks embodied in the Software. All Software Licenses are granted subject to the provisions and for the purposes specified in the Contract and, except where otherwise specified in the Contract for a particular Software Product, are Perpetual Use Licenses. The term of the Licenses provided under the Contract shall commence no later than the date of Installation of the Software. In the event of a conflict between the Contract and any licensing terms or conditions supplied with or as part of the Software (hereinafter called an “End User License Agreement” or “EULA”), the provisions of the Contract shall prevail.

24.2 All Software Maintenance and Licensing costs necessary to fulfill the requirements of the Contract, to keep the System operating normally for the Useful Life Period in accordance with the Supply Requirements, and to maintain compliance with any Supplemental

Agreements, whether payable to the Firm or to a third party or subcontractor, that are not specifically indicated in the Price Schedules shall be deemed to be included in the Contract Price in accordance with GCC Clause 18.4 and shall be paid promptly by the Firm when due.

24.3 Renewable Software Licenses provided under the Contract are subject to the following provisions, except as otherwise specified in the SCC:

- (a) Renewable Licenses shall be renewed automatically on an annual basis, and, for Licenses restricted to particular Software version(s), renewal automatically upgrades the License to include the latest version of the Software available on the renewal date, and also any subsequent Maintenance Releases therefor.
- (b) The Purchaser may terminate a Renewable License at any time upon 30 days prior written notice, and shall promptly receive a full refund of any unused portion of the License fee from the Firm.
- (c) The Firm may revoke a Renewable Software License in the event of non-payment of the Renewable License fee indicated in the Maintenance Costs Form no less than thirty (30) days after receipt by the Purchaser of an invoice therefor and no less than sixty (60) days from the date of expiration of the term of the Renewable License.
- (d) Upon termination or expiration of a Renewable Software License, all use of the Software by the Purchaser shall cease within thirty (30) days from the effective date of such termination or expiration.

24.4 All Software Licenses provided under the Contract are subject to the following provisions, as may be modified or added to (for all or specific Software) in the SCC:

- (a) The geographic scope of the Licenses extends to the Purchaser's country and all of its territories, except for any Software, which the Supply Requirements specify is for use on portable computers or to be accessed over the Internet, the geographic scope of which shall be worldwide.
- (b) The Purchaser may configure, customize, adapt, integrate or combine the Software Product with other Software,

provided that derivative Software incorporating any substantial part of a supplied Software Product shall be subject to same restrictions as are set forth in this Contract therefor.

- (c) The Purchaser may make a reasonable number of backup copies of the Software Media for safekeeping only.
- (d) The Purchaser may transfer the Software License to a replacement computer without charge and at its own discretion, provided that, if the Contract restricts the License to a specific type of computer, the replacement computer is of the specified type. Use of the Software on the original and replacement computers may be simultaneous during a reasonable transitional period of no more than 60 days.
- (e) With the exception of operating system and special purpose disaster recovery Software, the Purchaser may install the Software on a designated backup computer for purposes of providing disaster recovery preparedness without charge, as long as the Software is only used on the backup computer in the event that the primary computer for which the Software License was purchased, or its replacement, is inoperative; during a reasonable transitional period when use is being transferred between primary and backup, or for reasonable test periods for verifying the state of preparedness.
- (f) The Purchaser may install and test the operation of the Software, in the event of receipt of a new Release of the Software or of underlying or prerequisite System Software, on a development or test machine for up to 30 days, at no additional charge and at its own discretion.
- (g) The Firm shall promptly supply any license keys or authorization codes necessary to facilitate installation, testing and use of the Software on the Purchaser's replacement, backup or test computers as described in GCC Clauses 24.4(d), 24.4(e) and 24.4(f), respectively.
- (h) The Firm may not revoke a Software License, except as provided under GCC Clause 24.3©, nor disable or otherwise impair the operation of the Software without the Purchaser's prior written consent.
- (i) The Firm warrants that each copy of the Software Distribution Media that it provides to the Purchaser under

the Contract shall be free from physical defects and will promptly replace any defective Media at its own expense.

- (j) The Firm warrants that for the Useful Life Period no copy of the Software contains or will contain any Surreptitious Code.
- (k) The Purchaser's usage of the Software Products identified in the SCC shall be subject to audit by the Firm to verify compliance with the terms of the Software License Agreement, at the Firm's expense and no more frequently than once per calendar year, to be scheduled at mutually convenient times and under mutually agreeable procedures, subject to the audit provisions specified in the SCC. For such audit provisions to be effective, appropriate software enabling the Purchaser to track and manage Software License quantities, types and Maintenance agreements purchased, and to automatically monitor software usage and version releases against those licenses and agreements must be included in the tender and installed and configured by the Firm prior to the expiration of the Warranty Period for the relevant Software Product.

24.5 Unless otherwise specified in SCC for a specific Software Product, User-Based Licenses for personal computer Software shall permit its installation and use on more than one computer normally used by the same user at different times, provided that the Software will not be used simultaneously on more than one computer system.

24.6 Except as otherwise specified in the SCC, Software Licenses provided under the Contract shall permit the Software to be disclosed to and reproduced for use by service providers and their subcontractors exclusively in performance of their obligations under the Contract or any Supplemental Agreements associated therewith, subject in all cases to the same restrictions as are set forth in this Contract.

24.7 Subject to a supplementary Source Code Escrow Agreement for the Application Software identified in the SCC as being critical to the Purchaser's operations, which shall be entered into with a reputable escrow agent of the Firm's choosing and acceptable to the Purchaser prior to the Start Date per GCC Clause 3.1, the Firm and/or the Software Producer shall maintain current and verifiably complete Source Code on deposit with the escrow agent throughout the entire Useful Life Period, with provisions for the following

- (a) The events triggering prompt release of the Source Code to the Purchaser shall include insolvency or bankruptcy of the Software Producer, its cessation of or notice of intent to cease fulfilling its obligations under any Maintenance Agreement for the Software to which it is party, or its default under the Escrow Agreement.
- (b) In the event that any of the trigger events occur, the Purchaser shall promptly receive and have the right to access, modify, retain and use the Source Code to enable its continued use of the Software in accordance with the License Agreement for the rest of the Useful Life Period.
- (c) The Source Code shall be released in favor of the owner thereof at the end of the Useful Life Period, or in the event of normal or Purchaser-initiated termination of the License and Supplemental Agreements for Maintenance Services for the Software, whichever comes first.
- (d) Inclusion of any additional provisions specified in the SCC.

25. Confidential Information

- 25.1 The Firm shall not, without the Purchaser's prior written consent, disclose the Contract, or any provision thereof, or any specification, plan, drawing, schedule, design, or information furnished by or on behalf of the Purchaser in connection therewith, to any person other than a person employed by the Firm in the performance of the Contract. Disclosure to any such employed person shall be made in confidence and shall extend only so far as may be necessary for purposes of such performance.
- 25.2 The Firm shall not, without the Purchaser's prior written consent, make use of any document or information enumerated in GCC Clause 25.1 except for purposes of performing the Contract.
- 25.3 Any document, other than the Contract itself, enumerated in GCC Clause 25.1, or the development of which under the Contract is paid for by the Purchaser shall remain the property of the Purchaser and shall be returned (all copies) to the Purchaser on completion of the Firm's performance under the Contract, if so required by the Purchaser.
- 25.4 Each party shall keep confidential and use only for the purposes of the contract any information identified by the other in the SCC as being of a confidential nature, except that either party may disclose

under confidence to subcontractors or financiers confidential information necessary to perform the Contract

E. Guarantees and Liabilities

26. Warranty

26.1 The warranty provisions in this GCC Clause 26 shall remain valid for the Warranty Period specified in GCC Clause 1.1(hh), except where otherwise specified in the specific sub-clause. Provisions of the Firm's or Producer's standard commercial warranty for the Products shall also apply to the extent that they do not conflict with the provisions of the Contract.

26.2 The Firm warrants that the Products supplied under the Contract are new, unused, the most recent or current models, and that they incorporate all recent improvements in design and components, unless provided otherwise in the Contract. The Firm also warrants that all Products form part of the Producers' current product lines, have been previously released to the market, and that all Off-the-shelf Software has been in production use for at least three months. The Firm also warrants that the Products comply with the Supply Requirements and the Firm's tender.

26.3 The Firm further warrants that equipment supplied under this Contract shall have no defect arising from design, materials, or workmanship during Acceptance testing and for the duration of the Warranty Period that prevent the Products or the entire System from fulfilling the Supply Requirements, or that limit in a material fashion the performance, reliability or extensibility of the Product or System. This warranty excludes defects attributable to external factors beyond the Firm's control, including power supply fluctuations, or the Purchaser's alterations, misuse or negligence in proper maintenance of the equipment.

26.4 The Firm warrants for the Useful Life Period that the Products being supplied under this Contract are designed to be used prior to, during, and after the calendar year 2000 AD ("Year 2000"), including correct treatment of the year 2000 as a leap year, and that the System will operate during each such time period without error relating to date data, specifically including any error relating to, or the product of, date data that represents or references different centuries or more than one century. Without limiting the generality of the foregoing, the Firm further represents and warrants that, for the Useful Life Period, the Products will not abnormally terminate or provide invalid or incorrect results as a result of date data, specifically including date data that represents or references different centuries or more than one century; have

been designed to ensure Year 2000 compatibility, including, but not limited to, date data century recognition, calculations that accommodate same-century and multi-century formulas and date values, and date data interface values that reflect the century; and will manage and manipulate data involving dates, including single-century formulas and multi-century formulas, and will not terminate abnormally or generate incorrect data.

26.5 The Firm hereby represents and warrants that all Software provided under the Contract does not and will not infringe any Intellectual Property Rights held by any third party and that the Firm has all necessary rights, or at his sole expense shall have secured in writing all transfers of rights and other consents necessary to License for use and warrant the Software per the terms of this Contract for the Useful Life Period.

26.6 Unless otherwise specified in the SCC, the Firm hereby represents and warrants that, during the Implementation and Warranty Periods, Software provided under the Contract will operate in accordance with the associated documentation and with the claims related thereto included in the Firm's tender, which is hereinafter referred to as the "As-Documented Warranty." Instances of noncompliance with the As-Documented Warranty shall be reported to the Firm in writing prior to the expiration of the Warranty Period. Upon receipt of notice of such noncompliance, the Firm will promptly implement corrective measures for the defect in accordance with the provisions of GCC Clause 26.7 and any applicable Supplemental Agreement for Software Maintenance and/or Support Services.

26.7 The Firm shall effect repair or replacement of Products not meeting these performance guarantees and warranties promptly upon receipt of notice thereof at its own cost and expense and with all reasonable speed within such times and under such procedures, as are specified in the SCC. The Firm shall not be relieved of his obligations stated herein until the Purchaser is satisfied that the repaired and replaced Products are performing in accordance with the performance guarantees specified in the Contract. When he Firm fails to remedy the defect(s) within the times specified in the SCC, the Purchaser may proceed to take such remedial action as may be necessary, at the Firm's risk and expense and without prejudice to any other rights which the Purchaser may have against the Firm under the Contract.

26.8 Replacement equipment shall be covered under warranty for a three month period, or the time remaining in the Warranty Period for the

item replaced, whichever is greater. The Warranty Period for replacement Software shall be identical to the initial warranty period for the defective Software. Replaced Products shall be the property of the Firm.

**27. Intellectual
Property Rights
Indemnification**

27.1 The Firm shall indemnify and defend the Purchaser against all third party claims of infringement of Intellectual Property Rights of any kind arising from the supply or use by the Purchaser of the Products or any part thereof.

27.2 The Firm shall expeditiously extinguish any such claims and shall have full rights to defend itself therefrom. If the Purchaser is required to pay compensation to a third party resulting from such infringement, the Firm shall be fully responsible therefor, including all expenses and court and legal fees. If such a claim is made or appears likely to be made, the Purchaser agrees to permit the Firm to enable the Purchaser at the Firm's expense, to continue to use the Firm Product or to modify or replace it with a Firm Product which has published specifications equal or superior to the Firm product being replaced. If none of the alternatives is available, the Firm, at its own expense, will secure for the Purchaser the use of an equivalent product and bear all related conversion, installation, and training costs.

27.3 The Purchaser will give notice to the Firm of any such claim without delay, shall provide reasonable assistance to the Firm in disposing of the claim, and shall at no time admit to any liability for or express any intent to settle the claim.

27.4 Indemnities shall not apply if any claim of infringement or misappropriation is asserted by a parent, subsidiary or affiliate of the Purchaser's organization, if it is a direct result of a design mandated by the Purchaser's Technical Specifications and the possibility of such infringement was duly noted in the Firm's Tender, or if it results from the Purchaser's alteration of the Products in a manner not prescribed in the Contract.

**28. Limitation of
Liability**

28.1 Except in cases of criminal negligence or willful misconduct or in the case of IPR infringement pursuant to GCC Clause 27:

- (a) the Firm shall not be liable to the Purchaser, whether in contract, tort, or otherwise, for any indirect or consequential loss or damage, provided that this exclusion shall not apply to any obligation of the Firm to pay liquidated damages to the Purchaser; and

- (b) the aggregate liability of the Firm to the Purchaser under the Contract shall not exceed the total Contract Price, provided that this limitation shall not apply to the cost of repairing or replacing defective Products, or to the obligation of the Firm to indemnify the Purchaser pursuant to GCC Clause 27.

F. Distribution of Risk

29. Property Damage, Accident or Injury Indemnities

- 29.1 Each party, whether the Purchaser or the Firm, shall indemnify and hold harmless the other party and its employees and officers from and against any and all losses, liabilities and costs (including losses liabilities and costs incurred in defending a claim alleging such a liability) which the other party or its employees or officers may suffer as a result of the death or injury of any person or loss of or damage to any property, arising in connection with the activities under the Contract and by reason of the negligence of the first party or its Subcontractors, or their employees, officers or agents.

30. Insurance

- 30.1 The Firm shall arrange and pay or otherwise provide for and maintain in effect the insurance set forth below, unless otherwise specified in the SCC. The identity of the insurers and the form of the policies shall be subject to the approval of the Purchaser, which approval shall not be unreasonably withheld.

- (a) Cargo Insurance During Transport. The Products supplied under the Contract shall be fully insured to 110% of their price in a freely convertible currency, covering the Products against loss or damage during shipment through receipt at the Project Site. The policy shall name the Purchaser as the beneficiary.
- (b) Third Party Liability Insurance covering bodily injury or death suffered by third parties (including the Purchaser's personnel) and loss of or damage to property (including the Purchaser's property and any Products that have been delivered or Installed by the Firm) arising out of the Firm's execution of the Contract.
- (c) Workers' Compensation and Employer's Liability Insurance in accordance with the statutory

requirements applicable in any country where the Contract or any part thereof is executed.

(d) Additional insurance as specified in the SCC.

30.2 The Firm shall provide coverage or otherwise ensure that its Subcontractor(s) shall take out and maintain in effect adequate insurance policies for their personnel and vehicles and for work executed by them under the Contract, including Workers' Compensation and Employer's Liability Insurance in accordance with applicable statutory requirements.

30.3 Unless otherwise provided in the Contract, the Firm shall prepare and conduct all and any claims made under the policies effected by it pursuant to this GCC Clause 30, and all monies payable by any insurers shall be paid to the Firm. The Purchaser shall give to the Firm all such reasonable assistance as may be required by the Firm in connection with any claim under the relevant insurance policies. With respect to insurance claims in which the Purchaser's interest is involved, the Firm shall not give any release or make any compromise with the insurer without the prior written consent of the Purchaser. With respect to insurance claims in which the Firm's interest is involved, the Purchaser shall not give any release or make any compromise with the insurer without the prior written consent of the Firm.

**31. Force
Majeure**

31.1 Notwithstanding the provisions of GCC Clauses 22, 34 and 35, the Firm shall not be liable for forfeiture of its performance security, liquidated damages, or termination for default if and to the extent that its delay in performance or other failure to perform its obligations under the Contract is the result of an event of Force Majeure.

31.2 For purposes of this clause, "Force Majeure" means an event beyond the control of the Firm and not involving the Firm's fault or negligence and not foreseeable. Such events may include, but are not restricted to, acts of the Purchaser in its sovereign capacity, wars or revolutions, fires, floods, epidemics, quarantine restrictions, and freight embargoes.

31.3 If a Force Majeure situation arises, the Firm shall promptly notify the Purchaser in writing of such condition and the cause thereof. Unless otherwise directed by the Purchaser in writing, the Firm shall continue to perform its obligations under the Contract as far as is

reasonably practical, and shall seek all reasonable alternative means for performance not prevented by the Force Majeure event.

G. Contract Administration

32. Implementation Schedule and Project Plan

32.1 Unless otherwise specified in the SCC, the Contract Implementation Schedule and the accompanying, detailed Project Plan included in the Firm's tender, and any amendments thereto shall conform to the Purchaser's requirements specified in the Supply Requirements. Any modifications to the Implementation Schedule shall be ratified by a written amendment to the Contract. The parties will use the Implementation Schedule to assess and manage the Firm's performance under the Contract, including but not limited to the assessment of liquidated damages in accordance with GCC Clause 22.

33. Change Orders

33.1 The Purchaser may at any time, by Notice given to the Firm pursuant to GCC Clause 5, make changes within the general scope of the Contract in any one or more of the following:

- (a) technical specifications of the Products
- (b) the method of shipment or packing
- (c) the place and schedule for Installation and Acceptance
- (d) the Services to be provided by the Firm
- (e) the obligations of either party under the Contract.

33.2 If any such change causes an change in the cost of, or the time required for the Firm's performance of the Contract, an equitable adjustment shall be made in the Contract Price or Implementation Schedule, or both, and the Contract shall be amended accordingly. Any claims by the Firm for adjustment under this clause must be asserted within thirty (30) days from the date of the Firm's receipt of the Purchaser's change order.

33.3 The Firm may at any time request a change to the same Contract elements listed in Clause 33.1, by means of written Notice to the Purchaser stating the cause for the request and the adjustments to Contract Price and Implementation Schedule that would result from the change. Within 15 days of receipt of the Notice, the Purchaser will make a determination as to whether to accept the change request

and amend the Contract accordingly, or to reject the change, and shall promptly inform the Firm.

34. Extensions of the Implementation Schedule

34.1 If at any time during performance of the Contract, the Firm should encounter conditions impeding timely delivery or Installation of the Products or Services, the Firm shall promptly notify the Purchaser in writing of the fact of the delay, its likely duration and its cause(s).

34.2 As provided under GCC Clause 31, or in the event of a delay in the Implementation Schedule resulting from failure of the Purchaser to provide necessary resources or timely completion of prerequisite tasks for Installation clearly identified in the Implementation Schedule or Project Plan as being the Purchaser's responsibility, or in the event of a delayed Installation that is a result of a delay in customs clearance that is not the fault of the Firm, the Firm shall be entitled to an extension in the Implementation Schedule without the assessment of liquidated damages

34.3 As soon as practicable after receipt of the Firm's notice of a delay, the Purchaser shall evaluate the situation and may extend the Firm's time for performance, with or without liquidated damages pursuant to GCC Clauses 22 and 34.2. Extensions to the Implementation Schedule shall be ratified by the parties by a written amendment to the Contract.

35. Termination for Default

35.1 The Purchaser, without prejudice to any other remedy for breach of Contract, by written notice of default sent to the Firm, may terminate this Contract in whole or in part:

- (a) if the Firm fails to deliver any or all of the Products within the period(s) specified in the Contract, or within any extension thereof granted by the Purchaser pursuant to GCC Clause 34; or
- (b) if the Firm fails to perform any other obligation(s) under the Contract.
- (c) if the Firm, in the judgment of the Purchaser, has engaged in corrupt or fraudulent practices in competing for or in executing the Contract. For the purposes of this Clause:

“corrupt practice” means the offering, giving, receiving or soliciting of any thing of value to influence the action of a

public official in the procurement process or in contract execution.

“fraudulent practice” means a misrepresentation of facts in order to influence a procurement process or the execution of a contract to the detriment of the Purchaser, and includes collusive practices among Tenderers (prior to or after tender submission) designed to establish tender prices at artificial non-competitive levels and to deprive the Purchaser of the benefits of free and open competition.

36. Termination for Convenience

36.1 The Purchaser, by written notice sent to the Firm, may terminate the Contract, in whole or in part, at any time for its convenience. The notice of termination shall specify that termination is for the Purchaser’s convenience, the extent to which performance of the Firm under the Contract is terminated, and the date upon which such termination becomes effective.

36.2 The Products or Services that are complete and ready for shipment or delivery within thirty (30) days after the Firm’s receipt of notice of termination shall be accepted by the Purchaser at the Contract terms and prices. For the remaining Products and Services, the Purchaser may elect:

(a) to have any portion completed and delivered at the Contract terms and prices; and/or

(b) to cancel the remainder and pay to the Firm an agreed amount for partially completed Products and Services and for materials and parts previously procured by the Firm.

36.3 The Purchaser shall pay to the Firm the cost of satisfying all other obligations, commitments and claims that the Firm may in good faith have undertaken with third parties in connection with the contract.

37. Termination for Insolvency

37.1 The Purchaser may at any time terminate the Contract by giving written notice to the Firm if the Firm becomes bankrupt or otherwise insolvent. In this event, termination will be without compensation to the Firm, provided that such termination will not prejudice or affect any right of action or remedy which has accrued or will accrue thereafter to the Purchaser.

38. Assignment

38.1 Neither the Purchaser nor the Firm shall assign, in whole or in part, their obligations under this Contract, except with prior

written consent of the other party. Such consent shall not relieve the Firm from any of its obligations, duties, or responsibilities under the Contract.

Section IX. Special Conditions of Contract

The following Special Conditions of Contract (SCC) shall supplement and / or amend the General Conditions of Contract (GCC). Whenever there is a conflict, the provisions herein shall prevail over those in the GCC. The Corresponding clause number of the GCC is indicated in the left hand column.

GCC 1.1(g&h)	The coverage period of assignment is Completion for test run and piloting shall not exceed eighteen (18) calendar months after award of contract.
GCC 1.1 (v)	The Purchaser is: Ghana Gold Board (GoldBod)
GCC 1.1 (r&s)	The Purchaser's Project Manager is: To be determine during Contract Award
GCC 1.1 (hh)	The Warranty Period shall but not be limited to the following: (Not less than 1 year or more after post go-live) (a) Software (b) Servers and communications equipment (c) Workstations and all other Products, equipment and facilities.etc. Firms must indicate their warranty for the Software and equipments etc.
GCC 1.1	The following additional provisions apply to the definitions provided
GCC 3.1 (e)	The following additional conditions must be met prior to the Start Date: 1. Performance Security 2. Inception Report 3. Requirement analysis report 4. Plan of activities and timelines

GCC 4.1	The Contract language shall be English .
GCC 5.1	For <u>notices</u> , the Purchaser's address shall be: Executive Secretary Attention: Head, Procurement Department Street Address: High Street City: Accra Region: Greater Accra Country: Ghana Telephone: <u>+233 209989440 / +233 208702970</u> Electronic mail address: frank.ademan@goldbod.gov.gh
GCC 5.1	For <u>notices</u> , the Firm's address shall be: N/A
GCC 6	The Contract shall be governed by the Laws of Ghana
GCC 7.2(b)	The formal mechanism for the resolution of disputes shall be as follows: <i>(a). For contracts entered with Foreign Firms:</i> In case of a dispute between the Purchaser and the Firm, the dispute shall be settled by arbitration in accordance with the provisions of the United Nations Commission on International Trade Law (UNCITRAL) arbitration Rules. <i>(b). For contracts entered with Firms from Ghana</i> In the case of a dispute between the Purchaser and a Firm from Ghana, the dispute shall be referred to adjudication or arbitration in accordance with the laws of Ghana.
GCC 8.2	The Supplemental Agreements shall also include provisions substantially equivalent to the following:
GCC 11.3	Any transportation costs required after initial delivery to the designated Project Site will be the responsibility of the Firm and <u>not</u> the Purchaser.

GCC 12	Acceptance test procedures and success criteria are specified in the Technical Specifications and the Project Plan. Acceptance test support Services required from the Firm are specified in GCC Clause 15. The procedures for responding to and correcting any reported Acceptance testing failures are as follows: Provision of on-site technical and operational support for each Installation Batch at each Project Site for the time periods specified in the Supply Requirements, with at least one expert systems engineer knowledgeable in the overall systems design and the major Products comprising the Systems, with backup support and expertise available off-site and/or at other Project Sites as necessary to provide coverage for the full suite of Products included in the Installation Batches being tested together, as specified in the Supply Requirements.
GCC 13.1	The Firm's obligations shall include the following additional requirements: Provision of all requirements in the Terms of Reference (TOR)
GCC 14.1	The Purchaser's obligations shall include the following additional requirements: 1. <i>Timely response to request for data and information</i> 2. <i>Provision of data set for application development (if available)</i> 3. <i>Facilitation of access to site under the control of the GoldBod</i> 4. <i>Provide contact person[s] for the project (if available)</i>
GCC 15.1	The following additional or modified Project Implementation Services shall be provided by the Firm (As provided GCC 15.1)
GCC 16	The following additional or modified Software Maintenance and Technical Support Services shall be provided by the Firm: As provided in GCC 16
GCC 17	The following additional or modified Hardware and Maintenance and Operational Support Services shall be provided by the Firm: As provided in GCC 17
GCC 18.2	There is no applicable price adjustment formula.
GCC 18.3(a)	The contract of carriage for CIP Products shall include payment by the Firm of the cost of any customs formalities, duties, taxes or other charges payable on the Products for or during their transit through any country other than Ghana, and shall exclude the cost of customs formalities, duties, taxes or other charges payable on the Products upon import into Ghana.
GCC 18.3(b)	No changes to Contract provisions for Products supplied from within Ghana.
GCC 19.1	The method and conditions of payment to be made to the Firm under this Contract shall be as follows: Subject to Negotiation

GCC 21.3	The Performance Security shall be for an amount equal to: <i>Ten (10) percent <u>Performance Guaranty</u> of Contract Sum from a reputable Bank regulated by Bank of Ghana and</i> <i>Twenty-Five (25) percent <u>Insurance Bond</u> of Contract Sum from an reputable Insurance Company regulated by National Insurance Commission</i> <i>The Performance Security shall be <u>valid</u> until successful completion and deployment of CAPS system.</i>
GCC 21.5	No changes to GCC provisions regarding reduction of performance security.
GCC 22	Liquidated damages for late response and provision of required services shall be assessed at 0.15 percent of contract sum per day.
GCC 24.3	Renewable Software Licenses provided under the Contract are subject to the following additional provisions: As provided in GCC 24.3
GCC 24.4	Software Licenses provided under the Contract are subject to the following additional provisions: As provided in GCC 24.4
GCC 24.4 (k)	The following Software shall be subject to license audits as specified below: As provided in GCC 24.4 (k)
GCC 24.5	The following restrictions shall apply to User-Based Licenses for the particular personal computer Software indicated below: As provided in GCC 24.5
GCC 24.6	No additional stipulations shall apply regarding the Purchaser's right to disclose or provide access to the Software to contracted service providers.
GCC 24.7	No Application Source Code Escrow arrangements are required. Source code must be provided directly to the purchaser.
GCC 25.1	For the purposes of the confidentiality agreement between the Parties the following special documents are identified as being of a confidential nature: By the Purchaser: Ghana Gold Board By the Firm: Service Provider
GCC 26.6	Exceptions, limitations or additional performance standards for the Documented Warranty for Software are as follows: All facilities necessary for the provision of TAS.

GCC 26.7	The Firm shall effect Warranty repair or replacement within the following time periods and under the following procedures: As provided under GCC 26.7
GCC 30.1	No modifications to GCC provisions regarding Insurance.
GCC 30.1(d)	No additional insurance is required
	No modifications to GCC Implementation Schedule provisions.

I. Actions requiring the employer's prior approval are;

- 1. Revision of scope of work***
- 2. Subletting of aspects of scope of work***
- 3. Revision of timelines***
- 4. Revision of specification***
- 5. Change of key personnel***

II. Prohibited activities include:

- 1. Transfer of contract to another firm***
- 2. Use of personal data in contravention of the Data Protection Act***
- 3. Falsification of information***
- 4. Unauthorised disclosure of Client's information***

III. Assistance that the Employer may provide under

- 1. Timely response to request for data and information (if any, will be provided at discretion)***
- 2. Provision of data set for application development (if any, will be provided at discretion)***
- 3. Facilitation of access to various institutions (if any, will be provided at discretion)***
- 4. Provide contact person[s] for the project (if any, will be provided at discretion)***

Section X. Contract Forms

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INSTRUCTIONS FOR THE CONTRACT FORMS

1. The Purchaser or the successful Tenderer, as appropriate, shall fill in and complete the Contract Forms in accordance with these instructions, the Instructions to Tenderers and the Contract. Tenderers shall not submit these forms with their tenders.
2. The **Contract Form** shall be completed by the Purchaser and then sent to the successful Tenderer for signature. Instructions on how to complete this form are included in the User's Guide.
3. The successful Tenderer shall fill in and complete the **Performance Security** form in accordance with ITT Clause 46 and the annotated version of the form provided in the User's Guide. Tenderers shall not submit this form with their tenders.
4. The successful Tenderer shall fill in and complete the **Bank Guarantee for Advance Payment** form in accordance with GCC Clause 21.1 and the annotated version of the form provided in the User's Guide. Tenderers shall not submit this form with their tenders.
5. Specify currencies in accordance with ITT Clause 14 using ISO (standard 4217) currency codes.
6. Specify countries using official ISO (standard 3166) country codes..

CONTRACT FORM

THIS AGREEMENT made the _____ day of _____, 20____,
between _____ of _____
(hereinafter "the Purchaser"), of the one part, and _____
of _____ (hereinafter "the Firm"), of the other part:

WHEREAS the Purchaser invited tenders for certain Products and Related Services, viz., _____ and has
accepted a Tender by the Firm for the supply of those Products and Related Services in the
sum of _____ (hereinafter
"the Contract Price").

NOW THIS AGREEMENT WITNESSETH AS FOLLOWS:

1. In this Agreement words and expressions shall have the same meanings as are respectively assigned to them in the Contract referred to.
2. The following documents shall be deemed to form and be read and construed as part of this Agreement, viz.:
 - (a) this Contract Form;
 - (b) the Special Conditions of Contract;
 - (c) the General Conditions of Contract;
 - (d) the Schedule of Supply;
 - (e) the Technical Specifications;
 - (f) the Tender and the Price Schedules submitted by the Firm;
 - (g) the Purchaser's Notification to the Firm of award of Contract; and
 - (h) _____.
3. In consideration of the payments to be made by the Purchaser to the Firm as indicated in this Agreement, the Firm hereby covenants with the Purchaser to provide the Products and Related Services and to remedy defects therein in conformity in all respects with the provisions of the Contract.
4. The Purchaser hereby covenants to pay the Firm in consideration of the provision of the Products and Related Services and the remedying of defects therein, the Contract Price or such other sum as may become payable under the provisions of the Contract at the times and in the manner prescribed by the Contract.

IN WITNESS whereof the parties hereto have caused this Agreement to be executed in accordance with the laws of Ghana on the day, month and year indicated above.

Signed by _____ (for the Purchaser)

Signed by _____ (for the Firm)

PERFORMANCE SECURITY

Date: _____

ICT No.: _____

Invitation for Tender No.: _____

Tender & Lot Nos.: _____

To: _____

WHEREAS _____ (hereinafter "the Firm") has undertaken, pursuant to Contract No. _____ dated _____, _____ to supply _____ (hereinafter "the Contract").

AND WHEREAS it has been stipulated by you in the aforementioned Contract that the Firm shall furnish you with a security _____ issued by a reputable guarantor for the sum specified therein as security for compliance with the Firm's performance obligations in accordance with the Contract.

AND WHEREAS the undersigned _____, legally domiciled in _____, (hereinafter "the Guarantor"), have agreed to give the Firm a security:

THEREFORE WE hereby affirm that we are Guarantors and responsible to you, on behalf of the Firm, up to a total of _____ and we undertake to pay you, upon your first written demand declaring the Firm to be in default under the Contract, without cavil or argument, any sum or sums within the limits of _____ as aforesaid, without your needing to prove or to show grounds or reasons for your demand or the sum specified therein.

This guarantee is valid until the _____ day of _____, _____.

Signed _____ in the capacity of _____

Name _____

Duly authorized to sign the authorization for and on behalf of _____

Dated on _____ day of _____, _____.

Bank Guarantee for Advance Payment

Date: _____

ICT No.: _____

Invitation for Tender No.: _____

Tender & Lot Nos.: _____

To: _____

In accordance with the payment provision included in the Contract, relation to advance payment, _____ (hereinafter called "the Firm") shall deposit with the Purchaser a bank guarantee to guarantee its proper and faithful performance under the said Clause of the Contract in the amount of _____.

We, the undersigned _____, as instructed by the Firm, agree unconditionally and irrevocably to guarantee as primary obligator and not as surety merely, the payment to the Purchaser on its first demand without whatsoever right of objection on our part and without its first claim to the Firm, in the amount not exceeding _____.

This guarantee shall remain valid and in full effect from the date of the advance payment received by the Firm under the Contract until _____, _____.

Name _____ In the capacity of _____

Signed _____

Duly authorized to sign for and on behalf of _____

Dated on _____ day of _____, _____.